

Cosmology

Core track, 2009 Winter Legacy Cosmology

Lectures by Leonard Susskind

Companion edition by [LazyingArt LLC](#) with [Video2Book](#)

About These Notes

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Chapter 1

Geometry of the Expanding Universe

The point is the expanding universe and its geometry. The aim is to describe what it means for space to expand before asking what drives the expansion. Newtonian $F = ma$ cosmology comes later, and only after that do geometry and dynamics get assembled into the full Friedmann–Robertson–Walker framework.

1.1 Comoving Coordinates on an Expanding Line

Start with the simplest possible example: one-dimensional space, not spacetime. Imagine an infinite line with equally spaced galaxies on it. The coordinate x does not yet measure a physical distance. It labels the galaxies. One may think of x as the name of a galaxy, and that name stays attached to the galaxy even if the whole arrangement stretches or contracts.

Let the physical distance from the galaxy called $x = 0$ to the galaxy called $x = 1$ be a . Then the physical distance between any two galaxies is

$$d = a \Delta x. \tag{1.1}$$

The coordinate difference Δx is only the difference between labels. The factor a converts that label difference into an actual distance measured by meter sticks.

Now imagine the line as an elastic thread being stretched. The labels stay with the galaxies, but the distance between neighboring galaxies changes with time. The same relation becomes

$$d(t) = a(t) \Delta x. \tag{1.2}$$

At this stage homogeneity means simply that neighboring galaxies are equally spaced everywhere, even though that common spacing may vary with time.

1.2 Hubble's Law and the Absence of a Center

To find the relative velocity of two fixed galaxies, keep the pair fixed so that Δx does not change with time. Differentiating gives

$$\dot{d} = \dot{a} \Delta x \quad (1.3)$$

$$= \frac{\dot{a}}{a} a \Delta x \quad (1.4)$$

$$= \frac{\dot{a}}{a} d. \quad (1.5)$$

If we define

$$H(t) \equiv \frac{\dot{a}}{a}, \quad (1.6)$$

then

$$\dot{d} = H(t) d. \quad (1.7)$$

This is Hubble's law.

The historical phrase “Hubble constant” is misleading. The quantity is constant in space, not in time. At a given moment it is the same for every galaxy, but in general it changes as the universe evolves. It is better thought of as the Hubble parameter.

The law says that more distant galaxies recede faster. It does not single out a center. Nothing in the argument privileges $x = 0$, or any other point. Every observer sees the same rule. The expanding universe in this model is therefore not a blast from one distinguished location.

Redshift enters immediately. If two galaxies exchange light signals while moving apart, the received light is reddened. The greater the recession speed, the greater the redshift. Spectral lines therefore tell us about recession speeds. Distances are harder to measure, and Hubble's early estimates were rough. His first plot was not a perfect straight line but a scattered set of points with large uncertainties. Even so, the geometric principle survived the noise: recession speed is proportional to distance.

Question from the audience. If the galaxies are far enough apart, does Hubble's law make them move faster than the speed of light?

^a **Response.** Yes, in this recession sense it does. What you are saying is correct. We will come back to that for sure. It is one of the most important things.

^aLecture timestamp: 00:20:23.

1.3 Higher Dimensions, Isotropy, and the Time Dependence of $a(t)$

Nothing essential changes when more spatial dimensions are added. In two dimensions imagine a square grid of galaxies labeled by (x, y) , and in three dimensions by (x, y, z) . The coordinates are again comoving labels. They move with the galaxies instead of measuring physical distance directly.

Why should the spacing in the x -direction be the same as the spacing in the y -direction? The answer is observational. If galaxies were systematically more crowded in one direction than another, the sky would look different in different directions. On sufficiently large scales it does not. That directional sameness is isotropy.

In two dimensions the distance between two galaxies is

$$d = a\sqrt{(\Delta x)^2 + (\Delta y)^2}, \quad (1.8)$$

and in three dimensions

$$d = a\sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}. \quad (1.9)$$

For a fixed pair of galaxies the coordinate differences remain constant, so differentiating again gives

$$\dot{d} = \frac{\dot{a}}{a} d. \quad (1.10)$$

Hubble's law is unchanged.

Homogeneity means sameness from place to place. Isotropy means sameness from direction to direction. Neither is exact on small scales. Looking through the Milky Way is different from looking out of its plane, and individual galaxies are not uniformly spread at short distances. But after averaging over many galaxies, the large-scale universe looks approximately the same everywhere and in every direction.

Question from the audience. If velocity is proportional to distance, and the distance is increasing, does that mean a fixed pair of galaxies must be accelerating away from each other?

^a **Response.** No. Hubble's law says that at a given moment the recession velocity is proportional to the distance. It does not by itself say whether the velocity of one fixed pair increases or decreases with time. That depends on the detailed time dependence of $a(t)$. If $a(t) \propto t$, neighboring galaxies separate at constant speed. If $a(t) \propto t^2$, the recession speed of a fixed pair increases with time. If $a(t) \propto \log t$, it decreases with time. The law $\dot{d} = (\dot{a}/a)d$ is always true as a function of distance, but acceleration and deceleration are separate questions.

^aLecture timestamp: 00:40:22.

For a long time cosmologists expected the universe to be decelerating, since gravity should slow an initial expansion the way gravity slows an object thrown upward. Here the point is only that Hubble's law by itself does not determine the behavior of $\dot{a}$ or $\ddot{a}$.

Question from the audience. How do you set the initial value of the scale factor a ?

^a **Response.** Its overall normalization is arbitrary. What matters physically is not the numerical size of a by itself, but how it changes. If one defines another scale factor $b = 3a$, then $\dot{b}/b = \dot{a}/a$. Rescaling a by a constant changes conventions or units, not the Hubble parameter.

^aLecture timestamp: 00:46:18.

1.4 From the Spatial Metric to the Spacetime Metric

Now replace Δx by dx . The physical distance between neighboring points is

$$ds = a dx, \quad (1.11)$$

so

$$ds^2 = a^2 dx^2. \quad (1.12)$$

This is the spatial metric in one dimension. The coefficient of dx^2 is

$$g_{xx} = a^2. \quad (1.13)$$

An expanding universe can therefore be described by saying that the spatial part of the metric depends on time.

To pass from space to spacetime, use the relativistic interval. In one spatial dimension,

$$d\tau^2 = dt^2 - a^2(t) dx^2. \quad (1.14)$$

For a comoving observer, $dx = 0$, so $d\tau = dt$. Ordinary cosmic time is the proper time measured by clocks moving with the galaxies.

The blackboard picture at this point is useful because it puts the comoving worldlines, the line element, and the metric notation together.

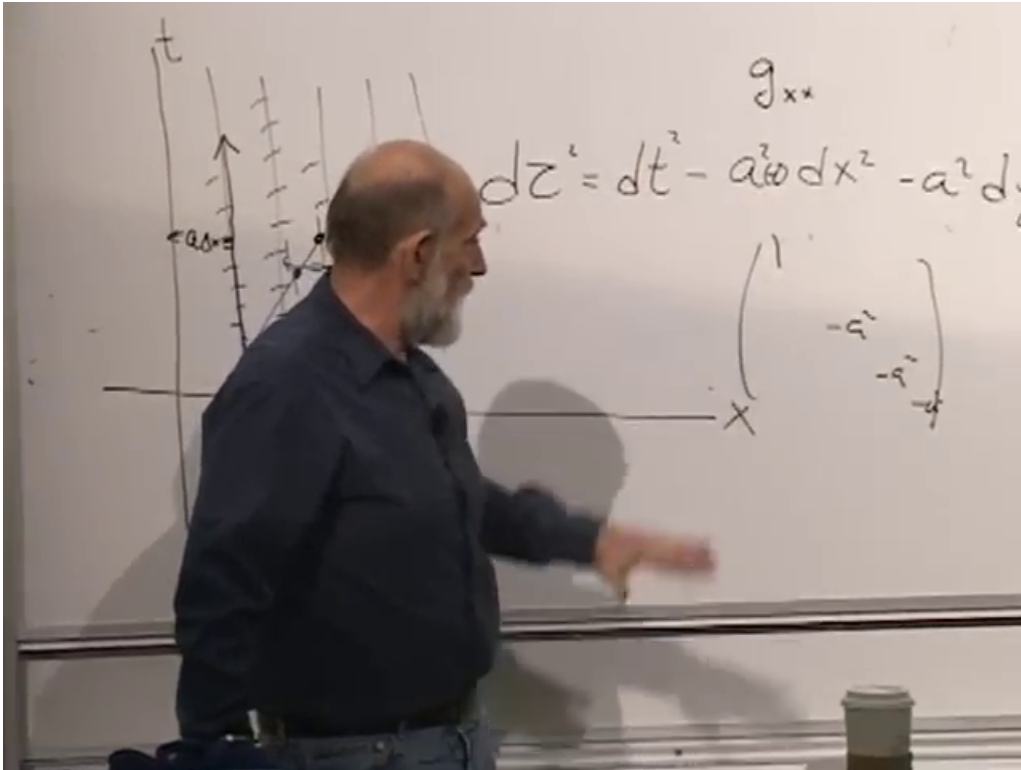


Figure 1.1: Comoving worldlines in an expanding one-dimensional universe together with the corresponding spacetime interval.

In three spatial dimensions the same pattern gives the cautious extension¹

$$d\tau^2 = dt^2 - a^2(t)(dx^2 + dy^2 + dz^2), \quad (1.15)$$

with diagonal metric

$$g_{\mu\nu} = \text{diag}(1, -a^2, -a^2, -a^2). \quad (1.16)$$

¹The board is partly hidden around 00:56:38–00:58:51. This three-dimensional form follows the spoken extension from one spatial dimension to three together with the visible diagonal pattern on the board.

The off-diagonal terms vanish,

$$g_{tx} = g_{ty} = g_{tz} = 0. \quad (1.17)$$

Homogeneity means that the metric coefficients may depend on time but not on spatial position. Isotropy means that the three spatial directions enter in the same way.

Question from the audience. Are we not supposed to think of dx as a distance?

^a **Response.** No. dx is not the distance. $a\,dx$ is the distance. dx is just a label interval. A useful way to think about it is to imagine that, at some moment, someone draws a coordinate mesh through the galaxies and then requires the mesh to follow them. The coordinate separation stays fixed by construction, while the physical distance changes. One could put the units of distance into x instead of into a ; that is a convention. What matters is the product $a\,dx$.

^aLecture timestamp: 01:01:18.

Question from the audience. The metric coefficient of dt^2 is equal to 1. Is that just by convention also?

^a **Response.** Yes. There is no physics in that by itself. If the coefficient depended only on time, one could change the time variable so that it became just dt^2 . Likewise one may choose units with $c = 1$. What is not a convention here is that the coefficient does not depend on position, although it may depend on time.

^aLecture timestamp: 01:03:24.

Question from the audience. If you put this into Einstein's field equations, what do you get for the energy-momentum tensor?

^a **Response.** That is the next kind of question, but it belongs to dynamics and not to the present discussion. One can put this metric into Einstein's equations and ask what energy-momentum distribution would make it a solution. That is very important, but for the moment the subject is geometry, not dynamics.

^aLecture timestamp: 01:05:59.

1.5 What the Horizon Hides

The large-scale universe is treated as homogeneous and isotropic only as an approximation. It is certainly not homogeneous on the scale of stars, planets, or even individual galaxies. It is not exactly isotropic in every local observation either.

A farmer's field gives the right scale dependence. On one scale it looks regular; on a much smaller scale it plainly does not; on a scale larger than the whole field it again ceases to look homogeneous. Cosmology works the same way. After averaging over many galaxies, the large-scale distribution looks roughly uniform. On sufficiently small scales it does not.

The universe is bigger than the part we can see, and bigger than the part we can ever see. On the scales that can actually be probed, it is isotropic and homogeneous apart from local variations. Beyond that there is no direct observational guarantee.

Question from the audience. You said we will never really directly find out. Does that imply that there might be an indirect way to find out?

^a **Response.** Perhaps. Most thinking these days is that on big enough scales the universe is not homogeneous, and my own bet is the same. But it is a bet I will never win and never lose, because for basic fundamental reasons we will not be able to see much farther than we can see now. That is the existence of a horizon.

^aLecture timestamp: 01:10:34.

Question from the audience. Couldn't you use gravity to look for masses beyond what you can see?

^a **Response.** No. A distant mass can simply put us into free fall. Sitting in a closed room, we are in the gravitational field of the Sun and orbiting it, but without seeing outside there is no easy way to tell that the Sun is there. Only tidal forces reveal it, and if the source is far enough away those tidal effects can be too small to detect.

^aLecture timestamp: 01:11:53.

The horizon is therefore an ultimate barrier to direct knowledge. He even says, with virtual certainty, that the universe is at least a thousand times larger in volume than the region we can ever see, and leaves the reason for that claim for later.

1.6 Bounded Without Boundary

Now come to closed geometries. The infinite line is open and unbounded. A line segment is bounded in the sense of having only a finite extent. Bounded does not mean a boundary, but finite extent.

A cube with walls will not do. It violates homogeneity, because somebody near a wall is not equivalent to somebody far from a wall, and it violates isotropy, because even somebody at the center sees different things in different directions.

The interior of a circle or ball also fails. Somebody at the center sees the same thing in every direction, but that symmetry belongs only to one special point. Move away from the center and the wall is near in one direction and far in another. The space is then neither homogeneous nor isotropic for generic observers.

On very large scales the working assumption is that the universe is homogeneous and isotropic. This is the cosmological principle. It is useful, but it is not protected from revision.

Question from the audience. Does isotropy everywhere imply homogeneity?

^a **Response.** Yes. If every observer sees the same thing in every direction, that implies homogeneity. But the converse is false. Homogeneity by itself does not imply isotropy. A universe can be homogeneous and still have a preferred direction, for example if it were filled with arrows all pointing the same way.

^aLecture timestamp: 01:18:57.

The right bounded model is the surface of a featureless sphere. A two-dimensional creature confined to that surface finds every place locally like every other place and every direction like every other direction, even though the world is finite. That is bounded without boundary.

1.7 A Closed One-Dimensional Universe on a Circle

To keep the algebra simple, reduce the sphere example to one spatial dimension: life on a circle. In one dimension isotropy means that left and right are equivalent. A circle has exactly that feature. Choose one point and call it $\theta = 0$. Every other point is labeled by an angular coordinate θ , measured in radians. A point just to one side of the origin can be described either by a small negative angle or by an angle just short of 2π . That periodicity is part of the geometry.

The inhabitants of the circle have access to the circumference, not to an external radius or center. They can measure the total distance around by marching along the circle until they return to the same point. If one draws the circle in an ambient plane, one may also introduce a radius-like quantity, but that belongs to the embedding picture, not to the intrinsic geometry they measure.

The blackboard picture is useful here because it keeps the angular labels and the total circumference on the same drawing.

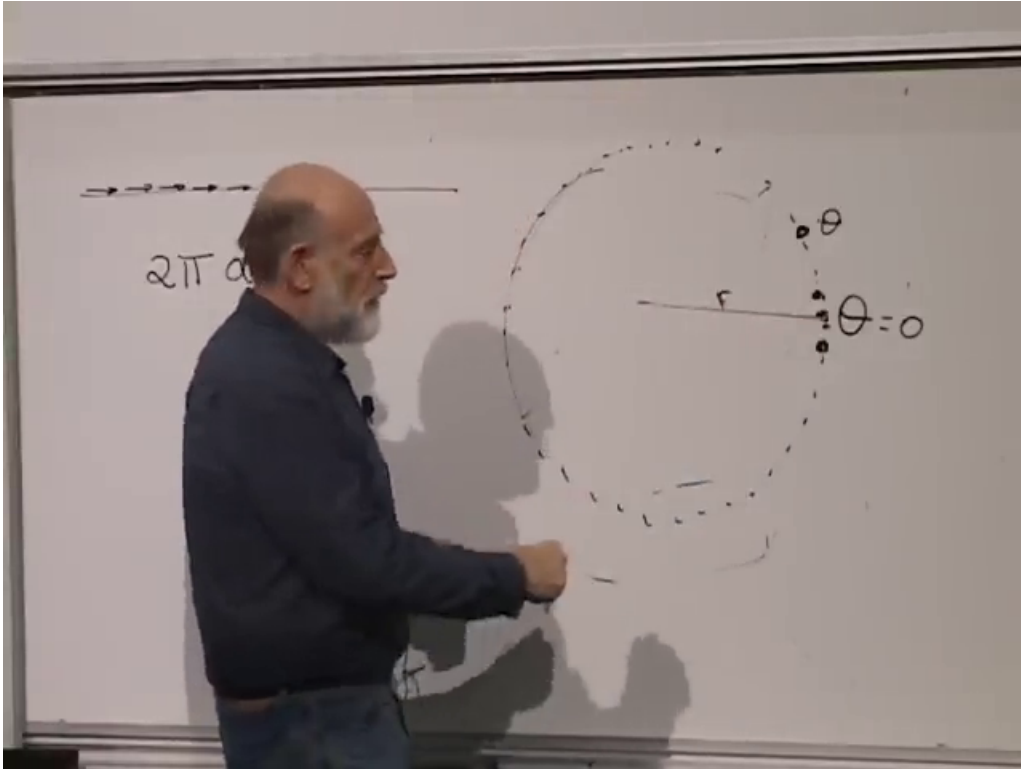


Figure 1.2: Angular description of a closed one-dimensional universe, with total circumference $2\pi a$.

A convenient convention is²

$$L = 2\pi a, \quad (1.18)$$

where L is the total circumference.

For a small angular separation $\Delta\theta$, the physical distance is

$$d = a \Delta\theta. \quad (1.19)$$

²Around 01:26:40–01:29:20 the notation shifts briefly between taking the whole circumference itself as the scale factor and writing it as $2\pi a$. The later convention is used here because it is the one carried into the arc-length formula.

Passing to infinitesimals gives

$$ds = a d\theta, \quad (1.20)$$

and therefore

$$ds^2 = a^2 d\theta^2. \quad (1.21)$$

This is the local metric on the circle.

If a depends on time, the closed one-dimensional universe expands or contracts. For a fixed pair of galaxies,

$$d(t) = a(t) \Delta\theta, \quad (1.22)$$

so

$$\dot{d} = \dot{a} \Delta\theta = \frac{\dot{a}}{a} d. \quad (1.23)$$

Hubble's law survives here as well, at least for pairs that are not so far apart that the other way around the circle becomes ambiguous.

The spacetime metric becomes

$$d\tau^2 = dt^2 - a^2(t) d\theta^2. \quad (1.24)$$

If a is constant, the spacetime picture is a cylinder. If $a \propto t$, it is a cone. If $a(t)$ changes more irregularly, the picture becomes vase-like.

Run the picture backward and the galaxies get closer and closer together. If there were 360 galaxies around the circle, they would be well spread out at late times, very close together earlier, then closer still. Eventually they would not be separate galaxies anymore, and earlier still not even separate atoms. But there is still no special point in space.

That is the cleanest answer to the question of where the Big Bang happened in the model. As $a(t) \rightarrow 0$, every separation $a(t) \Delta\theta$ goes to zero. The whole universe shrinks everywhere at once. The galaxies are not flying away from one privileged location in an external space. The geometry itself is changing scale. In that sense the answer is: everywhere.

Question from the audience. You said the spacetime is curved. Is it curved if that were a cylinder?

^a **Response.** No. If it were a cylinder, it would not be curved. That is the case where a is constant in time. It looks curved only because it is drawn in a higher-dimensional space. If one cuts it open, it can be laid flat without stretching.

^aLecture timestamp: 01:38:04.

From there the discussion turns to the cone. If $a \propto t$, the spacetime picture is a cone. Away from its tip the cone is also flat in the same intrinsic sense: it can be opened up into a plane. The tip is different. It is a conical singularity.

The next step is the geometry of the spherical universe and then the dynamics that determine how $a(t)$ grows or shrinks.

Chapter 2

Cosmology Lecture 2: From the Expanding Metric to Friedmann's Equation

We were talking last time about the metric of space. This lecture picks up exactly there and asks the next question: what equation tells the scale factor how to change with time? Susskind announces the plan immediately. First a little geometry, then dynamics; and the dynamics, at least for now, will be derived not from general relativity but from Newton's equations. That choice gives the lecture its shape. We begin with an expanding comoving grid, pass through a deliberately simple projectile problem, and then watch that one-body Newtonian problem turn into Friedmann's equation. The selected board frames are kept below as local mathematical evidence, following Leonard Susskind's lecture with supporting curation from LazyingArt LLC and Video2Book.

2.1 From Homogeneity to the Expanding Metric

The standing assumptions are homogeneity and isotropy. Under those assumptions Hubble's law is not something tacked on from observation after the fact; it is the natural kinematics of a uniformly expanding universe. At a fixed time,

$$v = H(t) d. \tag{2.1}$$

The proportionality is linear in the distance, and the important qualification comes immediately: the so-called Hubble constant is constant in space, not in time. Homogeneity means it does not vary from place to place. It does *not* mean it remains unchanged as the universe evolves.

Before writing any metric, Susskind pauses over the question of measuring rods. If space expands, why do meter sticks not expand? The lecture's answer is concrete. The forces binding ordinary matter are enormously stronger than the tiny effect of cosmological expansion on atomic scales. One may imagine adding a minuscule extra repulsion between atoms; it would shift an equilibrium separation a little, but it would not tear apart a tightly bound rod. The cosmological expansion is therefore seen not in bound objects but in the growing separation of sufficiently distant, sufficiently unbound systems such as galaxies.

This is the reason for comoving coordinates. We assign coordinates that move with the Hubble flow. A galaxy with no extra peculiar motion then keeps the same coordinate value for all time. What

changes is the relation between coordinate separation and physical distance measured in meters. That changing conversion factor is the scale factor $a(t)$.

For flat space the spatial line element is

$$ds^2 = a(t)^2 (dx^2 + dy^2 + dz^2). \quad (2.2)$$

Here x, y, z are comoving coordinates. If two galaxies have a fixed coordinate separation Δx along one axis, then their physical distance is

$$d(t) = a(t) \Delta x. \quad (2.3)$$

Differentiating with respect to time gives

$$v = \dot{d} = \dot{a} \Delta x = \frac{\dot{a}}{a} d. \quad (2.4)$$

So once the comoving metric is in hand, the Hubble parameter becomes

$$H(t) = \frac{\dot{a}}{a}. \quad (2.5)$$

That is already the first bridge from geometry to dynamics.

2.2 Flat Spatial Slices Inside Curved Spacetime

Now we shift from the metric of space to the metric of spacetime. The lecture insists on an important point: three-dimensional space may be flat while spacetime is nevertheless curved. If the geometry of space changes with time, then spacetime is no longer Minkowski even if each spatial slice remains flat. Susskind's image is a flat rubber sheet being stretched by distant giants. The sheet stays flat as a sheet, but the four-dimensional history is curved.

Starting from the special-relativistic form $d\tau^2 = dt^2 - ds^2$, we upgrade the spatial metric to

$$d\tau^2 = dt^2 - a(t)^2 (dx^2 + dy^2 + dz^2). \quad (2.6)$$

The coefficients do not depend on position in space; they depend only on time. That is the metric expression of homogeneity. In matrix language, completing the spoken board description in the standard way, one writes

$$g_{\mu\nu} = \text{diag}(1, -a(t)^2, -a(t)^2, -a(t)^2). \quad (2.7)$$

The off-diagonal terms vanish. There are no $dx dy$ or $dx dt$ terms. The metric is almost as simple as that of flat spacetime, except that the conversion between spatial separation and time is now time-dependent.

The picture is not decoration. It carries the lecture's intuition. Alice and Bob remain one coordinate unit apart, but one coordinate unit comes to represent more and more physical distance as time goes on. If $a(t)$ were driven to zero, then every physical separation would be driven to zero with it. In the lecture this is said in plain language: everything gets squashed on top of everything else, and the density becomes infinite. If $a(t)$ grows, the universe becomes correspondingly more diffuse.

There is another subtle point here. This metric has the stated form only in the frame moving with the galaxies. Rotations leave the spatial part unchanged, but a Lorentz boost does not preserve the same isotropic appearance. Matter itself picks out the preferred comoving frame.

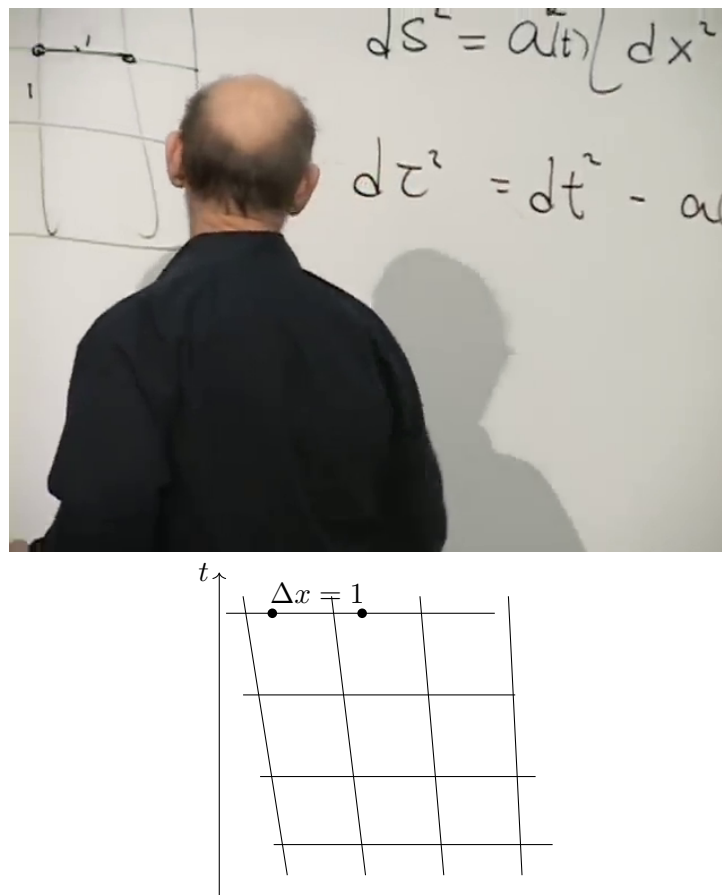


Figure 2.1: Spacetime interval and expanding-grid sketch. The lecture frame is retained as board evidence; the redraw beneath it makes the comoving grid legible as a geometric cartoon of fixed coordinate separation and increasing physical distance.

2.2.1 Question & Answer

Whose time is t in $a(t)$? It is the time measured by clocks moving with the galaxies. The lecture even says: imagine a cosmic clock factory making identical clocks, one carried by each galaxy. The coordinate time t is the time those comoving clocks measure.

This is also immediate from the metric. Along a galaxy worldline in the comoving frame we have

$$dx = dy = dz = 0, \quad (2.8)$$

so the interval reduces to

$$d\tau^2 = dt^2, \quad (2.9)$$

hence

$$dt = d\tau. \quad (2.10)$$

So the time appearing in $a(t)$ is cosmic proper time along the flow lines of the galaxies.

2.3 Newtonian Warm-Up: A Projectile in a Gravitational Field

With the geometry fixed, Susskind makes the lecture's second major pivot. Rather than deriving the dynamics of $a(t)$ from general relativity, he cools the problem down first and goes back to a familiar Newtonian one-body system. Before doing cosmology, let us understand a particle ejected radially outward from a heavy gravitating mass.

Take a heavy mass M and a much smaller mass m . Give the small mass an outward impulse and then let it move freely under gravity. Angular complications are not the point here, so the motion is taken to be radial. What matters for the cosmological analogy is energy conservation.

The kinetic energy is

$$\frac{1}{2}mv^2. \quad (2.11)$$

The gravitational potential energy has the product of the masses, Newton's constant, and one power of the distance in the denominator:

$$-\frac{GMm}{d}. \quad (2.12)$$

On the board the factors are written in the order mMG/D ; in the cleaned equations we restore the conventional form GMm/d , but the board order is kept visible in the screenshots below. The conserved total energy is then

$$E = \frac{1}{2}mv^2 - \frac{GMm}{d}. \quad (2.13)$$

Here “constant” means constant in time for the particular particle under discussion. Different projectiles may of course have different values of E .

Notice how the lecture builds the formula. First the kinetic term is written. Then the potential term is added. Only after that does the negative sign become the issue. The sequence matters because later the cosmological argument will lean on the same intuition: not just on the existence of a formula, but on what the sign of the gravitational term means.

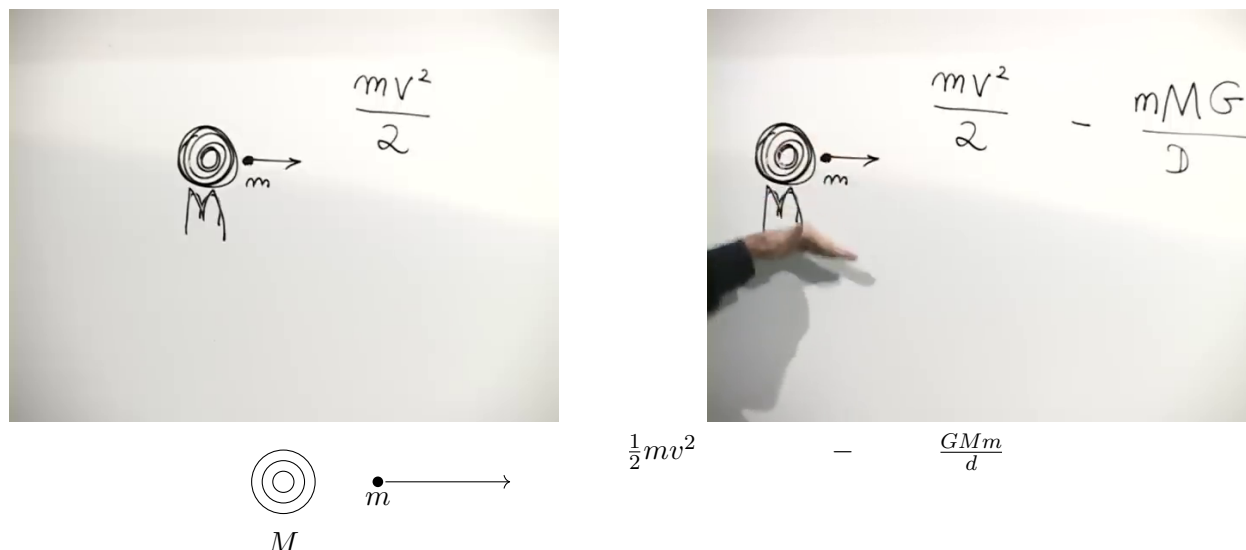


Figure 2.2: The Newtonian warm-up in the same order as the board. First the kinetic term, then the full energy balance with negative gravitational potential. The redraw regularizes the source–test-mass geometry without replacing the lecture frames.

2.3.1 Question & Answer

Why is the gravitational potential energy negative? The lecture does not answer this by abstract convention. It answers it by drawing the graphs of $1/D$ and $-1/D$. Forget the constants. The function $1/D$ decreases as D increases. The function $-1/D$ is the same curve upside down. Forces push in the direction of decreasing potential energy. Therefore an attractive force, which pulls inward toward smaller D , must correspond to the upside-down curve:

$$V(d) \propto -\frac{1}{d}. \quad (2.14)$$

Bringing the mass closer to the source creates a larger amount of negative potential energy. Moving it farther away makes the energy less negative. That is exactly what attraction means in energy language.

2.4 Escape Velocity and the Three Energy Regimes

Once the conserved energy is on the board, the lecture does not calculate right away. Instead it classifies the possible motions. There are only three cases.

- If $E < 0$, the motion is bound. The particle rises, slows, turns around, and falls back.
- If $E = 0$, the particle has exactly the escape velocity. The distance keeps increasing forever, while the speed decreases toward zero.
- If $E > 0$, the particle escapes completely. Gravity still decelerates it for a while, but not enough to prevent a nonzero asymptotic outward speed.

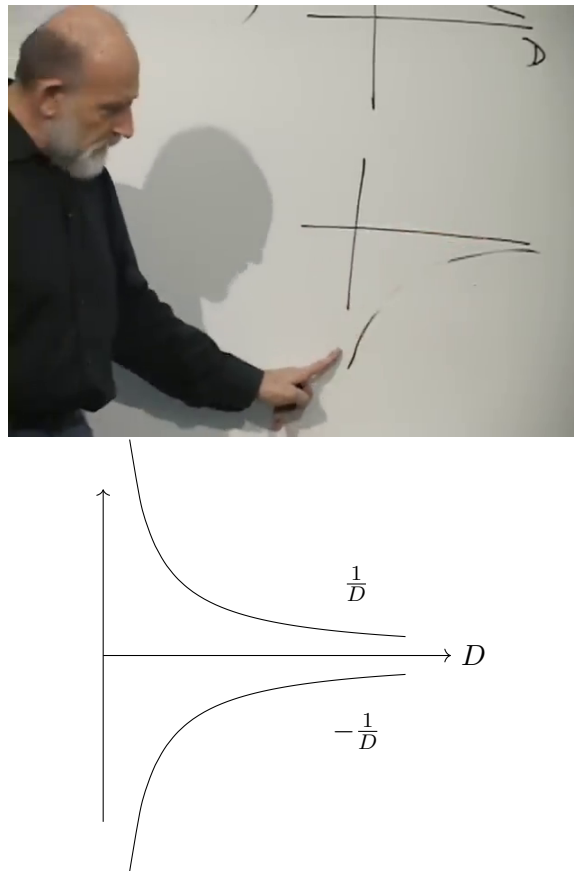


Figure 2.3: Qualitative potential-energy graphs. The lecture frame is kept because it records the actual board argument; the redraw regularizes the axes and the two comparison curves $1/D$ and $-1/D$.

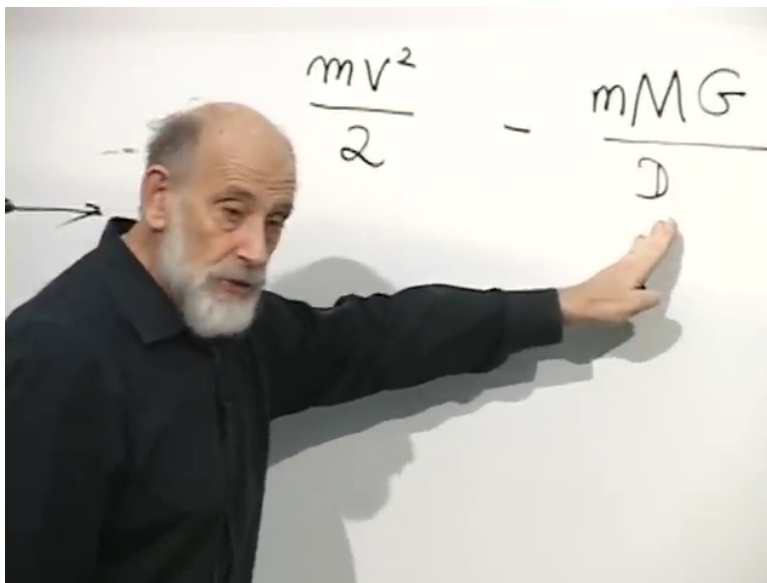


Figure 2.4: Later emphasis frame: at small enough initial outward speed, the negative potential term dominates the total energy.

The critical case $E = 0$ needs to be heard carefully. The distance is unbounded, but the speed goes to zero. It cannot actually come to rest at any finite time, because if it did, it would turn around and fall back. So the speed decreases forever without ever quite reaching zero at finite time.

For sufficiently small starting velocity the potential term dominates from the beginning, so the energy is negative:

This is the bridge to cosmology. The universe, in Newtonian language, behaves the same way. If the initial Hubble flow is large enough, galaxies escape from one another forever. If it is too small, gravity eventually wins and the universe recollapses. Between the two lies the critical case, the cosmological analog of escape velocity.

2.5 Newton's Shell Theorem and the Cosmological Reduction

The projectile analogy becomes a cosmological calculation only because of Newton's shell theorem. This is the lecture's third hinge, and Susskind takes time to make the audience trust it.

For a spherically symmetric mass distribution, the gravitational force on a particle at radius d is exactly the same as if all the mass inside radius d were concentrated at the center, while the mass outside radius d contributes no net force. The theorem is stronger than the familiar outside-a-sphere statement. A perfect spherical shell exerts no force on a particle anywhere inside it, even off-center. This is a surprising property, and in the lecture Susskind reminds us that it is special to the inverse-square law.

He even sketches a water-flow analogy: a point source produces a $1/r^2$ -type flux in three dimensions, and a symmetric shell of such sources produces no net interior flow. The details are postponed, but the moral is clear. Exterior shells cancel.

Now we use that theorem in cosmology. Pick any galaxy and, for the sake of calculation, call it the

center. Because the universe is homogeneous and isotropic, this choice cannot matter. Pick a second galaxy at comoving coordinate x . Draw a sphere centered on the first galaxy and passing through the second. Then only the mass inside that sphere matters for the motion of the second galaxy.

Two further facts are essential.

First, the center is arbitrary. We could have centered the calculation on Alice, Bob, Fred, or any other galaxy, and the same equation must come out.

Second, the sphere expands with the Hubble flow. It therefore encloses the same galaxies for all time, and so its enclosed mass is constant in time even though its physical radius changes.

2.6 From Galaxy Energetics to Friedmann's Equation

We are now ready to write for a galaxy exactly the same kind of Newtonian energy equation we wrote for the projectile. After dividing by the test mass, it is convenient to write

$$\frac{v^2}{2} - \frac{GM(d)}{d} = \varepsilon(x), \quad (2.15)$$

where $\varepsilon(x)$ is constant in time for the galaxy at comoving coordinate x . It is not required to be the same for every x .

The physical distance from the chosen center to that galaxy is

$$d = a(t)x, \quad (2.16)$$

and because x is comoving,

$$v = \dot{d} = \dot{a} x. \quad (2.17)$$

If $\rho(t)$ is the homogeneous density, then the enclosed mass is

$$M(d) = \frac{4\pi}{3} \rho d^3 = \frac{4\pi}{3} \rho a^3 x^3. \quad (2.18)$$

Substituting these into the energy equation gives

$$\frac{(\dot{a}x)^2}{2} - \frac{G \left(\frac{4\pi}{3} \rho a^3 x^3 \right)}{ax} = \varepsilon(x), \quad (2.19)$$

$$\frac{\dot{a}^2 x^2}{2} - \frac{4\pi G}{3} \rho a^2 x^2 = \varepsilon(x). \quad (2.20)$$

Here the lecture lingers on the factor of x^2 , and rightly so. Both terms on the left-hand side are proportional to x^2 . Everything else there is the same for every galaxy. Therefore the right-hand side must also be proportional to x^2 . That is the precise meaning of the seemingly casual statement that the “constant” must scale with x^2 . The farther galaxies carry larger energies, and that is exactly what one should expect since they are also moving faster.

So we write

$$\varepsilon(x) = -\frac{k}{2} x^2, \quad (2.21)$$

where the normalization of k is conventional. Dividing through by $x^2/2$, we obtain

$$\dot{a}^2 = \frac{8\pi G}{3} \rho a^2 - k. \quad (2.22)$$

Dividing once more by a^2 , we arrive at the standard historical form,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \quad (2.23)$$

This is Friedmann's equation in the Newtonian form emphasized in the lecture. One can also write it as

$$H(t)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}, \quad (2.24)$$

since $H = \dot{a}/a$. Susskind likes to pause here and point out the historical drama of the result: a purely Newtonian energy argument has landed on the same equation that later appears as a component of Einstein's equations.

The sign of k tracks the same three cases we already met in the escape-velocity problem. The critical case is $k = 0$. In units with $c = 1$, k is dimensionless.

2.7 Matter Density, the Critical Case, and the Age of the Universe

If we sent ourselves home at this point and tried to solve Friedmann's equation, the first objection would be immediate: ρ is not a constant. It changes as the universe expands. So the next task is to express that dependence cleanly.

Susskind does it with a unit comoving cube. Take a cube that is one unit on a side in coordinate space. The galaxies move with the comoving coordinates, so the same galaxies remain inside that cube for all time. Let the mass inside it be $m_\square$. Then the physical side length of the cube is $a(t)$, and the physical volume is $a(t)^3$. Therefore

$$\rho(t) = \frac{m_\square}{a(t)^3}. \quad (2.25)$$

Substituting this into Friedmann's equation gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G m_\square}{3a^3} - \frac{k}{a^2}. \quad (2.26)$$

Now choose the simplest case, $k = 0$, the analog of exact escape velocity. Then

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G m_\square}{3a^3}. \quad (2.27)$$

Before solving, the lecture points out a qualitative consequence. The right-hand side is always positive. Therefore the left-hand side cannot pass through zero, which means the model does not turn around. In the critical case the universe expands forever, but with a continually decreasing expansion rate.

Now comes the worked solution. Rather than grinding through the differential equation abstractly, Susskind tries a power law:

$$a(t) = ct^p. \quad (2.28)$$

Then

$$\dot{a} = cp t^{p-1}, \quad \frac{\dot{a}}{a} = \frac{p}{t}. \quad (2.29)$$

Substituting into the equation gives

$$\frac{p^2}{t^2} = \frac{8\pi G m_{\square}}{3c^3 t^{3p}}. \quad (2.30)$$

For this to hold for all t , the powers of t must match:

$$3p = 2. \quad (2.31)$$

So

$$p = \frac{2}{3}. \quad (2.32)$$

Then matching coefficients gives

$$\frac{4}{9} = \frac{8\pi G m_{\square}}{3c^3}, \quad (2.33)$$

hence

$$c^3 = 6\pi G m_{\square}. \quad (2.34)$$

The critical matter-dominated solution is therefore

$$a(t) = (6\pi G m_{\square})^{1/3} t^{2/3}. \quad (2.35)$$

More generally one may absorb the remaining integration constant into a shift of the origin of time and write

$$a(t) \propto (t - t_0)^{2/3}. \quad (2.36)$$

This is a more interesting result than it may look at first sight. The universe does not expand linearly with time. Linear growth would be the analog of unaccelerated motion. Here the growth is slower than linear:

$$a(t) \propto t^{2/3}. \quad (2.37)$$

So the expansion slows down, but in the critical case it never turns around.

The interpretation of $t = 0$ is also part of the lecture's rhythm. If $a(t)$ really went to zero, everything would be squashed on top of everything else and the density would diverge. Susskind immediately softens the literalness of that statement: the Newtonian model cannot be trusted all the way down to infinite density. But it does tell us that the universe emerged from an extremely dense early state. That is the Big Bang limit of the model.

The Hubble parameter now becomes explicit:

$$H(t) = \frac{\dot{a}}{a} = \frac{2}{3t}. \quad (2.38)$$

So in this model

$$t = \frac{2}{3H}. \quad (2.39)$$

This is one of the lecture's payoffs. The Hubble constant is not a constant at all; it falls like $1/t$. Susskind jokes that a final examination could consist of a single question: *Is the Hubble constant a constant?* The correct answer is no.

The same formula turns a present-day measurement of H into an age estimate. In the lecture Susskind quotes an age of roughly 13 to 14 billion years and remarks that $k = 0$ is observationally very realistic, with no known significant deviation from flatness at the level of precision he quotes.

2.7.1 Question & Answer

Why does the age estimate not depend on the arbitrary fiducial box mass? The lecture's late discussion of this point is subtle because the notation becomes slippery, but the underlying idea is clean. The fiducial comoving box is conventional. If we double its side length, then the mass inside it grows by a factor of 8, while $m_{\square}^{1/3}$ grows by a factor of 2. The corresponding normalization of a changes by the same factor. What is physical is the density

$$\rho = \frac{m_{\square}}{a^3}, \quad (2.40)$$

or equivalently the invariant combination $a/m_{\square}^{1/3}$, not a and $m_{\square}$ taken separately.

The cancellation appears directly in the solution. If

$$a(t) = ct^{2/3}, \quad (2.41)$$

then

$$\dot{a} = \frac{2}{3}ct^{-1/3}, \quad (2.42)$$

so

$$\frac{\dot{a}}{a} = \frac{\frac{2}{3}ct^{-1/3}}{ct^{2/3}} = \frac{2}{3t}. \quad (2.43)$$

The constant c , and therefore the dependence on the fiducial-box mass, drops out of $H = \dot{a}/a$. That is why the age relation

$$t = \frac{2}{3H} \quad (2.44)$$

does not depend on the arbitrary normalization of the comoving box in the critical model.

This is also the right place to hear the lecture's last reinterpretation. Rearranging Friedmann's equation gives

$$\frac{\dot{a}^2}{a^2} - \frac{k}{a^2} = \frac{8\pi G}{3}\rho. \quad (2.45)$$

The left-hand side is geometric: it is built from the scale factor and, next time, from spatial curvature. The right-hand side is matter density, or in relativistic language energy density. That is exactly the sort of equation Einstein's theory ought to produce.

2.8 Summary

The lecture unfolds in a very deliberate sequence. Homogeneity and isotropy fix the kinematics of expansion. The comoving metric gives the scale factor a precise meaning. A Newtonian projectile problem provides the right intuition for conserved energy and escape velocity. Newton's shell theorem then converts the many-body universe into an effective one-body problem, and that reduction yields

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \quad (2.46)$$

In the critical matter-dominated case $k = 0$, the solution is

$$a(t) \propto t^{2/3}, \quad H(t) = \frac{2}{3t}. \quad (2.47)$$

So the Newtonian model already gives a serious cosmological prediction: a slowing but never reversing critical expansion, together with an age estimate from the present Hubble parameter. The next lecture will reopen the equation from the relativistic side and show how curvature and Einstein's equations complete the story.

Chapter 3

Reviewing Friedmann, Adding Radiation, and Finding Dark Matter

This lecture is staged very deliberately. We begin in review mode, going back through the basic cosmological equation in the simplest possible setting, not because Newton is the last word, but because in this setting Newton and Einstein momentarily say the same thing. Then we change only one ingredient at a time: first the interpretation of ρ , then the dominant contents of the universe, then the observational consequences of the resulting scaling laws. Only after that cosmological spine is in place do we turn to dark matter and use the same Newtonian instincts again, now on galactic scales. These notes follow Leonard Susskind's lecture order and rhythm, with transcript and frame curation indebted to LazyingArt LLC and Video2Book.

3.1 Homogeneity, Comoving Coordinates, and the Newtonian Setup

Susskind opens by reminding us that on scales larger than clusters of galaxies, the universe looks smooth enough to be treated as homogeneous. That does not mean it is literally featureless. It means that when we average over sufficiently large distances, the jagged astronomical picture gives way to something simple enough for cosmology.

With that approximation in hand, we choose ourselves to sit at the origin of coordinates. This choice has no physical privilege attached to it; it is merely a convenient orientation. We then place a representative galaxy, or if we want to strip away all unnecessary baggage, a purely imaginary test particle moving with the Hubble flow, at comoving coordinate $x = 1$. We are at $x = 0$.

Now the scale factor is introduced in the lecture in the most concrete possible way. The physical distance to the particle at $x = 1$ is, by definition,

$$a(t). \tag{3.1}$$

A particle at $x = 2$ is then at distance $2a(t)$, and so on. In other words, $a(t)$ is not first presented as an abstract metric coefficient; it is the physical distance to a conveniently chosen comoving particle.

The corresponding Newtonian picture is just as simple. We draw a sphere centered on us, with radius equal to the particle's distance $a(t)$. The only gravitating material that matters for the

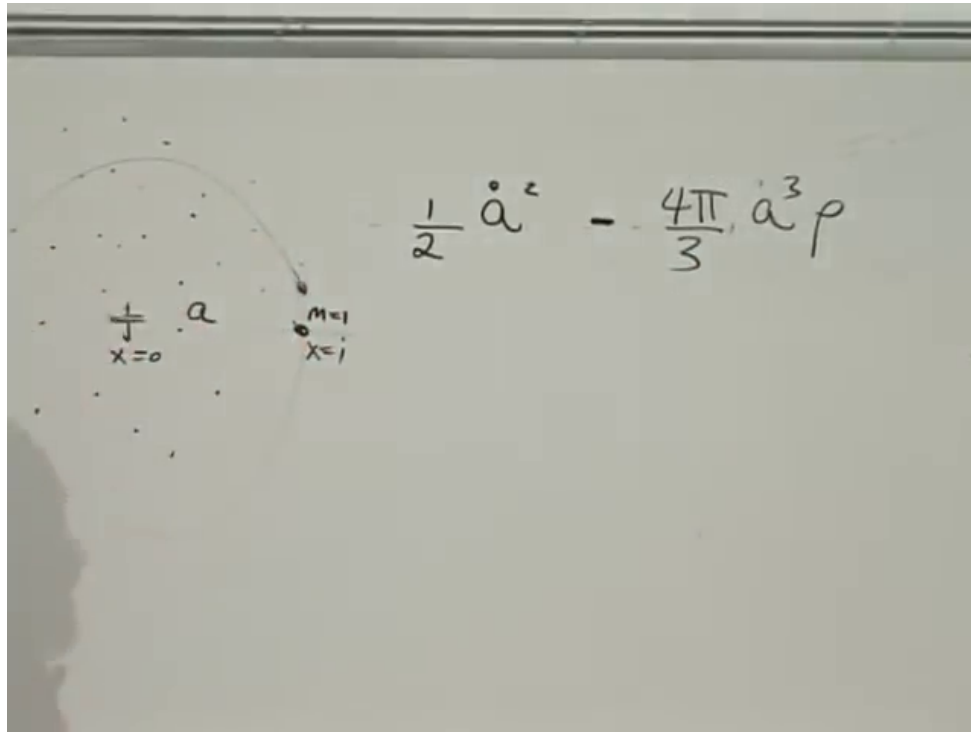


Figure 3.1: Energy-conservation term with spherical mass sketch

motion of the particle is the material inside that sphere. So the whole cosmological problem is reduced, at first pass, to the motion of one particle under the gravity of the homogeneous matter enclosed within a sphere of radius $a(t)$.

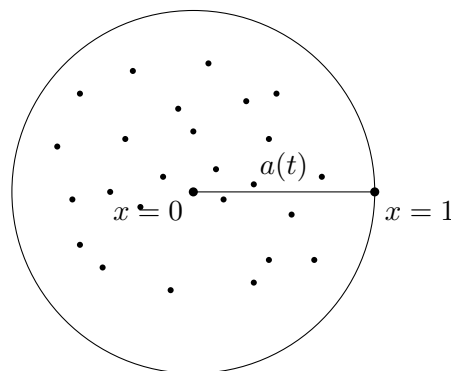


Figure 3.2: Clean reconstruction of the homogeneous sphere used in the lecture

3.2 Energy Conservation and the Friedmann Equation

At this point Susskind says, in effect: let us just go through the argument again, because it is really very simple. We assign the test particle unit mass, so that we do not have to keep writing factors

that cancel out anyway. Its kinetic energy is then

$$\frac{1}{2}\dot{a}^2. \quad (3.2)$$

The dot denotes the derivative with respect to cosmic time.

The mass enclosed inside the sphere of radius $a(t)$ is

$$M_{\text{enc}} = \frac{4\pi}{3}a^3\rho. \quad (3.3)$$

Here ρ is initially being spoken of in Newtonian language as a mass density. The gravitational potential energy of the particle is therefore

$$-\frac{GM_{\text{enc}}}{a} = -\frac{4\pi G}{3}a^2\rho. \quad (3.4)$$

The first blackboard frame only secures this term partially; the factor of G and the precise power of a are restored from the spoken derivation rather than from the chalk alone.

So the basic energy balance is

$$\frac{1}{2}\dot{a}^2 - \frac{4\pi G}{3}a^2\rho = \text{const.} \quad (3.5)$$

Newton tells us that this constant does not depend on time. In the lecture Susskind first names the constant $-k$, then immediately remarks that the standard normalization really introduces k only after multiplying by 2. The clean way to write the result is therefore

$$\dot{a}^2 = \frac{8\pi G}{3}a^2\rho - k, \quad (3.6)$$

and then, dividing by a^2 ,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \quad (3.7)$$

This is the Friedmann equation in the form that will drive the entire rest of the lecture.

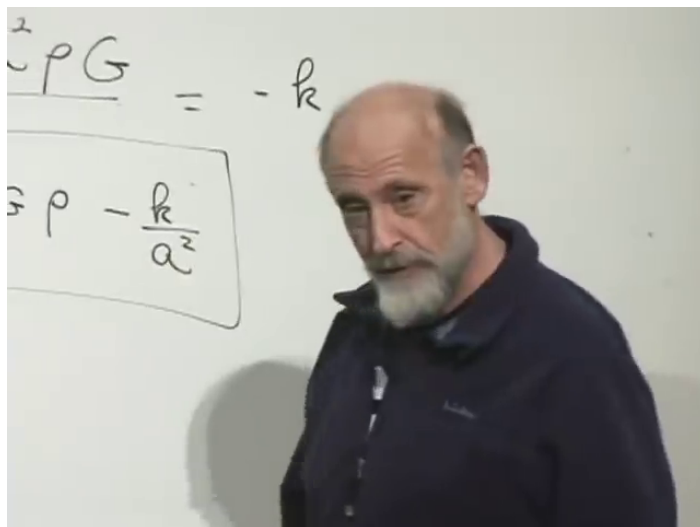


Figure 3.3: Boxed curvature term in the Friedmann equation

The board frame above is useful in a narrow but important way: it shows that the curvature term $-k/a^2$ was explicitly boxed and emphasized. Susskind also notes that k is empirically very close to zero. It is related to the spatial curvature, but observationally it is so small that, at the level of this lecture, it is a very good approximation simply to ignore it.

This is also the point at which he briefly pivots back toward Einstein. The aim is not to rederive general relativity, but to orient us. In schematic form the time-time Einstein equation looks like

$$R_{00} - \frac{1}{2}g_{00}R \propto T_{00}. \quad (3.8)$$

If we insert an expanding homogeneous metric, the geometric side produces the combination of $\dot{a}/a$ and k/a^2 , while the matter side produces the energy density. So this Newtonian-looking equation is, in fact, the 00-component of Einstein's equation for a homogeneous universe. In Susskind's phrase, we have not abandoned Albert for Isaac; at this stage Isaac and Albert are saying the same thing.

That remark also changes the meaning of ρ . From now on it is better to call ρ the energy density. For nonrelativistic matter at rest in the comoving frame, energy density and rest-mass density are the same thing once we set $c = 1$. But the distinction will matter as soon as radiation enters.

3.3 Matter-Dominated Cosmology

The next move in the lecture is: now let us assume a form for ρ . Since our coordinates move with the material, the matter is at rest in the coordinate frame we are using. Under those conditions, ρ is simply the rest-energy density.

Take a cube one coordinate unit on a side. Its physical side lengths are all $a(t)$, so its physical volume is $a^3(t)$. Suppose the total mass-energy inside that comoving cube is M , and suppose it stays there: matter moves with the flow, but it does not cross the boundary of the comoving box. Then

$$\rho_m = \frac{M}{a^3}. \quad (3.9)$$

This is the matter-dominated case.

Susskind is careful about what that phrase means. Matter-dominated does not mean merely that there are particles present. It means the dominant contribution to the energy density comes from particles whose energies are essentially their rest energies in the comoving frame. In that limit the pressure is approximately zero. In the language of the lecture, pressure comes from particles crossing the boundaries of the box; here they do not. Strictly speaking, our present universe is not matter-dominated because dark energy now dominates the cosmic energy budget, but the ordinary matter sector and a long stretch of cosmological history are accurately described this way.

With $k \approx 0$, the Friedmann equation becomes

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} \frac{M}{a^3}. \quad (3.10)$$

Now comes the “honorable” method of guessing a solution. We try a power law,

$$a(t) = Ct^p. \quad (3.11)$$

Then

$$\dot{a} = Cp t^{p-1}, \quad (3.12)$$

$$\frac{\dot{a}}{a} = \frac{p}{t}, \quad (3.13)$$

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{p^2}{t^2}. \quad (3.14)$$

Substituting into the matter-dominated equation gives

$$\frac{p^2}{t^2} = \frac{8\pi G}{3} \frac{M}{C^3 t^{3p}}. \quad (3.15)$$

The powers of t must match, so

$$3p = 2, \quad \text{hence} \quad p = \frac{2}{3}. \quad (3.16)$$

Therefore

$$a(t) \propto t^{2/3}. \quad (3.17)$$

If we also match the coefficients, we recover Susskind's bookkeeping constant,

$$C^3 = 6\pi GM. \quad (3.18)$$

This is the first concrete cosmological law of the lecture: in a matter-dominated universe the scale factor grows not linearly in time, but like the two-thirds power of time. In the lecture he also sketches the curve and stresses that it rises more slowly than a straight line. Historically, this law describes a long intermediate era, roughly from about 10^5 years after the Big Bang until a few billion years later.

3.4 Radiation-Dominated Cosmology

Now we run the universe backward. Earlier than the matter-dominated era, the lecture says, the energy density was dominated not by matter but by radiation. The reason is not mysterious once we phrase it correctly: the universe contains an enormous number of photons, roughly 10^9 photons for every proton, and when we compress the universe those photons become much more energetic.

Susskind pauses to say something important. A photon has no rest mass, but that does not mean it has no energy. We are not talking about rest energy for photons. If one tried to bring a photon to rest, there would be no particle left over. But a moving photon certainly has energy, and it is that energy that matters cosmologically.

So again we take a comoving unit box. Photons move through the box, but on average as many enter as leave, so the average number inside stays fixed. The energy of a single photon is

$$E_\gamma = h\nu = \frac{hc}{\lambda}. \quad (3.19)$$

The key physical assumption, which the lecture treats as well justified by wave dynamics, is that the wavelength stretches with the expansion:

$$\lambda \propto a. \quad (3.20)$$

Therefore

$$E_\gamma \propto a^{-1}. \quad (3.21)$$

If the average number of photons in the comoving box stays fixed, then the total energy in the box scales the same way:

$$E_{\text{box}} \propto a^{-1}. \quad (3.22)$$

Dividing by the physical volume a^3 , we get

$$\rho_r \propto a^{-4}. \quad (3.23)$$

This extra factor of a^{-1} is the new ingredient. Matter dilutes only because the universe gets bigger. Radiation dilutes for that reason and also because each individual photon is redshifted.

It is useful to collect the scaling chain in one place:

$$E_\gamma = \frac{hc}{\lambda}, \quad (3.24)$$

$$\lambda \propto a, \quad (3.25)$$

$$E_\gamma \propto a^{-1}, \quad (3.26)$$

$$E_{\text{box}} \propto a^{-1}, \quad (3.27)$$

$$\rho_r = \frac{E_{\text{box}}}{a^3} \propto a^{-4}. \quad (3.28)$$

The radiation-dominated Friedmann equation is therefore

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} \frac{M_r}{a^4}, \quad (3.29)$$

where M_r is simply a constant setting the scale of the radiation energy density. In the lecture the same spoken symbol gets reused here, but it is no longer the matter constant from the previous section; it is cleaner to rename it.

We solve in exactly the same way as before. Try

$$a(t) = Ct^p. \quad (3.30)$$

Then the left-hand side scales as t^{-2} , while the right-hand side scales as t^{-4p} . Matching powers gives

$$4p = 2, \quad \text{so} \quad p = \frac{1}{2}. \quad (3.31)$$

Thus the radiation-dominated universe expands as

$$a(t) \propto t^{1/2}. \quad (3.32)$$

And if we keep the coefficient,

$$C^4 = \frac{32\pi G}{3} M_r. \quad (3.33)$$

3.4.1 Question & Answer

Question. Photons have no rest mass. Why is there still a factor of G in the radiation equation?

Answer. Because what gravitates is energy, not rest mass by itself. When particles move slowly, the distinction is easy to miss because their energy is mostly just rest energy. But photons are the example that forces the point. They have no rest mass and yet they carry energy, momentum, and pressure. Susskind answers this in the lecture by going back to Einstein’s old box-of-photons argument: a box full of radiation has more inertia than the same box without the radiation, so by the equivalence principle it also has more weight. In cosmology that statement is summarized by putting energy density on the right-hand side of the Friedmann equation.

Question. If each photon loses energy as the universe expands, what happened to conservation of energy?

Answer. The lecture’s answer is not that conservation failed, but that one has to keep the whole Friedmann bookkeeping in view. If we restore the full energy balance,

$$\frac{1}{2}\dot{a}^2 - \frac{4\pi G}{3}a^2\rho = \text{const}, \quad (3.34)$$

then the redshifting of the photon bath is only one part of the story. There is also the kinetic energy of the expansion and the gravitational part of the energy balance. The energy stored in radiation decreases, but the full cosmological energy accounting is still encoded in the same conserved equation.

3.5 Matter-Radiation Equality and the Change of Expansion Law

Having extracted the two scaling laws, Susskind stops computing and turns to the board. This is an important shift in rhythm. The issue is no longer how to solve each toy model separately, but how the real universe passes from one regime to the other.

For matter we have

$$\rho_m \propto a^{-3}, \quad (3.35)$$

and for radiation

$$\rho_r \propto a^{-4}. \quad (3.36)$$

So when a is large, matter dominates; when a is small, radiation dominates. The lecture’s point is that the early universe must therefore have been radiation-dominated.

The frame above preserves the blackboard moment at which he begins this plot. The completed picture shows exactly what the algebra already told us: radiation lies above matter at small a , matter lies above radiation at large a , and the two cross at an equality scale factor.

In time language, the two limiting laws are

$$a(t) \sim t^{1/2} \quad \text{early}, \quad a(t) \sim t^{2/3} \quad \text{later}. \quad (3.37)$$

The lecture-level estimate is that matter-radiation equality happened after about 10^4 years, perhaps a bit more.

Susskind then draws the complementary plot, this time of scale factor versus time, to make the same point dynamically rather than statically.

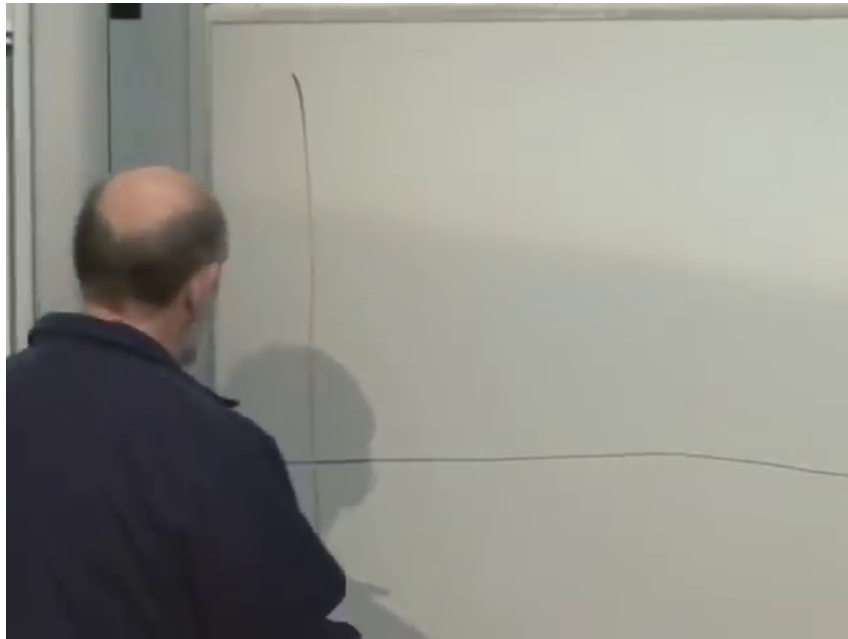


Figure 3.4: Axes for the energy-density plot

3.5.1 Question & Answer

Question. Does the power p jump from $1/2$ to $2/3$?

Answer. No. The lecture makes a point of correcting precisely that misconception. The exponents $1/2$ and $2/3$ describe the two clean limiting eras. The real universe interpolates continuously between them over a relatively narrow transition region. The effective law changes, but it does not jump discontinuously.

3.6 Temperature, Ionization, and the Surface of Last Scattering

Before estimating the crossover in more detail, Susskind suddenly says: before we talk about that, let us talk about temperature. This is a characteristic move in the lecture. The algebra has given us the scaling laws; now he wants to translate them into the thermal history of the universe.

Because the energy of each photon scales like $1/a$, the temperature of the radiation scales the same way, up to an unimportant numerical factor:

$$T \propto a^{-1}. \quad (3.38)$$

If a doubles, the radiation temperature is cut in half. Running the universe backward makes the photons hotter.

The lecture then uses a rough but effective classroom estimate. Take the present cosmic radiation temperature to be about 3 K, and take the ionizing temperature to be about 3000 K, the rough surface-of-the-sun scale repeatedly invoked in the lecture. Then

$$\frac{a_{\text{today}}}{a_{\text{ion}}} \sim \frac{3000 \text{ K}}{3 \text{ K}} \sim 10^3. \quad (3.39)$$

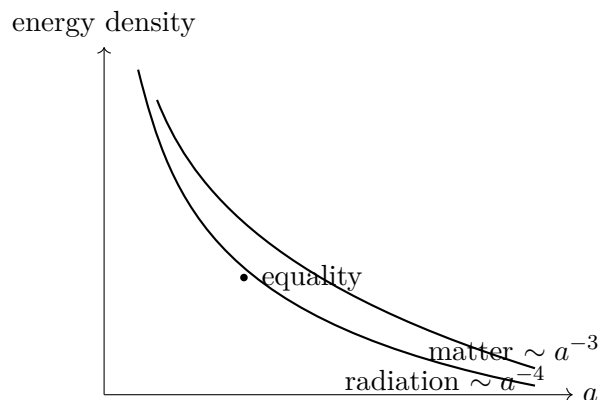


Figure 3.5: Transcript-derived reconstruction of the energy-density comparison drawn in the lecture

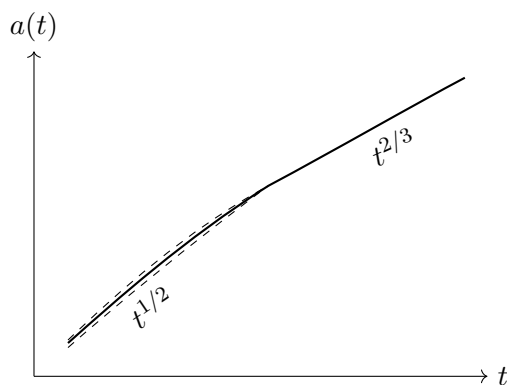


Figure 3.6: Transcript-derived sketch of the smooth transition from radiation-dominated to matter-dominated expansion

From recombination to the present the universe is well approximated as matter-dominated, so $a \propto t^{2/3}$. Therefore

$$\left(\frac{t_{\text{today}}}{t_{\text{ion}}} \right)^{2/3} \sim 10^3, \quad (3.40)$$

$$\frac{t_{\text{today}}}{t_{\text{ion}}} \sim 1000^{3/2} \sim 3 \times 10^4. \quad (3.41)$$

If we now use the lecture's order-of-magnitude value $t_{\text{today}} \sim 10^{10}$ years, we get

$$t_{\text{ion}} \sim 3 \times 10^5 \text{ years}. \quad (3.42)$$

This is not meant as a modern precision cosmological parameter. It is a lecture estimate, and it is good enough to locate the key physical event.

What is the event? Earlier than this, the universe was hot enough to ionize hydrogen and helium. It was a plasma. In a plasma, electromagnetic waves drive electric charges back and forth; in the lecture's language, the medium behaves like a conductor, and conductors are opaque. That is why the surface of the sun is opaque, and it is why the early universe was opaque. Light could not travel very far without scattering.

As the universe cooled, electrons recombined with nuclei to make neutral atoms. Once the plasma turned into ordinary neutral gas, the universe became transparent enough for photons to travel essentially freely. The transition is not infinitely sharp. Susskind stresses that it is a fuzzy place, or equivalently a fuzzy time interval, not a mathematical surface with zero thickness. But it is sharp enough cosmologically that we give it a name: the surface of last scattering.

Now the observational picture becomes vivid. An astronomer looking outward is looking backward in time. Nearby galaxies show us the relatively recent past. Farther galaxies show us earlier times. Eventually every line of sight reaches back to the epoch when the universe was still ionized and opaque. Beyond that, electromagnetic radiation cannot give us a clear view. So the cosmic microwave background is literally the radiation arriving from that last-scattering shell.

The lecture pushes the idea one step further. A neutrino telescope, if one could build a sufficiently sensitive one, would see deeper into the past, because neutrinos decouple earlier. A graviton telescope would see earlier still. But for ordinary electromagnetic astronomy the last-scattering surface is the horizon.

A final clarification comes from the question period. The temperature tracks the energy of a typical photon, but the full radiation energy density grows faster because the photon number density also changes:

$$n_\gamma \propto T^3, \quad \rho_\gamma \propto T^4. \quad (3.43)$$

The first relation counts photons per unit volume; the second combines photon number density with energy per photon.

3.6.1 Question & Answer

Question. If the early universe was as hot as the surface of the sun in every direction, why do we not look up and see a blinding sky?

Answer. Because the last-scattering surface is not sitting still relative to us. It is receding, and the photons that left it have been redshifted on the way in. In the lecture's rough estimate, they have been redshifted by about a factor of 10^3 , exactly the factor that turns a 3000 K radiation field into a 3 K microwave background. So we are indeed seeing that hot surface, but after its photons have been stretched into microwaves. Without that redshift, as Susskind says, the sky would be bright enough to burn our eyeballs.

3.7 Dark Matter from Galactic Dynamics

Only now does the lecture return to the promise made at the very beginning: dark matter. The change of scale is explicit. Dark matter was not discovered by studying cosmology in the very large-scale sense. It was discovered by astronomers looking at galaxies and measuring their rotation curves.

So Susskind goes back, once again, to the simplest Newtonian mechanics. Consider a light object of mass m orbiting a central mass M at radius r . The gravitational force is

$$\frac{GMm}{r^2}, \quad (3.44)$$

and for circular motion the centripetal acceleration is v^2/r . Newton's law gives

$$m \frac{v^2}{r} = \frac{GMm}{r^2}. \quad (3.45)$$

Cancel the small mass m , multiply through by r , and we get

$$v(r) = \sqrt{\frac{GM}{r}}. \quad (3.46)$$

Thus if the gravitating mass is concentrated near the center, the orbital speed falls like $r^{-1/2}$. This is the Keplerian law familiar from the solar system.

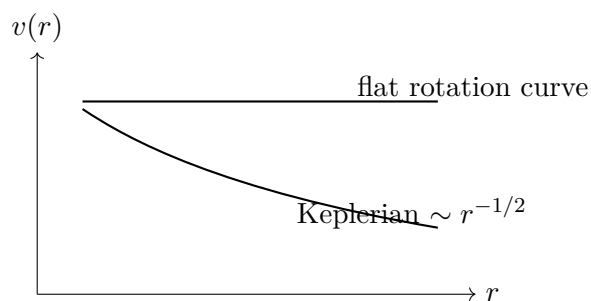


Figure 3.7: Transcript-derived comparison between Keplerian falloff and the approximately flat rotation curves seen in large galaxies

But the measured galactic rotation curves do not fall that way. If one assumes that mass follows light, so that most of the mass sits where most of the luminous matter sits, near the galactic core, then the outer stars should move more slowly. They do not. Their orbital speeds remain roughly constant out to large radii.

So the lecture changes the question. Perhaps mass does not follow light. Let $M(r)$ be the total mass enclosed inside radius r . Then the orbital relation becomes

$$\frac{v^2}{r} = \frac{GM(r)}{r^2}. \quad (3.47)$$

If $v(r)$ is approximately constant, then

$$M(r) \propto r. \quad (3.48)$$

That is already a powerful conclusion: the total gravitating mass keeps growing with radius even where the visible light does not.

Assume now, as Susskind does for simplicity, that the hidden matter distribution is roughly spherical. Then

$$M(R) = \int_0^R 4\pi r^2 \rho(r) dr. \quad (3.49)$$

If $M(R) \propto R$, differentiating with respect to R gives

$$4\pi R^2 \rho(R) \propto 1, \quad (3.50)$$

and hence

$$\rho(r) \propto \frac{1}{r^2}. \quad (3.51)$$

So the lecture arrives at a simple mathematical picture: a visible galaxy sits inside a much larger dark halo whose density decreases gradually rather than collapsing sharply with the light distribution.

At the numerical level Susskind plays loosely between factors of 7 and 10, and he is explicit that he is speaking heuristically. The stable statement is that the total gravitating mass of a galaxy is of order several to ten times larger than what one would infer from luminous matter alone.

From here the lecture shifts from derivation to diagnosis. Whatever this material is, it does not shine, and it evidently does not dissipate the way ordinary matter does. Ordinary gas clouds collide, radiate, and keep shrinking. This stuff stays in a large halo.

3.7.1 Question & Answer

Question. If the halo is so massive, why does it not dissipate and collapse together with the luminous matter?

Answer. Because collapse requires dissipation. Gravity by itself can pull a cloud inward, but to settle into a smaller structure the system must lose energy. Ordinary matter has many channels for doing that: collisions, friction-like processes, radiation, and electromagnetic interactions. Dark matter evidently does not. In the lecture Susskind emphasizes that it is very nearly collisionless and does not radiate appreciably. So the gravitational contraction converts potential energy into kinetic energy, but there is no efficient mechanism for dumping that kinetic energy away. What keeps the halo extended is the same thing that keeps the Earth from falling straight into the sun in the absence of friction: motion.

This is also where the lecture draws a careful boundary. The inferences $M(r) \propto r$ and $\rho(r) \propto r^{-2}$ are solid lecture mathematics. But Susskind does not claim a simple first-principles derivation of the detailed halo profile, or a detailed understanding of halo rotation, flattening, or intergalactic distribution. On those points he repeatedly says, quite openly, that he does not know and that large-scale simulations do much of the work.

The last observational beat of the lecture is the Bullet Cluster. When galaxy clusters collide, the hot ordinary gas interacts strongly and can be seen in X-rays, while the dominant gravitating component inferred from lensing passes through much more weakly interacting matter. The resulting separation between the lensing mass and the luminous baryonic material is exactly the sort of thing one would not expect if the whole phenomenon were merely a modification of Newton's force law tied to visible matter. In the lecture, this serves as the decisive piece of evidence that there really is dark matter.

Susskind ends on a note of bounded confidence. Very little is known in detail about what dark matter is, but nothing about it requires a radical rewriting of physics. The conservative possibility is that it consists of new, very weakly interacting particles. In that sense it is mysterious, but not in a way that forces us to throw away the basic principles that got us here.

3.8 Summary

The lecture begins with a single equation and then changes the meaning of its source term step by step. Starting from a homogeneous Newtonian sphere centered on us, we recover the Friedmann

equation,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}, \quad (3.52)$$

and are reminded that this is also the time-time Einstein equation for an expanding homogeneous universe. For matter at rest in a comoving box, $\rho_m \propto a^{-3}$ and $a(t) \propto t^{2/3}$. For radiation, because photon wavelengths stretch with the expansion, $\rho_r \propto a^{-4}$ and $a(t) \propto t^{1/2}$.

From those two scaling laws come the next pieces of the story: radiation domination in the early universe, matter domination later, temperature scaling as $1/a$, recombination after roughly 3×10^5 years at lecture level, and the surface of last scattering as the electromagnetic horizon. Only then does the lecture narrow from cosmology to astronomy and use orbital mechanics to infer dark matter from flat galactic rotation curves. The mathematical punch line there is just as sharp: $v \approx \text{const}$ implies $M(r) \propto r$, hence a large dark halo with $\rho(r) \propto r^{-2}$. The details remain open, but the structure of the argument does not.

Chapter 4

Pressure, Equation of State, and Vacuum Energy

This lecture opens in the form of questions, not yet in the form of equations. Susskind first clears away several conceptual obstacles: the microwave background does define a preferred cosmological rest frame, superluminal recession does not violate local relativity, and a horizon is the place where expansion defeats return signals. Only after that warm-up does he deliberately reset the lecture and turn to the technical core: pressure, energy density, the parameter w , and the special role of vacuum energy. The aim here is to follow that unfolding as closely as possible, while keeping the mathematics clean and the board evidence visible. These notes are based on Leonard Susskind's lecture, with supporting curation by LazyingArt LLC and Video2Book.

4.1 Opening Questions: Preferred Frame, Recession, and the Microwave Background

The first question concerns the cosmic microwave background. If the background radiation looks isotropic only in one frame, does that not define a preferred frame of reference? Susskind's answer is yes, it does. That is not a contradiction of relativity. It is a statement about the large-scale state of the universe. There is a special cosmological frame in which the microwave background looks the same in all directions, and this is also, on average, the frame in which the local galaxies are at rest.

If we move through this bath of photons, the Doppler effect immediately breaks the isotropy. Photons coming from the direction of motion are blueshifted; photons from the opposite direction are redshifted. In that sense there is a preferred cosmological rest frame, but the local laws of physics remain perfectly relativistic.

The next question asks whether distant galaxies can recede faster than light. Here the essential distinction is between local motion and global recession. Relativity forbids one object from overtaking another right in front of us at superluminal speed. It does not forbid sufficiently distant galaxies from separating according to

$$v = Hd. \tag{4.1}$$

For large enough d , the recession speed v can exceed c . In the units of the lecture we set $c = 1$, so the horizon is the distance at which $Hd = 1$.

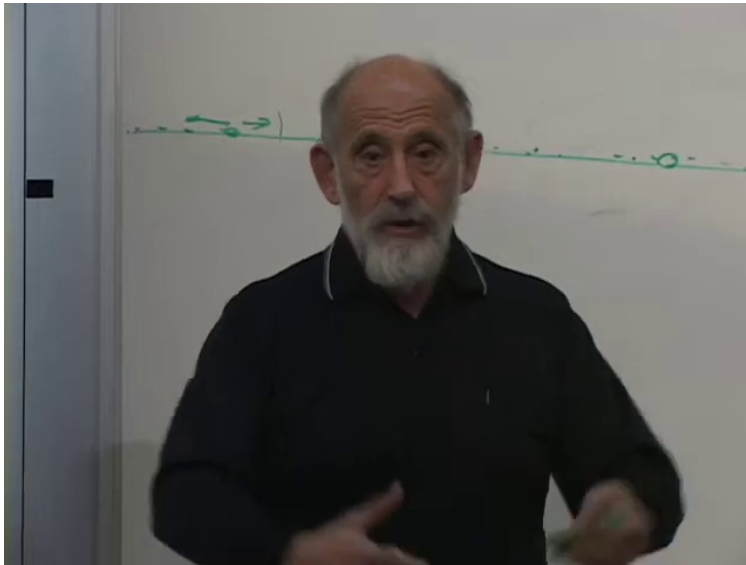


Figure 4.1: One-dimensional expansion sketch on the board. The visual evidence is sparse but important: a long line, directional arrowing, and marked points used to introduce the intuition of an expanding metric.

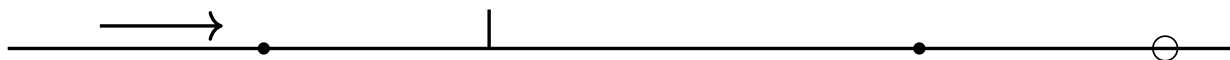


Figure 4.2: Minimal redraw of the line analogy. The lecture uses the picture qualitatively, so the redraw is kept deliberately spare.

This immediately leads to the observable universe. The cosmic horizon is the boundary beyond which light cannot get back to us. The surface of last scattering is not exactly the same thing, but it lies very close to it. Radiation from the last-scattering surface is enormously redshifted, by a factor of order 10^3 , but not infinitely redshifted. For that reason the surface of last scattering sits slightly inside the horizon, while still serving in practice as the edge of what we can directly see.

4.2 Expanding Metric and the One-Dimensional Line Analogy

At this point Susskind insists that ordinary special-relativistic intuition is no longer enough. The problem is not simply one of adding velocities. In general relativity the metric itself can expand.

His favored picture is a one-dimensional line. If we stretch an ordinary material line, it thins out and changes its local properties. That is not the right analogy. The right analogy is a line with a hidden reservoir of new points. Whenever neighboring points separate too much, new points are inserted. The line grows, but its local character does not thin out.

What matters is the rule for signals. Locally, a signal always moves at the limiting speed relative to nearby points. Globally, however, the total distance it must overcome may keep increasing because more space is being inserted between source and destination. The lecture returns to this distinction again and again: local speed limits survive, even while global recession can exceed the speed of light.

Susskind also uses the same picture to explain redshift. If we draw a wave on the line, then as the

line expands the wave stretches with it. In the full theory this is not merely an image. It is what one finds by writing wave equations in an expanding geometry.

4.2.1 Question & Answer

How can a light signal fail to get back if nothing locally moves faster than light? Because the signal is not racing against a receding material object in a fixed background. It is moving through a geometry whose distance scale is itself increasing. The light ray always propagates locally at the limiting speed, but the proper distance to its destination can grow faster than the ray can eat it up. In Susskind's line picture, the signal advances from point to point while new points are being inserted in front of it. That is the cosmological horizon in a nutshell.

4.3 Reset: Friedmann Equation and the Three Components

After the opening discussion Susskind explicitly resets the lecture. He says, in effect, that this is the theoretical minimum: just enough mathematics to follow a serious cosmology argument. He dislikes jargon, but he also knows that one eventually has to understand what astronomers mean when they say that they are measuring w . So he begins again from the basic expansion equation.

The central relation is

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \quad c = 1. \quad (4.2)$$

Here $a(t)$ is the scale factor, and

$$H \equiv \frac{\dot{a}}{a} \quad (4.3)$$

is the Hubble parameter. The right-hand side contains the total energy density.

The lecture then reviews the three dynamically important contributions to ρ :

- matter, meaning objects effectively at rest in the comoving frame,
- radiation, meaning photons and other effectively massless components,
- vacuum energy, written ρ_0 , which does not dilute when the universe expands.

For matter and radiation the earlier results are recalled:

$$\rho_m = \frac{C_m}{a^3}, \quad a(t) \sim t^{2/3}, \quad (4.4)$$

$$\rho_r = \frac{C_r}{a^4}, \quad a(t) \sim t^{1/2}. \quad (4.5)$$

The third component is introduced as the odd one:

$$\rho_0 = \text{constant}. \quad (4.6)$$

Susskind does not yet derive this from first principles. He first wants to explain pressure, because pressure is what will ultimately tell us how ρ changes with a .

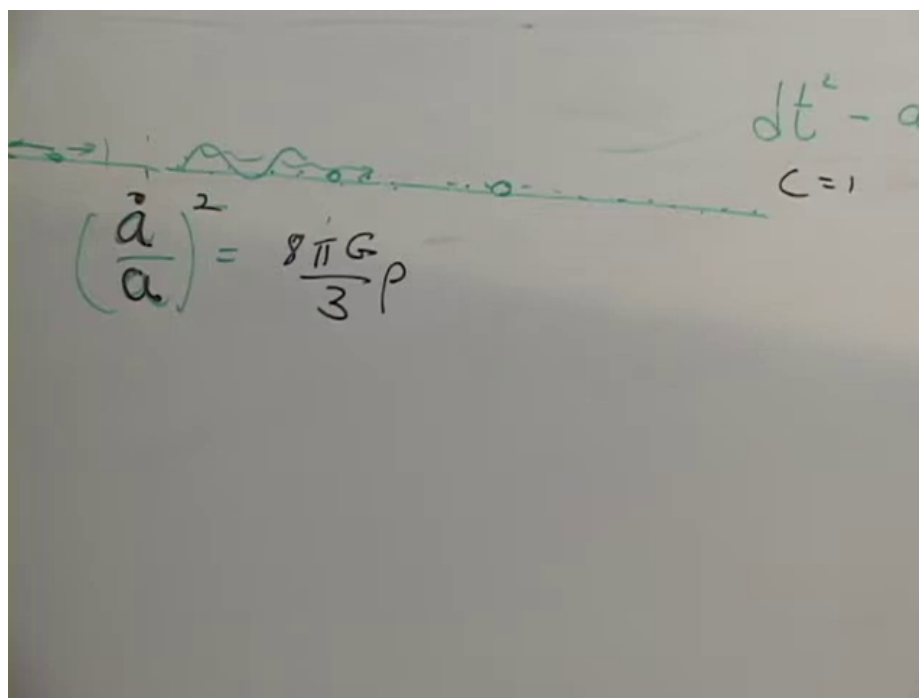


Figure 4.3: Friedmann equation with line analogy. The cropped notation on the far right only certifies that $c = 1$ was written nearby; the operative equation on the board is the Friedmann equation shown clearly here.

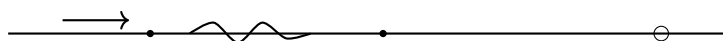


Figure 4.4: Redraw of the same line analogy, now with the small disturbance visible above the basic expansion equation.

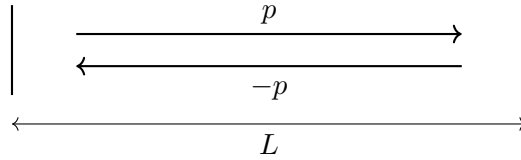


Figure 4.5: One-dimensional photon-in-a-box argument. The wall receives momentum $2p$ at each reflection, and the average force becomes energy divided by length.

4.4 Pressure from Radiation and Matter

We now come to the first real derivation of the lecture. Susskind begins with radiation, because there pressure is visible. Light bouncing off walls exerts a force, and so pressure can be computed directly from momentum transfer.

The one-dimensional box. Start with a one-dimensional box, that is, a line segment of length L , containing photons moving back and forth. In one dimension force and pressure are effectively the same thing, so this is the cleanest place to begin.

Take a single photon with momentum magnitude p . When it reflects from a wall, its momentum reverses. The change in momentum at each collision is therefore

$$\Delta p = 2p. \quad (4.7)$$

The time between successive collisions with the same wall is

$$\Delta t = \frac{2L}{c}, \quad (4.8)$$

since the photon must traverse the box and return. Hence the average force on the wall is

$$F = \frac{\Delta p}{\Delta t} = \frac{pc}{L}. \quad (4.9)$$

Now use the photon relation

$$E = pc. \quad (4.10)$$

Then

$$F = \frac{E}{L}. \quad (4.11)$$

But E/L is precisely the one-dimensional energy density. In this toy model,

$$P = \rho, \quad (4.12)$$

with the important caveat that here ρ means energy per unit length, not energy per unit volume.

With many photons the jolts are simply added and averaged. The same relation survives:

$$P = \rho \quad (4.13)$$

in one dimension.

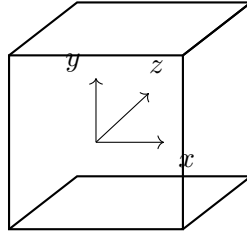


Figure 4.6: Three-dimensional schematic used to motivate the isotropic result $P_r = \rho_r/3$.

From one dimension to three. Now pass to a three-dimensional box. If all photons moved only along the x -direction, then the same reasoning would give a pressure on the walls perpendicular to x , but no pressure on the other faces. That is obviously not the correct picture of thermal radiation. Susskind deliberately makes this wrong assumption first, precisely to expose what has to be corrected.

In an isotropic radiation bath, the momentum is distributed over all directions. The pressure on any given wall depends only on the component of momentum normal to that wall. Averaged over directions, one third of the energy density contributes to each Cartesian pressure component. The result is

$$P_r = \frac{\rho_r}{3}. \quad (4.14)$$

Cold matter. The matter-dominated case is the opposite limit. Take a comoving box, meaning a box that moves with the cosmological flow. Cold matter is at rest relative to that box. The particles still contribute energy density, because each one carries rest energy, but they do not transfer momentum to the walls. Therefore

$$P_m = 0. \quad (4.15)$$

It is useful to write the board summary in clean form:

$$\text{matter:} \quad a(t) \sim t^{2/3}, \quad P = 0, \quad (4.16)$$

$$\text{radiation:} \quad a(t) \sim t^{1/2}, \quad P = \frac{\rho_r}{3}. \quad (4.17)$$

4.5 Equation of State and Density Scaling

Only now does Susskind introduce the jargon he had apologized for. The point is simple: if pressure is proportional to energy density, then the proportionality constant is a useful label for the substance filling the universe. He writes the board relation as

$$P = w\rho. \quad (4.18)$$

On the board the parameter is written as ω ; in these notes we normalize to the standard w . The examples already obtained give

$$w = 0 \quad \text{for cold matter}, \quad w = \frac{1}{3} \quad \text{for radiation}. \quad (4.19)$$

One can imagine intermediate values as well. The lecture mentions, for example, the possibility of a warm component with $0 < w < 1/3$.

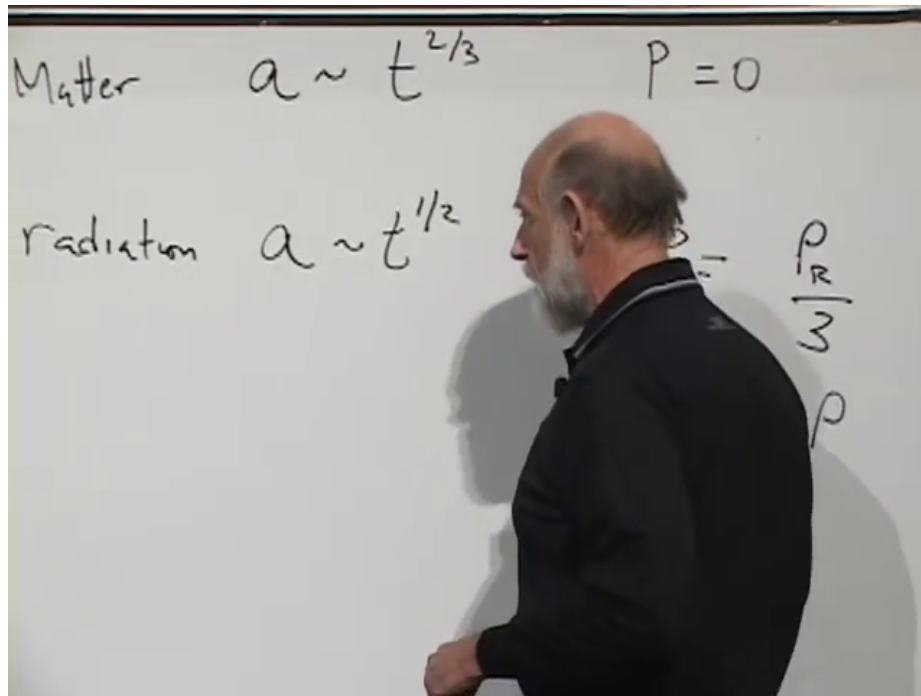


Figure 4.7: Matter and radiation equation-of-state summary on the board. The lecture uses the row-by-row layout itself as part of the argument.

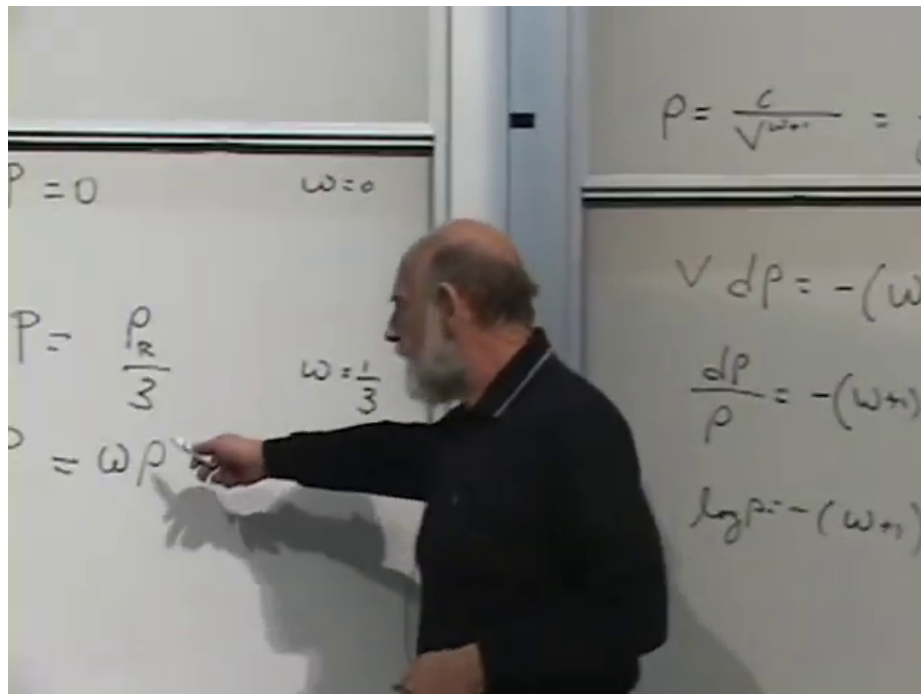


Figure 4.8: Equation-of-state parameter on the board. The left panel highlights $P = w\rho$, while the right panel carries the density-scaling derivation in partial form.

At this point the lecture poses the natural question. Matter has $w = 0$ and $\rho_m \propto a^{-3}$. Radiation has $w = 1/3$ and $\rho_r \propto a^{-4}$. Is this a coincidence, or is there a general relation between w and the way ρ scales?

4.5.1 Question & Answer

Why should the pressure-to-density ratio determine how the density changes with the scale factor? Because expansion does work. This is the thermodynamic heart of the lecture.

Take a box of volume V and let it expand by a small amount dV . If the contents exert pressure P , then the work done on the walls is $P dV$. That work must come from the internal energy of the contents, so

$$dE = -P dV. \quad (4.20)$$

Now write the internal energy as

$$E = \rho V. \quad (4.21)$$

Its differential is

$$dE = \rho dV + V d\rho. \quad (4.22)$$

Equating the two expressions for dE gives

$$\rho dV + V d\rho = -P dV. \quad (4.23)$$

Rearranging,

$$V d\rho = -(\rho + P) dV. \quad (4.24)$$

Now insert the equation of state $P = w\rho$:

$$V d\rho = -(1 + w)\rho dV. \quad (4.25)$$

Divide by ρV :

$$\frac{d\rho}{\rho} = -(1 + w) \frac{dV}{V}. \quad (4.26)$$

Integrating,

$$\ln \rho = -(1 + w) \ln V + \text{const}, \quad (4.27)$$

and therefore

$$\rho \propto V^{-(1+w)}. \quad (4.28)$$

This is the main result of the section. The rest is the subtle step that Susskind corrects on the fly during the lecture: in cosmology the comoving box has volume proportional to a^3 , not to a . Thus

$$V \propto a^3, \quad (4.29)$$

and so

$$\rho \propto a^{-3(1+w)}. \quad (4.30)$$

Equivalently,

$$\rho = \frac{C}{a^{3(1+w)}} \quad (4.31)$$

for some constant C .

Now the earlier cases fall out immediately:

$$w = 0 \implies \rho \propto a^{-3}, \quad (4.32)$$

$$w = \frac{1}{3} \implies \rho \propto a^{-4}. \quad (4.33)$$

Matter and radiation are no longer separate facts. They are two special points on the same curve of possibilities.

Once $\rho(a)$ is known, one can feed it back into the Friedmann equation and solve for $a(t)$. Susskind does not derive the general power law in detail here; he leaves that as an exercise in trying $a \sim t^\alpha$. What matters for the lecture is the logic: w determines $\rho(a)$, and $\rho(a)$ determines the expansion history.

4.6 Vacuum Energy as the Special Case $w = -1$

The lecture has been steering toward one especially important value, namely

$$w = -1. \quad (4.34)$$

If we put that into the general scaling law, the exponent vanishes:

$$\rho \propto a^{-3(1+w)} = a^0. \quad (4.35)$$

Thus

$$\rho = \rho_0 = \text{const.} \quad (4.36)$$

The energy density does not dilute at all. At the same time the equation of state becomes

$$P = -\rho. \quad (4.37)$$

This is exactly the vacuum-energy case.

4.6.1 Question & Answer

How can positive energy density come with negative pressure, and why does that not make the theory inconsistent? Because negative pressure is simply tension. Positive pressure pushes on the walls of a box. Negative pressure pulls them inward. A stretched rubber band is the everyday model: it has positive energy stored in it, but it pulls its ends together. There is nothing pathological about that sign. The energy density remains positive; the pressure is negative.

Now feed ρ_0 back into the Friedmann equation:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_0. \quad (4.38)$$

Since the right-hand side is constant, the Hubble parameter becomes a true constant,

$$H = \sqrt{\frac{8\pi G\rho_0}{3}}. \quad (4.39)$$

This is the special case in which $\dot{a}/a$ really deserves to be called a constant. The equation becomes

$$\dot{a} = Ha. \quad (4.40)$$

Its solution is exponential:

$$a(t) = a_* \exp(Ht) = a_* \exp\left(\sqrt{\frac{8\pi G\rho_0}{3}} t\right). \quad (4.41)$$

This is the late-time accelerated expansion. The scale factor does not merely keep increasing; its rate of increase is proportional to the scale factor itself. The characteristic timescale is H^{-1} , the e-folding time.

Susskind also remarks that the combination of ρ_0 with gravitational constants is often packaged into the cosmological constant Λ . Historically the numerical normalization is conventional and a bit untidy. For the present lecture the important physics is simpler than the terminology:

$$\Lambda \propto \rho_0. \quad (4.42)$$

Vacuum energy and cosmological constant are the same physical ingredient, up to the numerical factors attached by historical convention.

As always, the squared Friedmann equation admits two branches when one takes a square root. There is an expanding branch and a time-reversed contracting branch. The lecture, of course, follows the expanding one.

4.7 Three Eras, Observation, and the Late-Time Horizon

With the general framework in hand, Susskind steps back and looks at the whole history. The universe passes through three broad eras:

- a radiation-dominated era at small a ,
- a matter-dominated era at intermediate a ,
- a vacuum-dominated era at large a .

This follows immediately from the scaling laws

$$\rho_r \propto a^{-4}, \quad \rho_m \propto a^{-3}, \quad \rho_0 = \text{const.} \quad (4.43)$$

The time history of the scale factor reflects the same sequence:

- early on, $a(t) \sim t^{1/2}$,
- later, $a(t) \sim t^{2/3}$,
- at late times, $a(t)$ bends upward toward exponential growth.

The lecture gives the crossover times only as order-of-magnitude statements. Radiation gives way to matter very early, after roughly 10^4 years in the lecture's rough accounting, well before the universe becomes transparent. Matter gives way to vacuum much later, after several billion years, roughly halfway through cosmic history on the scale Susskind has in mind.

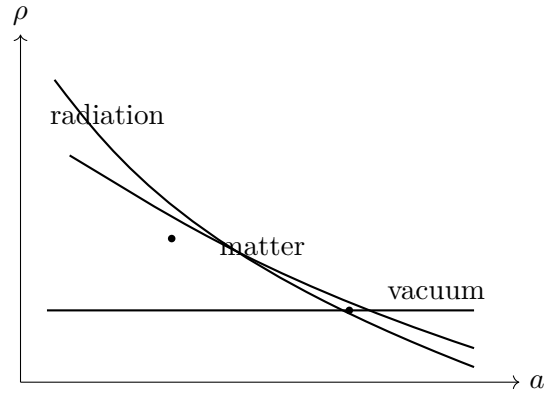


Figure 4.9: Qualitative energy-density history. Radiation dominates at very small a , matter later, and vacuum at large a .

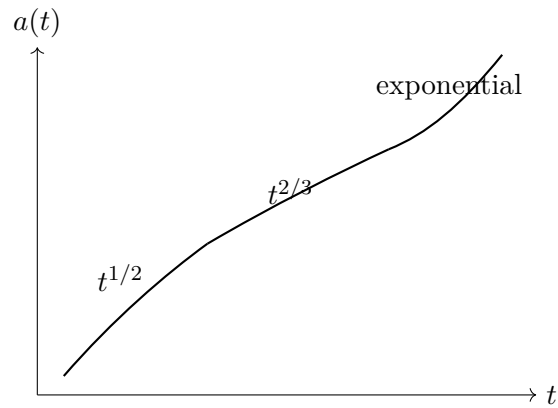


Figure 4.10: Qualitative time history of the scale factor: radiation era, matter era, then late accelerated expansion.

How do we know this sequence? We do not see the future, but we do see the past. Looking farther away means looking to earlier times. By observing distant galaxies, measuring their recession, and correlating distance with lookback time, one reconstructs the Hubble parameter as a function of time and therefore the history of $a(t)$. The radiation era is too early to be observed directly by ordinary galaxy astronomy, but the matter-dominated era and the transition to late acceleration are visible in the reconstructed expansion history.

At sufficiently late times, when vacuum energy dominates almost completely, the Hubble parameter approaches the constant value

$$H = \sqrt{\frac{8\pi G\rho_0}{3}}. \quad (4.44)$$

Using again the Hubble law $v = Hd$, the horizon distance is defined by the condition $v = 1$, that is,

$$1 = Hd_H. \quad (4.45)$$

Therefore

$$d_H = \frac{1}{H} = \frac{1}{\sqrt{8\pi G\rho_0/3}}. \quad (4.46)$$

This is the late-time cosmic horizon. Once the vacuum-dominated exponential era sets in, the horizon distance is fixed. Galaxies continue to move outward and pass through it one by one, disappearing not by abruptly vanishing, but by becoming ever more redshifted and fainter.

This is why Susskind spends time on the melancholy future of astronomy. In the remote future, only nearby gravitationally bound structures remain visible. Distant galaxies cross the horizon. The microwave background is stretched to such long wavelengths that it effectively fades away. Astronomers living then may infer that they inhabit a single isolated island of matter in an empty universe.

The lecture then briefly restores the curvature term that had been omitted in the main derivation:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{K}{a^2}. \quad (4.47)$$

Numerically this term appears to be small, but conceptually it matters. In the older cosmology, before vacuum energy was taken seriously, one imagined a matter-dominated universe with curvature of the right sign to halt the expansion and reverse it. The result was the old Big-Crunch picture: expansion, turnaround, recontraction. The mathematics is still legitimate, but it no longer appears to describe our universe.

The present-day composition, in the lecture's rough numbers, is

$$\frac{\rho_0}{\rho_{\text{total}}} \sim 0.7, \quad \frac{\rho_m}{\rho_{\text{total}}} \sim 0.3, \quad (4.48)$$

$$\frac{\rho_{\text{lum}}}{\rho_{\text{total}}} \sim 0.05, \quad \frac{\rho_{\text{dark}}}{\rho_{\text{total}}} \sim 0.25, \quad (4.49)$$

with radiation negligible today. Dark matter and luminous matter both scale as a^{-3} , so their mutual ratio stays roughly fixed even while the overall balance between matter, radiation, and vacuum changes from era to era.

The lecture closes by turning toward geometry. If we know the distance to the surface of last scattering, and if we have a standard lump there whose physical size can be estimated, then its observed angular size tests whether space is positively curved, negatively curved, or flat.

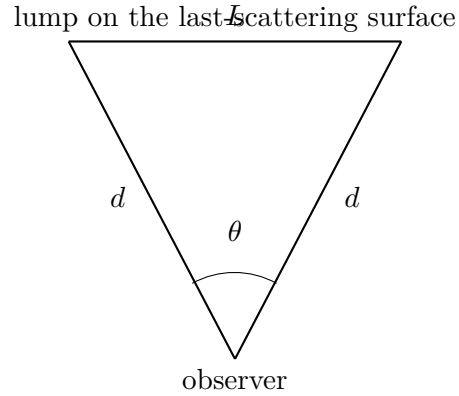


Figure 4.11: Preview of the next topic: the angular size of a standard lump on the last-scattering surface provides a test of spatial curvature.

That geometry argument belongs properly to the next lecture. Here it functions as an outlook. The work of the present lecture is already complete: pressure has been related to energy density, the equation-of-state parameter w has been introduced, and vacuum energy has emerged as the special case $w = -1$.

4.8 Summary

The lecture begins with preferred frames and horizons, but its mathematical center is the relation between pressure and energy density. Radiation gives

$$P_r = \frac{\rho_r}{3}, \quad (4.50)$$

cold matter gives

$$P_m = 0, \quad (4.51)$$

and both are unified by

$$P = w\rho. \quad (4.52)$$

Thermodynamics then shows that

$$\rho \propto a^{-3(1+w)}. \quad (4.53)$$

From that one immediately reads off the special case $w = -1$: constant vacuum energy, negative pressure, and exponential expansion. The three cosmic eras, the late-time fixed horizon, and the modern picture of an accelerating universe all follow from that single thread.

Chapter 5

Homogeneous Space, FRW Geometry, and the Meaning of k

Let us begin exactly where the lecture begins: not with expansion, but with some simple mathematics. The connection to Einstein's equations is the destination, but Leonard Susskind very deliberately refuses to go there at once. We first need a language for homogeneous curved spaces, then a way to look out from within them, and only then can the constant k reveal its double meaning. What follows is a companion account of the fifth cosmology lecture, prepared for the course notes with curation support from LazyingArt LLC and Video2Book.

5.1 Uniform Geometry and the Meaning of a Sphere

Susskind begins by reminding us that spheres are the paradigmatic *uniform* geometries. Uniform here means constant curvature: every point is like every other point. That is already the cosmological theme in miniature. Before one says anything about galaxies or expansion, one asks what kinds of large-scale spaces could be homogeneous.

The first description uses an embedding space as a crutch. A one-sphere is just a circle,

$$x^2 + y^2 = a^2, \tag{5.1}$$

with unit case

$$x^2 + y^2 = 1. \tag{5.2}$$

A two-sphere is described in the same way, only now the crutch is three-dimensional:

$$x^2 + y^2 + z^2 = a^2, \tag{5.3}$$

and the unit two-sphere is

$$x^2 + y^2 + z^2 = 1. \tag{5.4}$$

For the three-sphere one adds one more coordinate,

$$x^2 + y^2 + z^2 + w^2 = A^2, \tag{5.5}$$

with unit case

$$x^2 + y^2 + z^2 + w^2 = 1. \tag{5.6}$$

The essential point is that these ambient coordinates are not themselves the space in which the inhabitants live. The embedding space is only a device for writing an equation. The actual geometry is the sphere itself. If a creature lives on a one-sphere, it lives on a circle; it has no access to an “outside” or an “inside.” Likewise, if space were a two-sphere or a three-sphere, the geometry of interest would be the surface itself, not the surrounding Euclidean space we used to write it down.

This distinction matters immediately when the lecture turns toward cosmology. A spherical universe means that *space* is spherical. It does *not* mean that spacetime is some higher-dimensional sphere with time smuggled in as one more coordinate. Time will enter later, but it enters differently.

For amusement, and to show that the pattern continues backward, the lecture briefly asks about a zero-sphere. Extrapolating the same algebraic pattern gives

$$x^2 = 1, \tag{5.7}$$

whose solutions are the two points $x = \pm 1$. We will not need the zero-sphere later, but it makes the recursive picture of spheres feel less accidental.

5.2 Intrinsic Metrics and the Recursive Sphere Construction

The more useful description is intrinsic. We now stop leaning on the crutch and describe the sphere in coordinates that live on the sphere itself. That is the language cosmology will actually use.

For the unit one-sphere a single angular coordinate ϕ , with $0 \leq \phi < 2\pi$, is enough. The metric is simply

$$ds^2 = d\phi^2. \tag{5.8}$$

This is the first appearance of the notation

$$d\Omega_1^2 = d\phi^2, \tag{5.9}$$

where $d\Omega_n^2$ denotes the metric of the unit n -sphere.

Now we pass to the unit two-sphere. The lecture’s picture is not an abstract formula but a constructive one: think of the two-sphere as a sequence of one-spheres. Start at one pole with a circle of zero size, let the circle grow, reach a maximum, then let it shrink again to zero at the opposite pole.

Introduce a second coordinate θ , running from 0 to π . The metric follows almost by inspection:

1. Motion in the north-south direction contributes $d\theta^2$.
2. At fixed θ , motion in the transverse direction is motion around a one-sphere labeled by ϕ .
3. That transverse one-sphere does not have unit radius; its radius is $\sin \theta$.
4. Therefore the angular contribution is $\sin^2 \theta d\phi^2$.

So the metric of the unit two-sphere is

$$ds^2 = d\theta^2 + \sin^2 \theta d\phi^2. \tag{5.10}$$

In the Ω -notation this is

$$d\Omega_2^2 = d\theta^2 + \sin^2 \theta d\Omega_1^2. \tag{5.11}$$

The three-sphere is obtained in exactly the same way. One thinks of it as a family of two-spheres. Introduce one more coordinate α , and let the radius of the two-sphere at fixed α again grow and then shrink. Geometrically this requires $0 \leq \alpha \leq \pi$: the lower-dimensional sphere begins at zero size, reaches a maximum, and returns to zero. The metric is

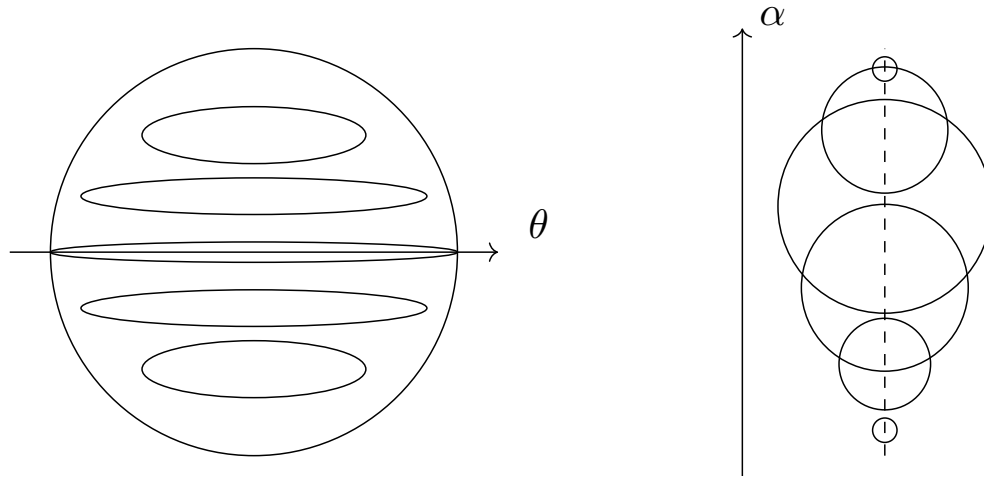
$$d\Omega_3^2 = d\alpha^2 + \sin^2 \alpha d\Omega_2^2. \quad (5.12)$$

If we expand the lower-dimensional piece explicitly, we get

$$d\Omega_3^2 = d\alpha^2 + \sin^2 \alpha (d\theta^2 + \sin^2 \theta d\phi^2). \quad (5.13)$$

Now the inductive rule is obvious:

$$d\Omega_n^2 = d\alpha^2 + \sin^2 \alpha d\Omega_{n-1}^2. \quad (5.14)$$



S^2 as a sequence of S^1 's S^3 schematically as a sequence of S^2 's

Figure 5.1: The lecture's constructive picture: a two-sphere built from one-spheres whose radius grows and shrinks, and a three-sphere built the same way one dimension higher.

This is the real geometric backbone of the lecture. Once the pattern is seen, the four-dimensional embedding picture becomes unnecessary. The geometry is already in the metric.

It also becomes clear why the sphere is special. We could replace $\sin^2 \theta$ by some other function that vanishes at both ends and still obtain a smooth closed surface, but then the resulting geometry would not be uniform. An ellipsoid, for example, is perfectly smooth, but it is not homogeneous; its curvature varies from place to place. The sphere is special precisely because every point looks the same.

5.3 From a Static Three-Sphere to a Positively Curved FRW Universe

Once the unit-sphere metric is in hand, the next move is almost embarrassingly simple. To obtain a sphere of fixed radius A , one multiplies the unit metric by A^2 :

$$ds^2 = A^2 d\Omega_3^2. \quad (5.15)$$

The square is important. A metric measures squared distances, so the radius must enter as A^2 , not as A .

Now the lecture turns from space to spacetime. We no longer consider a static three-sphere of fixed radius, but a three-sphere whose radius varies with time. At this point one must keep the warning sharp: time is not another embedding coordinate. The fact that space is spherical does not mean that spacetime is a higher-dimensional sphere. The cosmological degree of freedom is instead the time-dependent radius $a(t)$.

The lecture also assumes, as FRW cosmology does, that the metric has no mixed space-time terms such as $dt dx$. With that assumption the positively curved FRW spacetime metric is

$$d\tau^2 = dt^2 - a(t)^2 d\Omega_3^2. \quad (5.16)$$

Equivalently,

$$d\tau^2 = dt^2 - a(t)^2 \left(d\alpha^2 + \sin^2 \alpha d\Omega_2^2 \right). \quad (5.17)$$

This is the first genuinely cosmological metric of the lecture. It describes a universe whose spatial slices are three-spheres, and whose size changes with time. One may think of the Big Bang, in this language, as a three-sphere that begins very small and then expands.

To make the geometry intelligible from the inside, Susskind chooses one pole and places us there. That does not make us special; it merely chooses coordinates convenient for an observer. With that choice, α can be interpreted as a radial coordinate away from us on the unit sphere. It is not a radial distance in kilometers. It is an angular measure, hence dimensionless, but it plays the role of radial distance in the geometry.

In two dimensions the fixed- α sets are circles around us; in three dimensions they are two-spheres. The striking fact is that as we look farther away, these two-spheres first get larger, then eventually smaller. That is the positive-curvature intuition the lecture will soon turn into an observational statement.

This spacetime geometry is what Susskind calls FRW, following common usage even though one sometimes inserts an L for Lemaître. The spherical case is labeled by

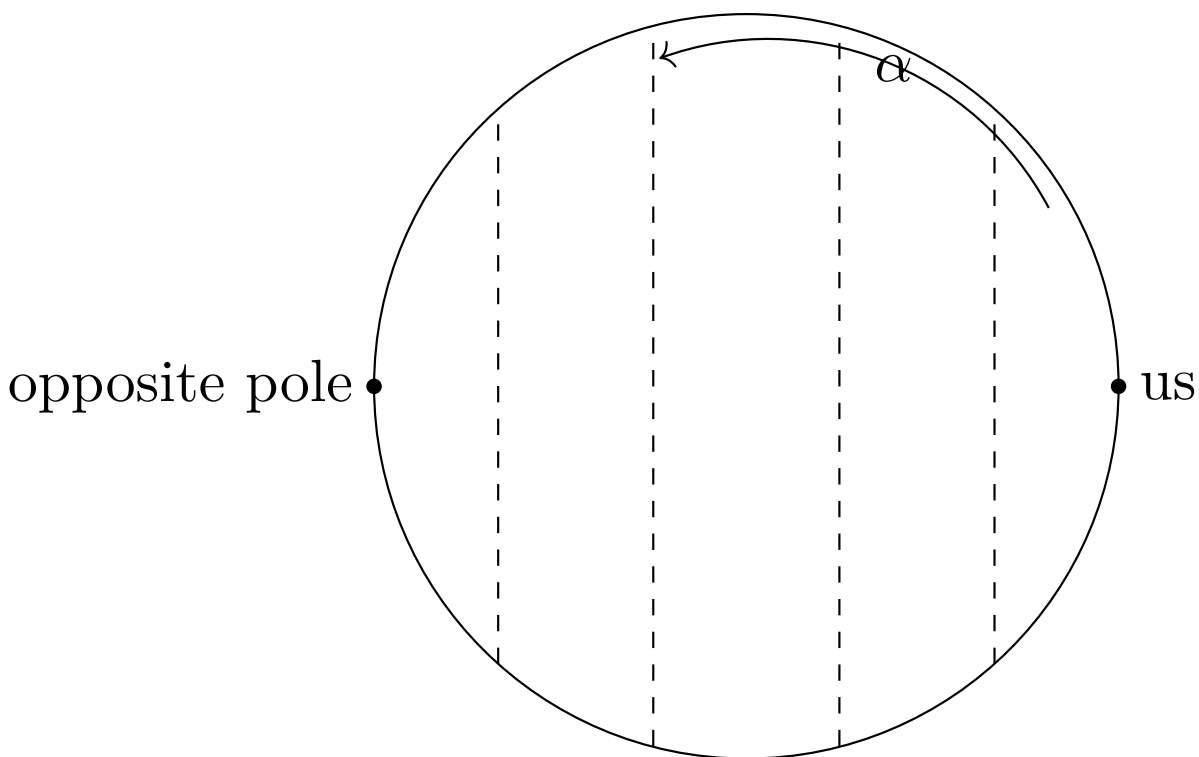
$$k = +1. \quad (5.18)$$

At this stage k is first of all a sign: it means positive *spatial* curvature, not positive spacetime curvature.

5.4 Looking Outward: Stereographic Projection and Angular Size

At this point the lecture deliberately pauses the formalism. A metric by itself is not yet a picture. We want to know what a positively curved space looks like to an astronomer who sits at one point and looks outward. This is why Susskind spends so much time drawing projections: curvature should be translated into something one might see in the sky.

The stereographic projection is the key construction. Take a two-sphere resting on a plane tangent at the south pole. To map a point P on the sphere to the plane, draw the straight line from the north pole through P , and continue it until it hits the plane. The intersection point is the image P' .



Fixed- α spheres grow, then shrink

Figure 5.2: Observer-centered spherical geometry in two-dimensional cross-section. In three dimensions the fixed- α surfaces are two-spheres around us: they first grow with distance and then, after a maximum, begin to shrink.

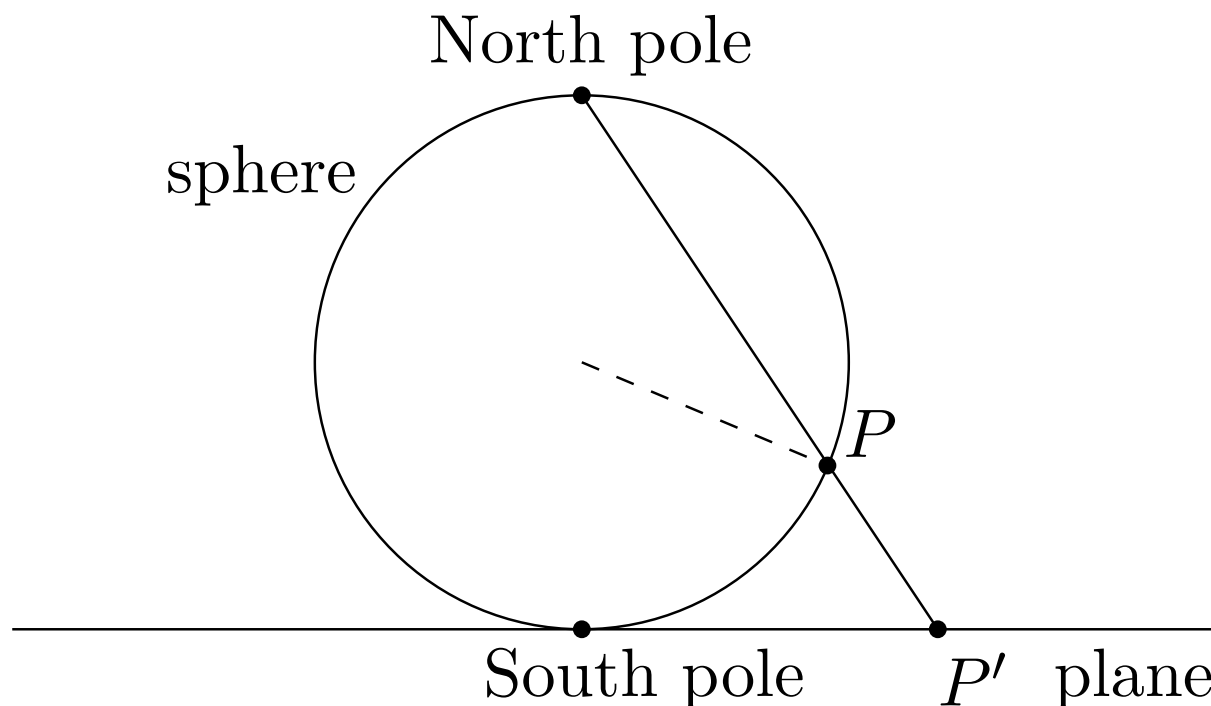


Figure 5.3: Stereographic projection from the north pole onto the plane tangent at the south pole. The north pole itself is sent to the point at infinity.

This mapping is already telling us something. Near the south pole, where we have placed ourselves, the projection is gentle. But points near the north pole are sent very far away. In fact the north pole itself becomes the point at infinity. Thus the projection of a finite sphere occupies an infinite plane.

Now come the observational consequences. If equal “continents” are placed farther and farther away on the sphere, their stereographic images become larger and larger. Translating that into astronomy: in a positively curved space, a standard ruler at fixed distance subtends a larger angle than it would in flat space. Distant objects look too big. This is not merely a cartographic curiosity; it is the operational signature of positive curvature.

One can say the same thing in the language of triangles. On a positively curved surface, the sum of the angles of a triangle is greater than 180° . The sky remembers that. A curved geometry changes the relation between physical size, distance, and angular size.

The lecture also stresses a practical point. Angle alone is not enough. One needs both angular size and distance, together with some reason to believe the object’s actual size is known well enough for it to act as a standard ruler. In that sense the astronomer becomes a surveyor, measuring large triangles in the sky.

5.4.1 Question & Answer

Question. Why do circles stay circles, and what exactly does conformal mean here?

Answer. Stereographic projection has two different virtues. First, circles on the sphere project to circles in the plane, with straight lines understood as circles of infinite radius. That is a special property of this projection. Second, it is conformal: it preserves angles. Conformal does *not* mean that sizes are preserved. It means that sufficiently small shapes keep their local shape because the angles between directions are unchanged. Thus a small continent keeps the right shape while acquiring the wrong scale. The Mercator projection has the second property as well; stereographic projection has both.

A related warning belongs here. Because the projection stretches distances differently in different regions, signal propagation drawn on the map may appear to speed up or slow down across the page. That is a property of the projection, not a physical change in the speed of light.

5.5 Negative Curvature, the Hyperbolic Metric, and the Disk Picture

Having said, in effect, “that is about everything we need to know about spheres,” the lecture pivots to a new homogeneous geometry. It is not literally the opposite of a sphere, but it is the standard example of uniform negative curvature.

Susskind’s route is important. He does not begin with the picture. He begins with the metric, by analogy with the sphere:

$$d\mathcal{H}_n^2 = d\alpha^2 + \sinh^2 \alpha \, d\Omega_{n-1}^2. \quad (5.19)$$

Notice carefully what changes and what does not. The angular factor $d\Omega_{n-1}^2$ is still the metric of the ordinary unit sphere of one lower dimension. What changes is the growth law of its radius. We replace $\sin \alpha$ by $\sinh \alpha$.

The hyperbolic sine is

$$\sinh \alpha = \frac{e^\alpha - e^{-\alpha}}{2}. \quad (5.20)$$

The student correction in the lecture matters: the minus sign is the correct one. Also, α is no longer an angle running from one pole to another. In the hyperbolic case it runs from

$$0 \leq \alpha < \infty. \quad (5.21)$$

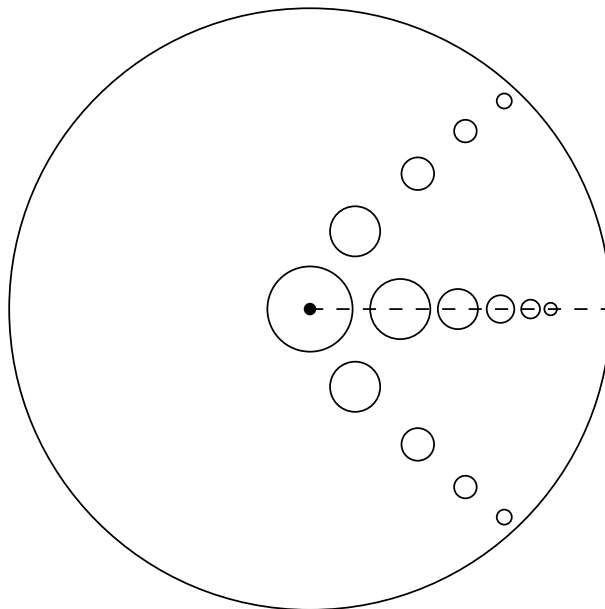
This single change is the whole story geometrically. For the sphere, $\sin \alpha$ starts at zero, rises, and returns to zero. For negative curvature, $\sinh \alpha$ starts at zero and then keeps growing. So the spheres around us do not first grow and then shrink. They grow without bound, and indeed exponentially for large α .

In three spatial dimensions we therefore write

$$d\mathcal{H}_3^2 = d\alpha^2 + \sinh^2 \alpha \, d\Omega_2^2. \quad (5.22)$$

Only after this algebraic setup is in place does the lecture turn to the picture.

The lecture likes to compare this with the stereographic projection of the sphere. There the page looked infinite, but the actual spherical distance was finite. Here the page looks finite, but the actual hyperbolic distance is infinite. Equal objects appear smaller and smaller as they approach the boundary. The drawing is deceptive in exactly the opposite way.



Poincaré disk: equal objects shrink toward the boundary

Figure 5.4: A schematic disk picture for negative curvature. The boundary looks finite in the drawing, but its intrinsic distance is infinite.

This is why distant objects in a negatively curved universe look too small relative to flat space. A standard ruler at fixed distance subtends too small an angle. Again the geometry has become an observational statement.

Only after the disk picture has been understood does the lecture write the cosmological spacetime metric:

$$d\tau^2 = dt^2 - a(t)^2 d\mathcal{H}_3^2. \quad (5.23)$$

This is the negatively curved FRW universe, often described in the lecture as the infinite FRW case. Its sign label is

$$k = -1. \quad (5.24)$$

Two more geometric remarks from the lecture should be preserved. First, this negatively curved space is still homogeneous. The drawing looks wildly inhomogeneous because the tiles near the edge are tiny, but any observer moved to another location sees the same intrinsic geometry. Second, one can draw small patches of negative curvature as saddle-shaped surfaces, but only locally. There is no global embedding of the entire hyperbolic space as a nice ordinary surface in Euclidean three-space.

5.5.1 Question & Answer

Question. Why does a finite-looking disk represent an infinite homogeneous space?

Answer. Because the Euclidean size in the drawing is not the intrinsic size in the metric. The little tiles or continents near the edge are geometrically the same size as those near the center; only their *pictures* are smaller. If one measures distance by counting equal geometric tiles, one crosses

infinitely many before reaching the boundary. The boundary therefore lies at infinite distance even though it looks finite on the page. Homogeneity survives because every tile is intrinsically like every other one, and the isometries of the space move one location into another without changing the geometry.

5.6 The Flat Limit and the Three FRW Cases

Only once the positively and negatively curved cases are on the table does the lecture turn to the intermediate case. Flat space is the boundary between them. One may think of it as a limit if one likes, but Susskind eventually says not to fuss over that. It is simply flat Euclidean space.

The corresponding FRW metric is

$$d\tau^2 = dt^2 - a(t)^2 (dx^2 + dy^2 + dz^2). \quad (5.25)$$

Its label is

$$k = 0. \quad (5.26)$$

At this point the three classical homogeneous cosmologies can be stated cleanly:

$$k = +1 : \text{space is spherical and positively curved}, \quad (5.27)$$

$$k = 0 : \text{space is flat}, \quad (5.28)$$

$$k = -1 : \text{space is hyperbolic and negatively curved}. \quad (5.29)$$

This is a regrouping moment in the lecture, not yet the final payoff. We have built the three geometries, understood how they distort angular sizes, and attached the sign labels $k = +1, 0, -1$. Only now are we ready to return to the earlier cosmological equations and ask what Einstein has to do with them.

5.7 Newtonian Cosmology Recalled, Einstein Cosmology Interpreted

After the break Susskind does not jump straight to general relativity. He backs up. That is important to the rhythm of the argument: the old Newtonian constant must be put back on the table before its geometric meaning can be explained.

The board at the transition still carries a fragment of the flat FRW metric:

The screenshot only certifies the visible spatial piece $dx^2 + dy^2 + dz^2$, but that is enough to witness the board continuity: the lecture is now moving from the three FRW metrics toward general relativity.

The Newtonian recap is brief and disciplined. Put ourselves at the center. Consider a galaxy at distance $r(t)$ moving away according to the Hubble flow. By Newton's shell theorem, only the mass inside the sphere of radius r contributes to the force on that galaxy. Writing the Newtonian energy

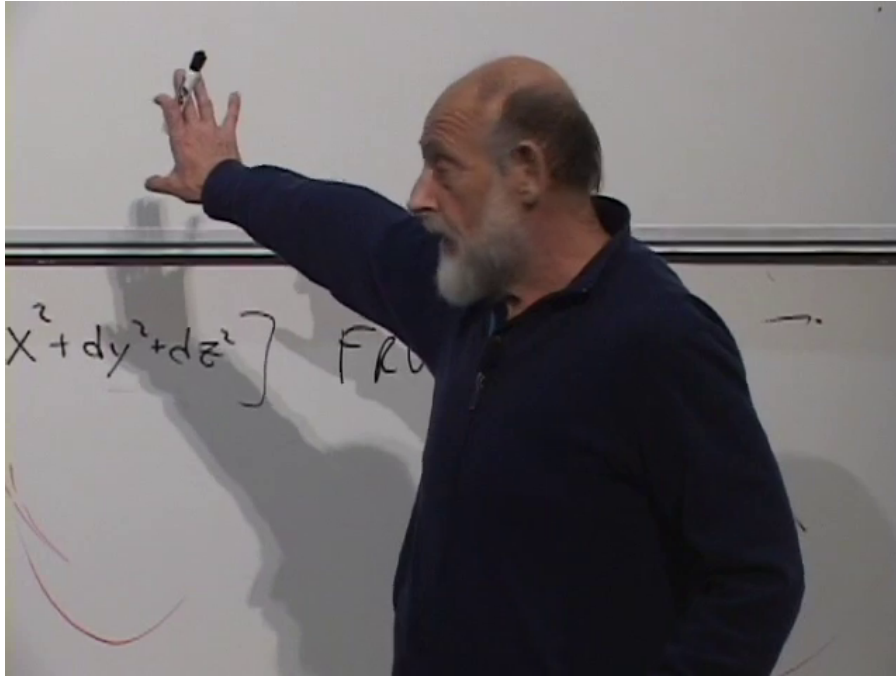


Figure 5.5: Flat-space metric fragment on the board, carried into the transition back from the break.

equation per unit mass,

$$\frac{1}{2}\dot{r}^2 - \frac{GM(r)}{r} = E, \quad (5.30)$$

$$M(r) = \frac{4\pi}{3}r^3\rho. \quad (5.31)$$

For a comoving galaxy we may write

$$r(t) = a(t)\chi, \quad (5.32)$$

with χ fixed. Then $\dot{r} = \dot{a}\chi$, and after dividing through by $a^2\chi^2$ one arrives at the Newtonian cosmology equation in the form

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}, \quad (5.33)$$

where the last term packages the previously omitted constant total-energy contribution. The knife-edge case studied earlier corresponds to setting that constant to zero, that is, exactly the escape-velocity case.

The lecture also reinstates the density scalings that accompany the matter content:

$$\rho_m \propto a^{-3}, \quad \rho_r \propto a^{-4}, \quad \rho_{\text{vac}} = \text{const.} \quad (5.34)$$

and, as before, pressure is written as

$$p = w\rho, \quad (5.35)$$

with the familiar cases

$$w = 0, \quad w = \frac{1}{3}, \quad w = -1 \quad (5.36)$$

for matter, radiation, and vacuum energy.

Only now does the lecture turn to Einstein's field equations. The board writes a compressed version, and the top-to-bottom staging is itself part of the rhetoric: first the general field equation, then the 00 component that cosmology actually needs.

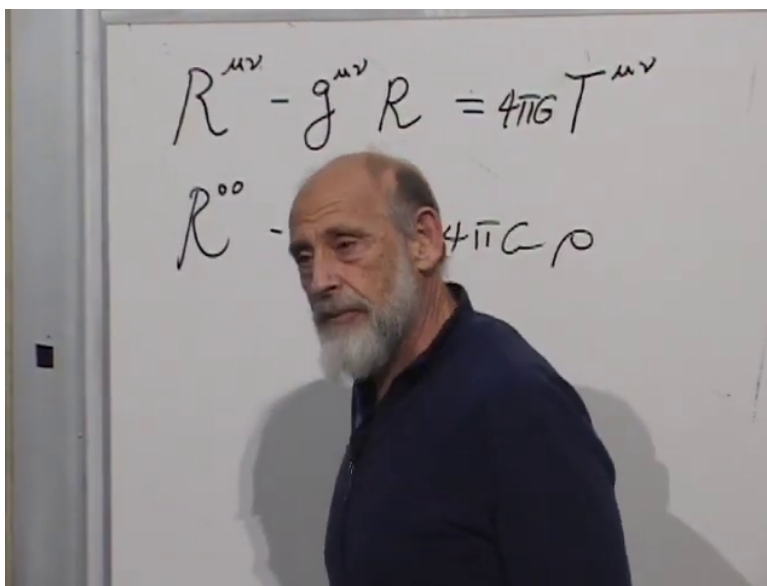


Figure 5.6: Einstein equation on the top line and its 00 component below it.

The board image does not show the factor $\frac{1}{2}$ multiplying $g^{\mu\nu}R$ clearly, but the lecture later acknowledges that a half belongs there. We therefore write the cleaned form while preserving the lecture's normalization:

$$R^{\mu\nu} - \frac{1}{2}g^{\mu\nu}R = 4\pi G T^{\mu\nu}. \quad (5.37)$$

The 00-component becomes

$$R^{00} - \frac{1}{2}g^{00}R = 4\pi G \rho. \quad (5.38)$$

The left-hand side is geometry, now spacetime geometry rather than merely spatial geometry. The right-hand side is matter, more precisely the energy-momentum tensor. The lecture also reminds us that the other components of $T^{\mu\nu}$ need not vanish. In particular, the space-space components encode pressure. That is why the equation of state $p = w\rho$ remains in the story even though the 00 equation is the one singled out to determine $a(t)$.

5.7.1 Question & Answer

Question. Why is one Einstein equation enough once FRW symmetry leaves only $a(t)$ as the unknown?

Answer. Because by the time we reach Einstein's equations we have already assumed an enormous amount. We have assumed homogeneity. We have assumed isotropy. We have assumed that the spatial slices are of one of the three FRW types. After all of that, the metric contains only one dynamical function, the scale factor $a(t)$. Many tensor equations remain formally present, but there is only one independent unknown function to solve for. The 00 component is the natural

choice because T^{00} is the energy density ρ , so this component most directly ties geometry to the cosmological matter content.

The lecture does not carry out the curvature calculation in detail on the board. Instead it tells us what kind of terms must appear when the FRW metric is substituted into the Einstein tensor.

First, there is a contribution that comes from the fact that the geometry is time dependent. It would be there even if space were flat. This term is proportional to

$$\left(\frac{\dot{a}}{a}\right)^2. \quad (5.39)$$

It represents spacetime curvature produced purely by the changing scale factor.

Second, there is a contribution that would remain even if the scale factor were frozen. This is the contribution of spatial curvature itself. For a homogeneous space of fixed shape, curvature scales like the inverse radius squared:

$$\text{curvature} \propto \frac{1}{a^2}. \quad (5.40)$$

With the sign convention already prepared by the geometry, that contribution is

$$\frac{k}{a^2}. \quad (5.41)$$

Positive curvature contributes with a plus sign, negative curvature with a minus sign, and flat space with zero.

The lecture's coefficient bookkeeping here is informal and partly self-correcting; Susskind checks the answer against the earlier Newtonian result rather than presenting a polished tensor derivation. The cleaned endpoint is

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho, \quad (5.42)$$

or equivalently

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \quad (5.43)$$

This is the real destination of the lecture. The same symbol k is now seen in two lights. From the Newtonian point of view, it classifies the motion: above escape velocity, at escape velocity, or below escape velocity. From the Einstein point of view, it classifies the shape of space itself: hyperbolic, flat, or spherical.

There is one final normalization subtlety that the lecture brings out in discussion. In the spherical and hyperbolic cases there is a natural curvature radius, so one may choose $a(t)$ so that $k = \pm 1$. In the flat case there is no distinguished spatial radius, so $a(t)$ may be rescaled without changing the statement $k = 0$. Thus the three values $k = +1, 0, -1$ are best thought of as a sign classification once the natural curvature scale has been absorbed into $a(t)$.

This is why general relativity does not simply replace the earlier Newtonian cosmology. It explains it. The old integration constant was already measuring something real; Einstein tells us that what it was measuring was the curvature of space.

5.8 Summary

The lecture unfolds in a very deliberate order. We begin with spheres, not because cosmology has become geometry for its own sake, but because homogeneous cosmology needs a precise language for homogeneous spaces. The recursive construction of S^1 , S^2 , and S^3 then becomes the spatial part of the FRW metrics once the fixed radius is promoted to $a(t)$.

The long visual detour is equally essential. Positive curvature makes standard rulers look too large, negative curvature makes them look too small, and stereographic or disk pictures teach us how to think from the observer's point of view. Only after that does the Einstein bridge become intelligible. The final Friedmann equation contains exactly the two geometric ingredients the lecture has prepared us for: the time-variation term $(\dot{a}/a)^2$ and the spatial-curvature term k/a^2 . The sign k is therefore both a dynamical classifier and a statement about the shape of space.

Chapter 6

From Last Scattering to Baryogenesis

We begin where Leonard Susskind begins: not with a speculative microphysical mechanism, but with the sky at different distances. Farther out means earlier, and earlier means denser and hotter. From that observational starting point the lecture moves through last scattering, then farther back into pair production and the quark–antiquark soup, then stops, turns the movie around, and asks why any matter was left over at all. These notes follow that order closely and retain selected board evidence from the lecture record prepared with LazyingArt LLC and Video2Book.

6.1 Looking Outward Means Looking Backward

The lecture opens with a very simple geometric idea. We put ourselves at the center, draw shells at larger and larger distances, and ask what we see. Nearby, the look-back time is negligible. The universe a year ago, or a hundred years ago, is not very different from the universe now. But once we look far enough, distance becomes time depth: the farther we look, the younger the universe we are seeing.

And once the universe is younger, it is denser; once it is denser, it is hotter. Susskind fixes the discussion with an order-of-magnitude landmark: if we look back far enough, we reach a place where the universe was about as hot as the surface of the sun. The sky does not *look* like the surface of the sun because the photons have been redshifted on their way to us. At the level needed in this lecture, we may summarize the relation by

$$T_{\text{obs}} \sim \frac{T_{\text{emit}}}{1+z}, \quad (6.1)$$

and for the last-scattering epoch the lecture uses the rough scale

$$1+z \sim 10^3. \quad (6.2)$$

So the temperature we observe in the sky is lower by about three orders of magnitude than the temperature at emission.

This is already the right cosmological attitude. We are not simply *told* that the early universe was hot. We are pushed toward that conclusion by the ordinary act of looking outward.

6.2 Last Scattering, Ionization, and Opacity

The first real threshold in the lecture is not baryogenesis but visibility. Hydrogen remains ionized above a temperature of order a few thousand kelvin. Thus when we look back beyond a certain distance, we do not find neutral atoms. We find an ionized gas of nuclei and free electrons, immersed in a sea of energetic photons.

Definition 6.1. The *surface of last scattering* is the transition surface at which the universe changes from an opaque ionized plasma to a transparent gas of neutral atoms.

The lecture's physical picture is vivid. The universe at that stage is mainly hydrogen, with a little helium. Above the ionization temperature, hydrogen does not remain bound into atoms. We have protons and electrons moving freely. In addition, the photons are shorter-wavelength and more energetic than they are today. If by accident an atom forms, it is quickly struck and reionized again.

That is why the medium is opaque. Ionized gas is an electrical conductor. Free charges respond to electric fields, slosh back and forth, and re-emit radiation. In that state the universe is not transparent. The transition is therefore

$$T \gtrsim \text{a few} \times 10^3 \text{ K} \implies \text{ionized plasma, opaque,} \quad (6.3)$$

$$T \lesssim \text{a few} \times 10^3 \text{ K} \implies \text{neutral atoms, transparent.} \quad (6.4)$$

Susskind compares this to the visible surface of the sun: not a hard material wall, but a place where one suddenly goes from opaque to transparent. Once the temperature drops below the hydrogen ionization scale, the universe clears and we can see through it.

That is the first observational boundary. Behind it, direct vision gives out. But the lecture does not stop there. Having explained why we cannot see farther, it immediately asks: how far can we nevertheless go by using known laws of atomic, nuclear, and particle physics?

6.3 Going Further Back: Pair Production and Thermal Thresholds

At the surface of last scattering the contents are still comparatively simple: mostly photons, with protons and electrons providing the charged matter content, and practically no positrons or antiprotons. But as we continue backward in time, the photons become more and more energetic. Eventually two photons can collide and produce an electron–positron pair:

$$\gamma + \gamma \leftrightarrow e^- + e^+. \quad (6.5)$$

The board records the threshold in its simplest lecture-level form:

$$E_{\gamma\gamma} \geq 2m_e c^2. \quad (6.6)$$

What matters here is not a fully invariant kinematic derivation, but the threshold logic. Below threshold, the photons do not have enough energy. Just above threshold, pairs can be made, but photons still dominate by number because electron rest mass is expensive. Far above threshold, however, that rest mass becomes less and less important relative to thermal kinetic energy, and the densities become comparable:

$$n_\gamma \sim n_{e^-} \sim n_{e^+}. \quad (6.7)$$

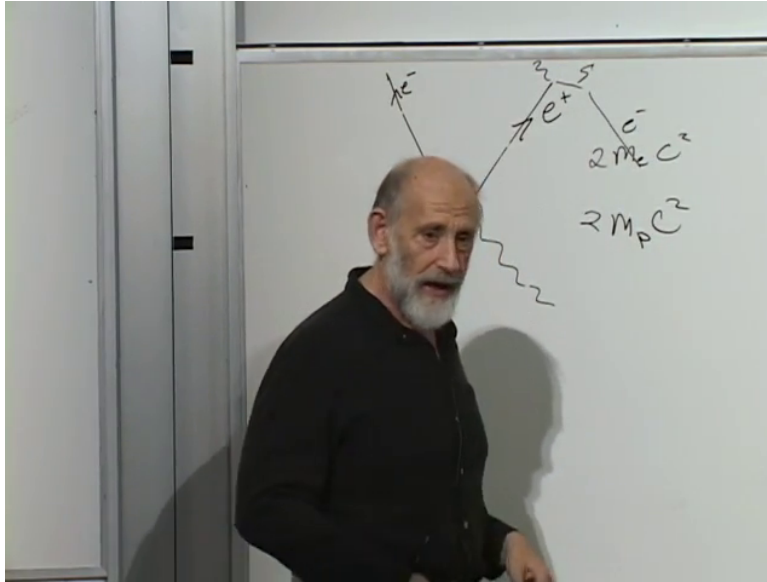


Figure 6.1: Electron–positron threshold sketch on the board. The upper threshold annotation $2m_e c^2$ is clear. The lower proton-scale note is less sharp in the frame, but the transcript makes the intended comparison unmistakable.

The lecture then asks the next natural question: what about protons? The answer is that the proton scale lies far higher,

$$E_{\gamma\gamma} \gtrsim 2m_p c^2, \quad m_p \approx 2000 m_e. \quad (6.8)$$

So there is a long interval in temperature where electrons and positrons are already copious while proton–antiproton pair creation is still negligible.

But even this proton language is only provisional. By the time the temperature is high enough to make proton–antiproton pairs in abundance, it is already high enough to shatter protons. That is why the lecture switches, at the hotter stage, from protons to quarks and antiquarks. The simplest schematic reaction he draws is

$$e^- + e^+ \rightarrow \gamma \rightarrow q + \bar{q}. \quad (6.9)$$

There are also gluons, neutrinos, and many other species, but the main point is the thermal logic: once we are far enough back, the soup contains nearly equal abundances of particles and antiparticles of each type, and in order-of-magnitude terms

$$n_q \sim n_{\bar{q}} \sim n_\gamma. \quad (6.10)$$

For the moment, however, the lecture does not want to drown in particle nomenclature. When it comes time to discuss the asymmetry, it deliberately simplifies back to protons and antiprotons as a stand-in for baryons and antibaryons.

6.4 Charge Neutrality and the Symmetric Hot Soup

Before the asymmetry problem can even be stated correctly, the lecture pauses over a distinction that is easy to miss.

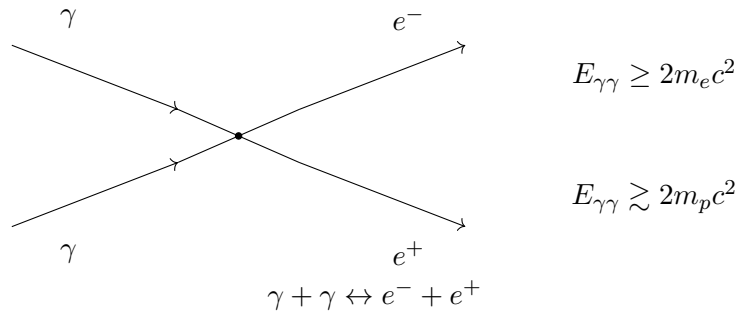


Figure 6.2: Clean reconstruction of the threshold comparison discussed in the lecture. The board image supplies the documentary evidence; the TikZ sketch only cleans up the process and the two mass scales.

The first statement is electrical neutrality. Today the universe appears electrically neutral. That means the total positive and total negative charge cancel. It does *not* mean that protons and antiprotons are equally numerous.

Susskind’s mnemonic for this is the flux picture on a closed spherical universe. If electric field lines cannot simply end in empty space, then a positive charge cannot sit by itself on a closed space without the flux refocusing somewhere and terminating on negative charge. Whether or not our universe is literally a closed three-sphere is not the point. The point is that net charge is extraordinarily constrained. In symbols,

$$Q_{\text{tot}} = 0. \quad (6.11)$$

But $Q_{\text{tot}} = 0$ is not the same statement as $N_p = N_{\bar{p}}$. Electrical neutrality means that protons, positrons, and all other positively charged species together balance electrons and all other negatively charged species together.

The baryon bookkeeping is different.

Definition 6.2. The baryon number is the net count of baryons minus antibaryons. For the simplified discussion in this lecture we may write

$$B = N_b - N_{\bar{b}}, \quad (6.12)$$

or more explicitly,

$$B = (N_p + N_n) - (N_{\bar{p}} + N_{\bar{n}}). \quad (6.13)$$

At sufficiently high temperature the pair-production processes drive us toward a hot soup in which particles and antiparticles are nearly equally abundant. Susskind is careful here. This is not a theorem about the beginning of the universe. It is the natural starting hypothesis suggested by the near symmetry of known particle physics and by the way thermal pair production works. Electrons and positrons are made together; quarks and antiquarks are made together. Thus, after simplifying back from quarks to protons for the sake of the story, the lecture asks us to imagine an early state with

$$N_p = N_{\bar{p}}. \quad (6.14)$$

This is now the real setup. We have a nearly symmetric hot soup, electrically neutral, and almost symmetric under exchange of particles and antiparticles. “Almost” is the right word. The laws

of physics are not exactly symmetric in that way, but they are close enough that the lecture can sharpen the problem without caricaturing it.

Up to this point we have been walking backward in time. The lecturer then stops and says, in effect: the real sequence is the other way around. Now let us run the movie forward.

6.5 Turning the Movie Around: Cooling, Annihilation, and Freeze-Out

The forward story is the cooling story. While the universe is hot enough, pair creation and annihilation both occur. Once the temperature drops below the relevant threshold, the inverse reaction shuts off first, while annihilation can continue. This is what the lecture calls the “one-way street.”

Consider first the proton channel. While the temperature is sufficiently high, we have

$$p + \bar{p} \leftrightarrow \gamma + \gamma. \quad (6.15)$$

As the universe expands, the photons are redshifted. Very soon they no longer have enough energy to climb back over the proton threshold, and what remains effectively is

$$p + \bar{p} \rightarrow \gamma + \gamma. \quad (6.16)$$

Exactly the same logic later applies to electrons and positrons:

$$e^- + e^+ \rightarrow \gamma + \gamma. \quad (6.17)$$

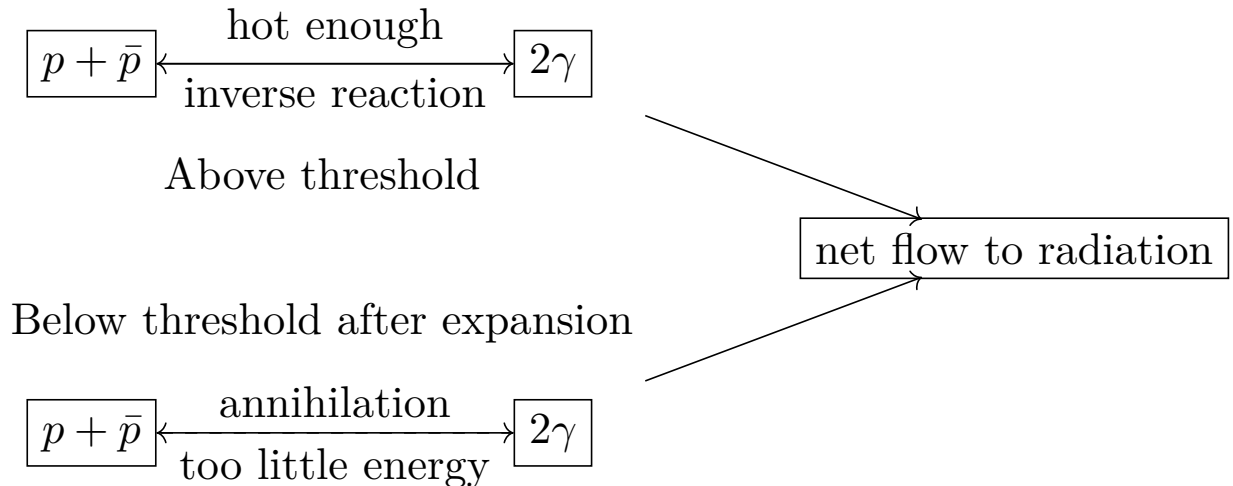


Figure 6.3: The “one-way street” of freeze-out. Above threshold the reaction runs both ways. After expansion lowers photon energies, annihilation still proceeds but inverse production is thermally suppressed.

There is a small but very useful piece of arithmetic here. A proton–antiproton annihilation event

changes the counts by

$$N_p \rightarrow N_p - 1, \quad (6.18)$$

$$N_{\bar{p}} \rightarrow N_{\bar{p}} - 1, \quad (6.19)$$

$$N_\gamma \rightarrow N_\gamma + 2. \quad (6.20)$$

Therefore the difference stays fixed:

$$(N_p - N_{\bar{p}}) \rightarrow (N_p - 1) - (N_{\bar{p}} - 1) = N_p - N_{\bar{p}}. \quad (6.21)$$

This is the mathematical heart of the freeze-out story. Ordinary annihilation does not *create* a proton excess. It merely strips away matched pairs and leaves any pre-existing difference exposed.

A concrete example makes the point completely transparent. Suppose at some early time

$$N_p^{\text{initial}} = N + \Delta, \quad N_{\bar{p}}^{\text{initial}} = N, \quad (6.22)$$

with $\Delta > 0$. After all possible annihilations have occurred, one finds

$$N_p^{\text{final}} = \Delta, \quad N_{\bar{p}}^{\text{final}} = 0. \quad (6.23)$$

What survives is precisely the initial excess.

The lecture then folds thermodynamics into the picture. The expansion is slow compared with microscopic reaction rates, so to a good approximation it is adiabatic:

$$S \approx \text{const}. \quad (6.24)$$

Strictly speaking it is the entropy that is conserved, not the particle count itself. But for a thermal gas of the sort being discussed, and especially for blackbody radiation, the entropy is, up to order-one factors, essentially the number of particles, in practice the number of photons. Thus the particles that disappear in annihilation are largely replaced by photons.

That is why the present photon bath is a fossil record of the earlier annihilation history. The lecture uses the present-day count

$$\frac{N_\gamma}{N_p} \sim 10^9\text{--}10^{10}. \quad (6.25)$$

Since today $N_{\bar{p}}$ is negligible, we may also phrase the same information as

$$\frac{N_p - N_{\bar{p}}}{N_\gamma} \sim 10^{-9}\text{--}10^{-10}. \quad (6.26)$$

So the asymmetry was not of order one. It was tiny: about one extra proton for every billion proton–antiproton pairs.

That numerical fact changes the entire tone of the discussion. We are no longer asking why a matter universe did not annihilate away. We are asking how such an absurdly small but nonzero difference could ever have been produced.

6.6 Baryogenesis and the Sakharov Conditions

The lecture now raises the central obstacle very explicitly: if we begin with a nearly symmetric hot soup, why is there any residue at all?

6.6.1 Question & Answer

Question. Suppose we start from a perfectly symmetric state,

$$N_p = N_{\bar{p}}, \quad (6.27)$$

and let the universe cool. Why can ordinary physics not generate a tiny extra proton population during cooling? And if not that, could the observed excess simply be a statistical fluctuation?

Answer. Ordinary cooling cannot do the job. We have already seen that pair annihilation preserves $N_p - N_{\bar{p}}$. So if we start from zero difference, annihilation alone leaves us with zero difference. To get one extra proton we would need a process that changes the net baryon number.

Electric charge is not the obstruction. One could imagine, schematically, a process of the form

$$\gamma + \gamma \rightarrow p + e^-, \quad (6.28)$$

which is electrically neutral overall. But that is precisely the kind of process that the tested standard model does *not* contain. Within the regime where laboratory particle physics is known to work, one cannot create a proton without also creating an antiproton. That empirical rule is baryon-number conservation.

In the lecture's simplified bookkeeping,

$$B = (N_p + N_n) - (N_{\bar{p}} + N_{\bar{n}}), \quad (6.29)$$

and within tested particle physics B is conserved. So once the temperature has fallen into the range where we trust the standard model, there is no ordinary low-energy route by which a proton excess can develop during cooling.

What about statistics? Could a random freeze-out leave a net excess just by fluctuation? The lecture answers this in the standard square-root-of- N way:

$$\Delta N_{\text{stat}} \sim \sqrt{N}. \quad (6.30)$$

If we count all particles in the observable universe, including photons, the lecture uses the rough scale $N \sim 10^{90}$, so

$$\sqrt{10^{90}} \sim 10^{45}. \quad (6.31)$$

But the actual proton count is of order 10^{80} . Thus the fluctuation estimate is hopelessly too small:

$$10^{45} \ll 10^{80}. \quad (6.32)$$

So the present matter excess is not a statistical accident of an otherwise unbiased world.

Once those escape routes are closed, the lecture turns constructive. If the asymmetry is real, what must the laws of nature have allowed at very high temperature?

The board now acquires its characteristic shape. First the symmetric starting point,

$$N_p = N_{\bar{p}}, \quad (6.33)$$

then the leftover difference,

$$N_p - N_{\bar{p}}, \quad (6.34)$$

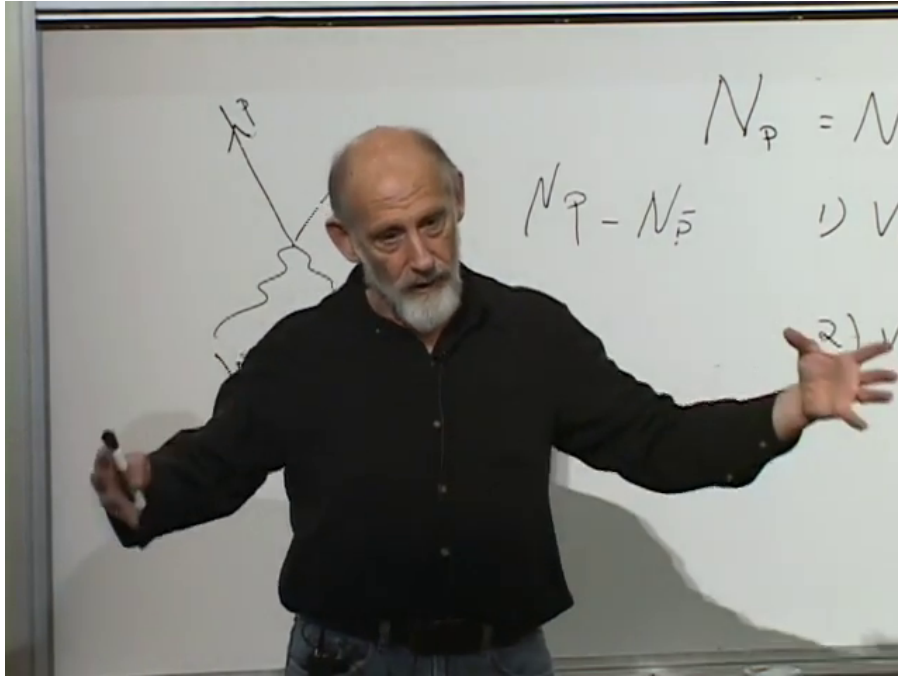


Figure 6.4: Board state during the transition from the residual asymmetry $N_p - N_{\bar{p}}$ to the developing numbered list of conditions. The screenshot is useful mainly as documentary evidence for the board's organization.

and then the numbered list of conditions under which one can get from the first expression to the second.

The first condition is baryon-number violation. If B is exactly conserved, then a baryon-neutral universe stays baryon-neutral forever.

The second condition is a bias between particles and antiparticles. The lecture first phrases this loosely as violation of charge-conjugation symmetry and later, correctly, sharpens it to CP violation. The reason is simple. Even if baryon number is violated, it could be violated symmetrically. Schematically, one might have

$$\gamma + \gamma \rightarrow p + e^-, \quad (6.35)$$

$$\gamma + \gamma \rightarrow \bar{p} + e^+. \quad (6.36)$$

If those two channels occur with equal average probability, the net asymmetry still vanishes. Baryon violation by itself is therefore not enough; one also needs a bias.

The third condition is departure from thermal equilibrium. Here the lecture invokes the CPT theorem in its conceptual form. In thermal equilibrium a movie run backward looks like the same movie run forward. If, in addition, the underlying theory is exactly symmetric under the combined CPT operation, then equilibrium effectively restores the particle–antiparticle symmetry that we are trying to break. A hot static pot cannot maintain a persistent proton excess. Therefore baryogenesis requires an epoch in which the universe is sufficiently out of equilibrium that forward and backward histories are microscopically distinguishable.

This is the Sakharov structure:

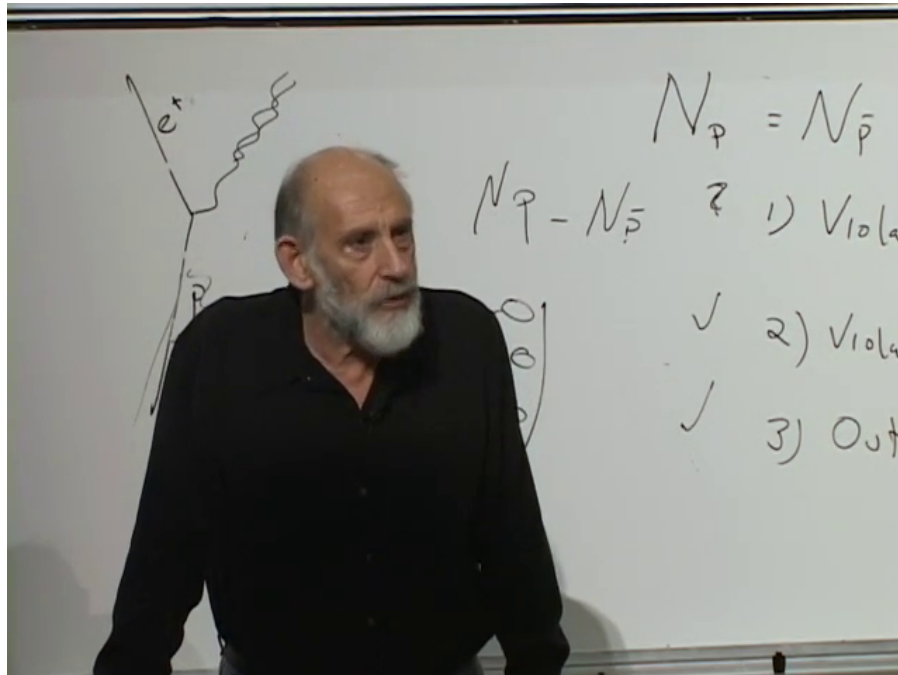


Figure 6.5: Asymmetry conditions and checklist on the board. The numbered items are reconstructed from the transcript rather than read directly from the incomplete pixels.

1. Violation of baryon number;
2. Violation of particle–antiparticle symmetry, more precisely CP violation;
3. Departure from thermal equilibrium.

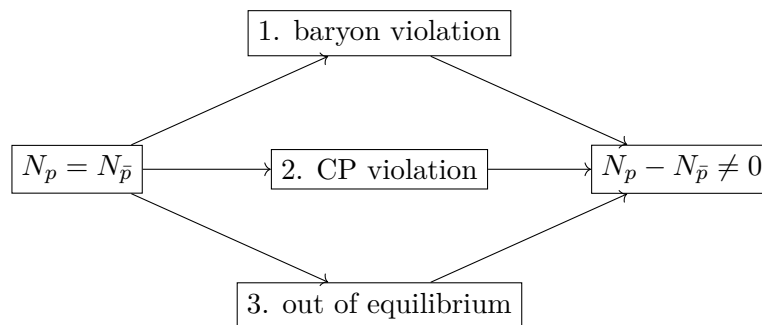


Figure 6.6: Minimal logical reconstruction of the board discussion. A symmetric starting point can tilt into a nonzero asymmetry only if the relevant laws violate baryon number, distinguish particles from antiparticles, and operate during a nonequilibrium epoch.

The lecture treats this combination as extraordinarily robust. If these conditions are present at sufficiently high energy, then a tiny leftover asymmetry is not mysterious. If they are absent, the asymmetry is extraordinarily hard to obtain.

6.7 Clarifications at the Edge of What We Know

The closing third of the lecture widens the ring of discussion. It does not abandon baryogenesis; it asks what part of the argument is theorem, what part is experiment, and what part is still conjecture.

First comes the distinction between CP, time reversal, and CPT. The relevant exact theorem is CPT, not plain charge conjugation. The lecture's point is subtle but important. Condition 3 is *not* a violation of CPT. To be out of equilibrium simply means that the actual history of the universe was not a static equilibrium history. The universe was expanding, and sufficiently early it was expanding rapidly enough to matter on microscopic time scales. That gives a real arrow in the history without implying that the time-reversed history would fail to satisfy the underlying equations. So one must not confuse

$$\text{out of equilibrium} \tag{6.37}$$

with any violation of the CPT theorem.

Second comes the empirical status of baryon-number violation. This is the question mark. We have never directly observed a reaction in nature in which

$$B = (N_p + N_n) - (N_{\bar{p}} + N_{\bar{n}}) \tag{6.38}$$

changes. That may only mean that the relevant reactions are important at very high temperature and fantastically improbable at low energy. But at present it remains unobserved.

That is why proton decay occupies so much of the late discussion. If baryon number is not exact, then the proton ought in principle to be unstable. Yet it is astonishingly stable. Ordinary particle processes occur on absurdly short time scales, but the proton must live at least as long as the age of the universe, and in fact much longer. The lecture quotes the experimental lower bound at the order-of-magnitude level as

$$\tau_p \gtrsim 10^{34} \text{ years.} \tag{6.39}$$

Huge underground detectors, typically giant volumes of water instrumented with photodetectors, have searched for precisely this sort of event. So far, nothing has been seen.

This empirical tension is one reason the story remains interesting. It is easy to write down extensions of the standard model that violate baryon number. The hard part is to do so without making the proton decay much too fast.

Third comes gravity. At the level of microscopic particle collisions, gravity is negligible and does not run the bookkeeping. But it enters the story in two deeper ways. One is obvious: without gravitational expansion of the universe there would be no cosmological departure from equilibrium. The second is the black-hole thought experiment emphasized in the lecture. If a star carrying a huge positive baryon number collapses to a black hole and later evaporates mostly into low-energy thermal radiation, it becomes very hard to believe that baryon number can remain an exact conservation law forever. That is not a laboratory observation of baryon violation, but it is powerful theoretical pressure against exact baryon conservation.

Electric charge is different. The lecture makes this contrast very sharply. A charged black hole cannot simply forget its charge, because electric flux lines continue to extend to infinity and keep track of it. That is why charge conservation is treated as exact here while baryon-number conservation is treated as provisional.

The lecture also briefly cleans up two side issues. The electroweak restoration scale, around 100–250 GeV, belongs to known physics and may matter in more complete models, but it is not the dramatic step in this story. Dark matter, in most standard discussions, is more or less a spectator here; it may carry its own history, but it does not alter the basic baryogenesis logic being isolated in this lecture.

There is also an important retrospective remark near the end. Older versions of the problem were often posed upside down: why are there so many photons compared with protons? But once the adiabatic picture is taken seriously, that formulation loses much of its force. The entropy of the universe, which in this context is essentially the number of photons up to an order-one factor, does not seem to have been manufactured in some gigantic late burst. The sharper question is the opposite one: why was the asymmetry between particles and antiparticles so tiny? Why was the excess only one part in a billion?

Neutron coda. The lecture closes with a short discussion of neutrons, and it is worth keeping only the central point. Neutron beta decay,

$$n \rightarrow p + e^- + \bar{\nu}, \quad (6.40)$$

does *not* violate baryon number; the neutron carries the same baryon number as the proton. A free neutron decays because its mass is slightly larger than that of the final state, which we may summarize schematically as

$$m_n \gtrsim m_p + m_e + m_\nu. \quad (6.41)$$

The available energy is small, and that small phase space contributes to the long lifetime of the free neutron. If the neutron is bound inside a nucleus, binding energy can remove the available decay energy:

$$M_{\text{bound}} \approx m_n + m_p - \frac{E_{\text{bind}}}{c^2}. \quad (6.42)$$

That is why a bound neutron may be stable even though a free neutron is not. This is not part of baryogenesis proper, but it neatly separates an allowed weak decay from the genuinely hypothetical baryon-violating reactions that baryogenesis would require.

6.8 Summary

The lecture begins with visibility and ends with asymmetry. Looking outward takes us to earlier and hotter epochs; last scattering marks the transition from opaque ionized plasma to transparent neutral matter. Going still farther back, known atomic, nuclear, and particle physics lead us to a hot soup in which electrons and positrons, and later quarks and antiquarks, are produced in nearly equal numbers.

When we then run the movie forward, the essential mechanism is freeze-out. As thresholds are crossed, inverse production shuts off while annihilation continues, and matched particle–antiparticle pairs disappear into photons. The difference $N_p - N_{\bar{p}}$ is not created by that process; it is preserved by it. The present photon excess,

$$\frac{N_\gamma}{N_p} \sim 10^9\text{--}10^{10}, \quad (6.43)$$

therefore tells us that the primordial asymmetry was only about one part in a billion.

That is why the lecture lands on baryogenesis. Ordinary cooling cannot generate the asymmetry within tested particle physics, and statistical fluctuations are far too small. To tilt a symmetric universe into a matter-dominated one, the laws must violate baryon number, distinguish particles from antiparticles, and act during a nonequilibrium epoch. Of these ingredients, the decisive missing empirical piece is still direct baryon-number violation. The structure of the story is compelling; the scale of the answer is plausible; but the most important experimental confirmation remains to be seen.

Chapter 7

Inflation, Vacuum Energy, and the de Sitter Mechanism

Inflation is the announced subject of this lecture, but the lecturer does not begin there. He first rebuilds the simplest cosmological mechanism that will carry the whole story: if the energy density is a true vacuum energy, then it does not dilute; if it does not dilute, then the Hubble rate is constant; and if the Hubble rate is constant, then the scale factor grows exponentially. Only after that does he pivot to the present accelerated universe, the cosmic horizon, the quantum meaning of vacuum energy, and finally to the conjecture that the same de Sitter mechanism once operated in the remote past with a vastly larger effective vacuum energy. From there the lecture turns to a scalar field ϕ , its potential $V(\phi)$, the slow drift along a high plateau, reheating at the cliff, and the survival of small quantum fluctuations that later become the seeds of galaxies.

Credit. These companion notes follow Leonard Susskind’s Lecture 7, with supporting frame extraction curated by LazyingArt LLC and Video2Book.

7.1 Vacuum Energy, de Sitter Space, and Exponential Expansion

The lecture opens by postponing inflation and going back to the Friedmann equation in flat space. Susskind speaks of $A(t)$; here we use the standard $a(t)$, with the same meaning:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho. \quad (7.1)$$

The earlier cosmological cases are then recalled. For matter one has $\rho \propto a^{-3}$; for radiation $\rho \propto a^{-4}$; but for vacuum energy the density does not dilute at all:

$$\rho = \rho_0 = \text{constant}. \quad (7.2)$$

That is the special case he wants to isolate. If the right-hand side of the Friedmann equation is constant, we may write

$$H^2 = \frac{8\pi G}{3}\rho_0, \quad H = \sqrt{\frac{8\pi G}{3}\rho_0}. \quad (7.3)$$

In this pure vacuum-energy cosmology the Hubble rate is truly constant:

$$\frac{\dot{a}}{a} = H. \quad (7.4)$$

Now the differential equation is as simple as it can be:

$$\dot{a} = Ha. \quad (7.5)$$

Its solution is exponential,

$$a(t) = a_0 e^{Ht}, \quad (7.6)$$

so the distance between sufficiently separated unbound galaxies grows like e^{Ht} . This exponentially expanding spacetime is de Sitter space.

The lecture also writes the metric in these expanding coordinates. Regularizing the spoken $d\tau^2$ notation into the standard line element, we have

$$ds^2 = dt^2 - e^{2Ht} (dx^2 + dy^2 + dz^2). \quad (7.7)$$

That is the basic machine. The later inflationary story will not introduce a different mechanism; it will re-use this one at a much larger scale.

One brief side issue appears here. The lecture assumes positive vacuum energy. In exactly flat space that is the case we need, because the left-hand side of the Friedmann equation is nonnegative. Negative vacuum energy can arise in more general cases once curvature terms are included, but that branch is not pursued in this lecture.

Remark 7.1. The lecture briefly recalls Einstein's original use of the cosmological constant as a balancing term against ordinary matter in an attempt to hold the universe static. It is worth remembering historically, but it is not part of the mathematical line we need here.

7.2 Present Acceleration and the Cosmic Horizon

Having rebuilt pure de Sitter space, the lecture asks what use it has in cosmology. The first use is contemporary. The universe today is not exactly de Sitter, but it is moving in that direction. Earlier, the scale factor followed the matter-dominated law

$$a(t) \propto t^{2/3}, \quad (7.8)$$

whereas now the observed expansion bends upward toward an exponential.

Acceleration means exactly what it should mean:

$$\ddot{a} > 0. \quad (7.9)$$

The lecture insists that this is not a speculation. The late-time data do not yet give a perfect exponential, because the universe still contains other forms of energy, but the trend is clear: vacuum energy is beginning to dominate.

From there the lecture moves to the horizon. In proper-distance form, Hubble's law is

$$v = HD. \quad (7.10)$$

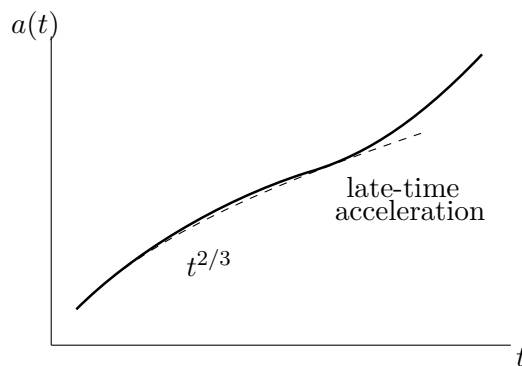


Figure 7.1: The lecture’s qualitative observational sketch: matter-dominated growth $a(t) \propto t^{2/3}$ gives way at late times to accelerated expansion approaching an exponential.

The horizon is the distance at which the recession speed becomes the speed of light. Setting $v = c$, we obtain

$$D = \frac{c}{H}, \quad (7.11)$$

or, equivalently,

$$HD = c. \quad (7.12)$$

Only the first of these is fully secure from the board image; the second is the natural completion of the partially written line.

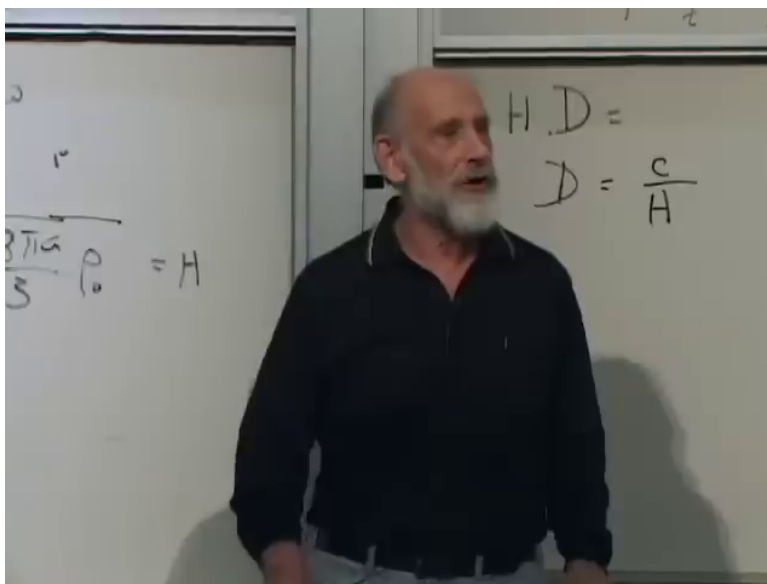


Figure 7.2: Hubble distance relation on the board. The right board cleanly shows $D = c/H$; the upper line suggests the equivalent form $HD = c$, while the left board preserves only a cropped residue of the preceding ρ_0 -to- H discussion.

This figure matters for more than the equation itself. It preserves the lecture’s board logic. On the left one is leaving the vacuum-energy derivation; on the right one has isolated the horizon formula. That is exactly the transition in the spoken argument.

7.2.1 Question & Answer

Why is $1/H$ numerically close to the age of the universe today, and why is that not a mathematical identity?

The lecture answers this by going back to a simple power law. If

$$a(t) = Ct^p, \quad (7.13)$$

then

$$\dot{a} = Cp t^{p-1}, \quad (7.14)$$

$$H = \frac{\dot{a}}{a} = \frac{Cp t^{p-1}}{Ct^p} = \frac{p}{t}. \quad (7.15)$$

So only for the special case $p = 1$ do we get $H = 1/t$. For a matter-dominated universe,

$$p = \frac{2}{3}, \quad H = \frac{2}{3t}. \quad (7.16)$$

Thus $1/H$ is of the same order as the age, but not equal to it.

The point becomes sharper in exact de Sitter space. There H is constant while the age continues to increase, so $1/H$ cannot possibly be the age of the universe. It is instead the e-folding time:

$$a(t + H^{-1}) = e a(t). \quad (7.17)$$

The lecturer says “double” a few times, but then corrects himself: the precise statement is multiplication by e . The true doubling time would be

$$t_{\text{double}} = \frac{\ln 2}{H}. \quad (7.18)$$

So the near coincidence today between the horizon timescale and the age of the universe is contingent, not a mathematical identity. The lecture leaves it as a real question why these two scales happen to be comparable in the era in which we live.

7.3 Vacuum Energy as Zero-Point Energy

At this stage a student asks the obvious question: what is vacuum energy actually? That question is not brushed aside. It becomes the next teaching unit.

The lecture answers first in the language of ordinary quantum mechanics. Consider a harmonic oscillator. Suppressing inessential constants, its energy is written schematically as

$$E_{\text{osc}} \sim p^2 + x^2. \quad (7.19)$$

Classically the ground state is the equilibrium point,

$$x = 0, \quad p = 0, \quad E = 0. \quad (7.20)$$

Quantum mechanically that state is forbidden by the uncertainty principle. One cannot simultaneously pin down position and momentum so sharply. Therefore the ground-state energy must be positive. For the harmonic oscillator one finds

$$E_0 = \frac{1}{2}\hbar\omega. \quad (7.21)$$

The lecture then lifts this picture from one oscillator to quantum fields. A field is full of modes, and each mode behaves like an oscillator. The electromagnetic field is used as the simplest example. Schematically, its energy density is

$$u_{\text{EM}} \sim E^2 + B^2. \quad (7.22)$$

The treatment is pedagogical rather than canonical, but the point is clear. Even when there are no real photons in the field, the ground state is not a zero-energy state. There remain zero-point oscillations.

Every field has such modes, and every field contributes zero-point energy. That is one lecture-level explanation of vacuum energy. It is not yet an explanation of why the cosmological constant is so tiny. The lecture explicitly postpones that harder problem. For now it wants only the essential property: this energy is there, and as the universe expands it does not dilute.

7.3.1 Question & Answer

How can “empty” space still carry energy?

Because the vacuum is not classical emptiness. It is the ground state of the quantum fields, and a quantum ground state need not have zero energy. The uncertainty principle forces irreducible fluctuations. So the vacuum can contain no ordinary particles and yet still carry zero-point energy.

The lecture also sharpens the distinction between vacuum energy and ordinary radiation. The three-degree microwave background is not vacuum energy. Radiation dilutes as the universe expands, and each photon redshifts as well. Vacuum energy is the residual field energy that remains even after all removable real quanta have been removed. That is why it does not dilute.

7.4 Energy, Horizons, and What Stays Fixed in de Sitter Space

Now the next objection arises. If the vacuum-energy density is constant while the universe expands, then does not the total vacuum energy increase? The lecture answers this cautiously and somewhat gesturally. It says, in effect, that in general relativity one is not dealing with ordinary mechanical energy alone; the expansion of space contributes a gravitational sector to the bookkeeping, and the concept of “the energy” is subtler than in elementary mechanics. It would be a mistake to convert this lecture comment into a polished Hamiltonian formula that was never actually derived on the board.

What the lecture does make precise enough is the horizon argument. In exact de Sitter space the horizon size is fixed:

$$D = \frac{c}{H} = \frac{c}{\sqrt{8\pi G\rho_0/3}}. \quad (7.23)$$

The volume inside that horizon is therefore fixed as well. If we estimate the vacuum energy accessible to a given observer, it scales like

$$E_{\text{hor}} = \frac{4\pi}{3} D^3 \rho_0. \quad (7.24)$$

Since D is constant and ρ_0 is constant, the amount of vacuum energy within the cosmic horizon does not change with time:

$$E_{\text{hor}} = \text{constant} \quad (\text{exact de Sitter space}). \quad (7.25)$$

This is the lecture's practical answer. From the observer's standpoint, the portion of the universe that can ever be seen contains a fixed horizon volume and therefore a fixed amount of vacuum energy. The lecture also remarks that objects can pass through this horizon in much the same sense that they can pass through a black-hole horizon: once outside, they are no longer in causal contact with us.

7.4.1 Question & Answer

If the vacuum-energy density is constant while space expands, where does the extra energy come from?

At the level of this lecture, the safest answer is: do not imagine a Newtonian box of ever-growing volume and then demand a simple conservation law for the energy inside that box. In exact de Sitter space the cosmic horizon is fixed, so the vacuum energy within the observable region is fixed. Beyond that, the lecture only wants to say that in general relativity the expansion sector participates in the bookkeeping, and that the idea of total energy is more delicate than in ordinary mechanics.

7.5 Why Inflation Was Proposed

Only after all of this does the lecture make its main turn. The same de Sitter mechanism, it says, appears to have operated in the remote past, but with a vastly shorter timescale:

$$H_{\text{inf}}^{-1} \sim 10^{-32} \text{ s}. \quad (7.26)$$

This is inflation. The universe in the remote past is conjectured to have undergone a period of exponential expansion before anything we can directly observe.

Why was such an idea introduced? The answer is not initially field-theoretic. It is geometric. The universe appears very flat and very homogeneous on large scales. More strikingly still, the surface of last scattering was already very smooth. So the smoothing cannot have happened only after the hot visible phase began. Something earlier must have stretched away the wrinkles.

The lecture's favorite analogy is the wrinkled balloon. Start with a completely deflated balloon and it is full of folds and texture. Inflate it enough and any local patch becomes nearly smooth. The raisin or prune analogy serves the same purpose. A wrinkled object, if stretched enormously, looks smooth on any visible patch. That is the intuition behind inflation.

The horizon problem then enters naturally. We see distant regions of the last-scattering surface that could not have exchanged signals with one another in the available time. So why do they have nearly the same density and temperature? Inflation answers by proposing a prehistory to what we ordinarily call the Big Bang: an earlier phase of exponential expansion stretches one small region into something vastly larger than the present observable universe.

The lecture is quite candid about the mixed blessing here. The good thing about inflation is that it erases the initial conditions. One need not know the detailed prehistory in order to explain why the visible universe is so smooth. The bad thing is the same: inflation erases much of the information about how things started.

This is also where the lecture reconnects with spatial curvature. Observationally the universe appears spatially flat, with the curvature parameter k consistent with zero to very good accuracy.

Inflation explains why that should be so. If the universe expanded enough before the visible hot era, then the curvature radius became so large that the region we can see now looks Euclidean.

7.6 Scalar Field ϕ and the Potential $V(\phi)$

So far inflation has been motivated geometrically. The next question is what could have supplied the much larger effective vacuum energy in the remote past. That is the point at which the scalar field enters.

A scalar field ϕ is a field with no directionality. At each point in space and time it simply takes a value:

$$\phi = \phi(x, y, z, t). \quad (7.27)$$

The lecture emphasizes that this is a hypothetical field. It is not identified with any known particle or any familiar observed field. It is introduced because the mathematics allows it and because the inflationary story seems to ask for something like it.

What matters is its potential-energy density $V(\phi)$. This is energy associated with the field value itself, not with the time derivative of the field. Different values of ϕ may carry different potential energies.

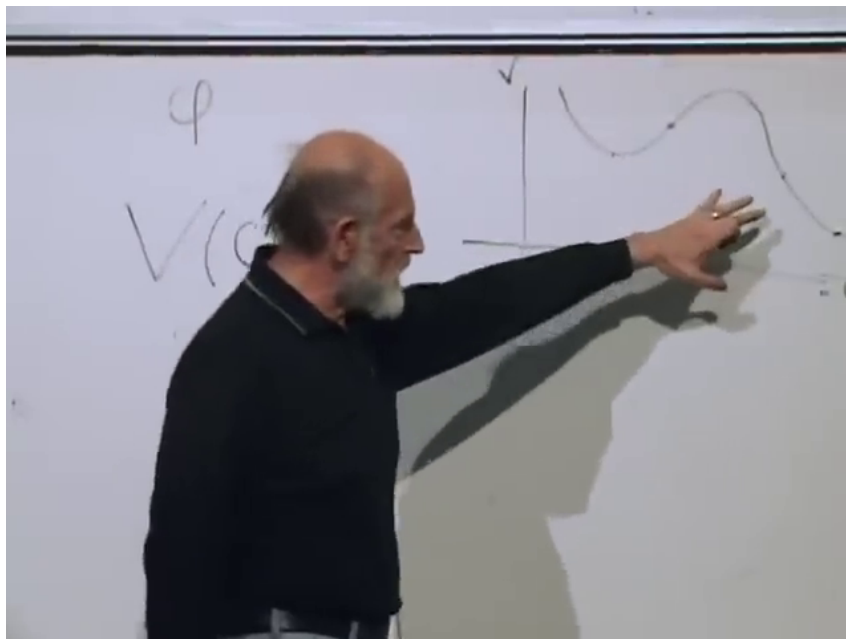


Figure 7.3: The first scalar-potential sketch used in the lecture. The board notation $V(\phi)$ is visible, and the curve makes the essential point that different values of ϕ correspond to different potential energies.

Now comes the decisive identification. If ϕ is approximately constant over the region we are studying, then $V(\phi)$ behaves just like vacuum energy:

$$\rho_{\text{eff}} = V(\phi). \quad (7.28)$$

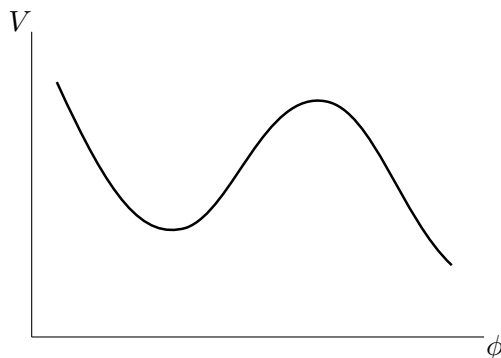


Figure 7.4: A clean redraw of the lecture’s generic $V(\phi)$ sketch. At this stage the point is simply that the potential energy changes with the field value.

Therefore the earlier de Sitter formula can be reused with the simple replacement

$$\rho_0 \longrightarrow V(\phi). \quad (7.29)$$

The corresponding Hubble rate is then

$$H(\phi) = \sqrt{\frac{8\pi G}{3} V(\phi)}. \quad (7.30)$$

This is the mathematical bridge between the first half of the lecture and the second. Inflation is not some additional cosmological law glued onto the Friedmann equation. It is the same Friedmann equation read with a different source.

The lecture then asks what happens if ϕ varies from place to place. In that case different regions expand with different local time constants. But exponential stretching tends to make any visible patch look more and more homogeneous. That is why inflation can start with a field that is only approximately uniform and still leave us living in a region where it appears nearly constant.

7.6.1 Question & Answer

What are the axes of the $V(\phi)$ graph, and in what sense can ϕ behave like a time parameter without being time itself?

The horizontal axis is the field value ϕ , and the vertical axis is the potential-energy density V . So this is not a graph of energy against time. It is a graph of energy against field value.

But if ϕ rolls monotonically down the potential, then the successive values of ϕ label successive moments along the cosmological history. In that restricted sense ϕ can act like a time parameter: later times correspond to later values along the curve. Still, ϕ is not time itself. It is a field value whose evolution happens to be monotonic during the relevant epoch.

7.7 Reheating, E-foldings, and Primordial Fluctuations

The lecture now sharpens into the picture it regards as observationally demanded. The potential is high on the left, almost flat over a broad plateau, then it comes to a cliff, falls sharply, and settles

into a lower minimum which is not exactly at zero.

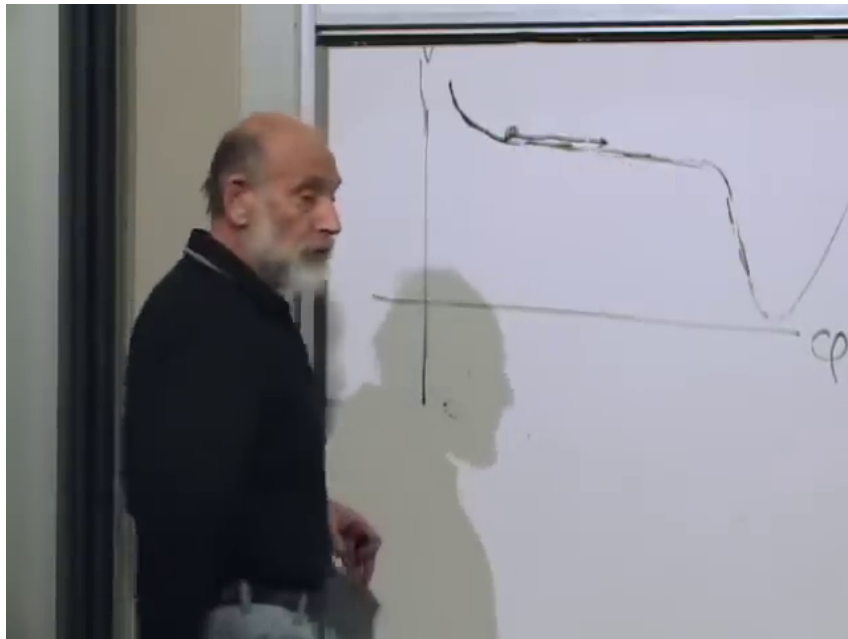


Figure 7.5: The inflationary potential as drawn late in the lecture: a high plateau, a sharp drop, and a lower minimum.

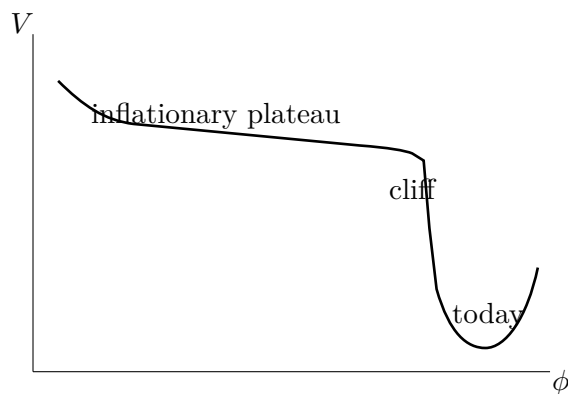


Figure 7.6: A schematic redraw of the lecture's high-plateau, cliff, and minimum structure. The large-scale shape, not the exact marker trace, is what matters.

The lecture is careful not to pretend that this is derived from first principles. It calls the picture contrived, even ridiculous-looking, but says that observational cosmology seems to demand something like it.

Now the inflationary stage becomes very vivid. High on the plateau the field moves slowly, because the slope is small and the cosmological friction is large. The lecture insists that the right analogy is viscosity rather than ordinary stick-slip friction. A rock falling slowly through honey is closer to the picture than a block on a rough table. The faster the universe expands, the larger this effective cosmological viscosity becomes. So the field drifts only very slowly:

$$\text{small slope} + \text{large cosmological viscosity} \implies \phi \text{ drifts slowly.} \quad (7.31)$$

While that happens, $V(\phi)$ changes only a little, so the expansion remains approximately exponential. That is the inflationary phase.

The lecture then turns this into a quantitative timescale. If the scale factor changes from a_i to a_f , the number of e-foldings is

$$N = \ln \frac{a_f}{a_i}. \quad (7.32)$$

One e-fold means multiplication by e . The lecture gives a lower bound of roughly fifty e-foldings, while noting that sixty is the standard figure often quoted:

$$N \gtrsim 50, \quad N \sim 60 \text{ as the usual benchmark.} \quad (7.33)$$

The e-folding time during inflation is of order

$$H_{\text{inf}}^{-1} \sim 10^{-32} \text{ s.} \quad (7.34)$$

The lecture also gives, cautiously, a rough ballpark for the plateau height, of order 10^{-14} in Planck units, though only as an estimate at the level of bounds.

At the bottom of the potential the geometry tells us the remaining vacuum energy today. Since

$$H^2 = \frac{8\pi G}{3} V, \quad (7.35)$$

the present minimum satisfies

$$V_{\text{today}} = \frac{3H_0^2}{8\pi G}, \quad (7.36)$$

while on the inflationary plateau one had

$$V_{\text{inf}} = \frac{3H_{\text{inf}}^2}{8\pi G} \gg V_{\text{today}}. \quad (7.37)$$

This is the lecture's way of reading the board sketch quantitatively: the top was much higher, the bottom is small but not zero.

Now the field reaches the cliff. Its potential energy turns into kinetic energy of motion, and then into heat, radiation, and particles. The lecture calls this reheating, while half-apologizing for the name. There is nothing particularly "re" about it unless one speculates about an even earlier prehistory. The essential point is that the universe is not hot on the plateau. Any pre-existing heat would have been diluted away by inflation. It becomes hot only after the fall.

This is also where the lecture deliberately plays with the phrase "Big Bang." One may, if one likes, speak of reheating as the Big Bang in a prehistory sense. But the visible Big Bang of ordinary cosmological language is much later. The surface of last scattering comes only after the universe has cooled enough to become transparent. In the lecture's potential sketch, everything we actually observe lies down near the bottom.

7.7.1 Question & Answer

Why must the plateau not be perfectly flat, and how do the remaining fluctuations become structure?

If the plateau were exactly flat, the field would have no classical force pushing it along. The lecture therefore insists that the plateau is not perfectly flat. It must have some tilt. At the same time it

cannot be too steep, or inflation would end too quickly. So one already sees a balance: enough tilt to move, little enough tilt to linger.

But there is a second balance. If the field moves too slowly for too long, then small quantum differences from one place to another become large differences in the times at which different regions reach the cliff. That would make the reheated universe too inhomogeneous. So the observed smallness of the primordial fluctuations constrains the flatness and height of the potential. The lecture is explicit about this: the observed $\delta\rho/\rho \sim 10^{-5}$ does not drop out of the qualitative picture automatically; it constrains the parameters of the picture.

The late part of the lecture then gives the central diagram. One plots ordinary space vertically and the field value ϕ horizontally. Surfaces of constant field are like contour lines on a map. If the field were perfectly homogeneous, those contours would be straight, and every region would fall over the edge together. Quantum mechanics spoils that exact homogeneity.

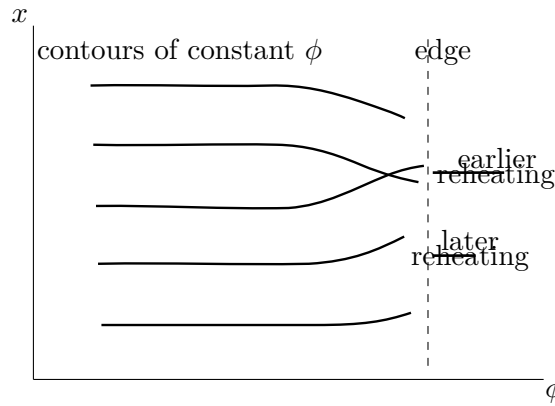


Figure 7.7: A transcript-driven contour sketch. Small quantum bumps in the field make different regions reach the cliff at different times, and those time delays become density inhomogeneities after reheating.

The lecture likes to describe this in several equivalent ways. One may imagine a little bump in an otherwise homogeneous field. One may imagine soldiers walking in lockstep toward a cliff, some a little ahead, some a little behind. Either way, the result is the same: some regions fall over earlier, others later.

Now the key point. The region that falls earlier reheats earlier and then has more time to expand and cool. The region that falls later is reheated later and remains hotter and denser. That is how a tiny fluctuation in ϕ becomes a fluctuation in energy density. The lecture summarizes the observed amplitude as

$$\frac{\delta\rho}{\rho} \sim 10^{-5}. \quad (7.38)$$

These are primordial density fluctuations, not dark energy. They are small, but without them there would be no galaxies.

After reheating, gravity amplifies them further. Over-dense regions attract more material; under-dense regions are depleted. So the tiny initial contrasts are magnified into the structure we see. The lecture stresses that this is not just a qualitative story. One can compute the fluctuation spectrum from quantum fields in an expanding background and compare it with the reconstructed primordial spectrum inferred from galaxies and the microwave background. The lecture's verdict is strong: it

fits very well.

And then the final pivot. These fluctuations do not just seed galaxies. They are processed into acoustic waves in the primordial plasma, and those acoustic waves provide a standard ruler on the sky. That ruler will be the subject of the next lecture, where geometry is measured from the angle subtended by structures of known physical size on the last-scattering surface.

7.8 Summary

The lecture begins by refusing to begin with inflation. Instead it rebuilds the de Sitter mechanism in its simplest form: constant vacuum-energy density implies constant H , and constant H implies

$$a(t) = a_0 e^{Ht}.$$

That mechanism is then used twice. First, to understand the present accelerated universe and the horizon scale $D = c/H$. Second, to motivate the idea that a much earlier de Sitter-like phase may have occurred in the remote past.

The student questions are not side remarks; they are part of the structure. They force the lecture to explain vacuum energy as zero-point energy and to think carefully about what stays fixed inside a de Sitter horizon. Only then does the main inflationary picture arrive. A hypothetical scalar field ϕ with potential $V(\phi)$ sits high on a nearly flat plateau, drifts slowly under a small slope and cosmological viscosity, falls off a cliff, reheats the universe, and finally settles near a low but nonzero minimum whose leftover energy is today's dark-energy-like vacuum energy.

The lecture's deepest payoff comes at the end. Inflation smooths the universe, but quantum mechanics prevents complete uniformity. Different regions reheat at slightly different times, producing primordial density contrasts of order 10^{-5} . Gravity then amplifies those contrasts into galaxies, while the microwave background preserves their early imprint. So the same mechanism that wipes out the initial wrinkles also leaves behind the small residue from which structure grows.

Chapter 8

Inflation, Structure, and the Fate of Initial Conditions

These notes follow Leonard Susskind's eighth lecture on cosmology, with the order of the argument and the strongest surviving board evidence kept close to the way the lecture actually unfolds. The classroom record is due to Leonard Susskind, LazyingArt LLC, and Video2Book. The lecture begins with a student's interruption about Ω , turns that interruption into a discussion of what we really know about the present universe, and then moves toward inflation as the proposed explanation of that knowledge. But the lecture never lets inflation stand as a slogan. At each stage a success creates a new problem: flatness demands inflation, inflation threatens structure, structure forces quantum fluctuations, and the explanatory power of inflation finally becomes an obstacle to recovering what came before it.

8.1 Omega, Vacuum Energy, and the Evidence for Flatness

A student asks what big Ω means, and Susskind immediately backs up to the Friedmann equation. There are several omegas in geometry and cosmology; this one is the cosmological bookkeeping device, not the Ω that denotes a sphere. The point is to understand how the expansion equation is organized in the late universe.

For the era of interest, well after radiation domination and including the present epoch, the Friedmann equation may be written as

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}(\rho_0 + \rho_m) - \frac{k}{a^2}, \quad (8.1)$$

where $H = \dot{a}/a$, ρ_0 is the vacuum-energy density, and ρ_m is the nonrelativistic matter density. Radiation could be added, but in the present universe it is so small that the lecture drops it at once. The important contrast is the way the two surviving energy densities scale:

$$\rho_0 = \text{constant}, \quad \rho_m = \frac{M}{a^3}. \quad (8.2)$$

Matter dilutes; vacuum energy does not.

At this point the transcript becomes rough, and the algebra on the board is not fully recoverable line by line. What the lecture is clearly doing, however, is dividing the Friedmann equation by H^2

and reading the resulting terms as normalized fractions of the expansion budget. Since the spoken notation wanders among Ω_0 , Ω_{vac} , and Ω_Λ , we standardize once and for all:

$$\Omega_\Lambda = \frac{8\pi G \rho_0}{3H^2}, \quad \Omega_m = \frac{8\pi G \rho_m}{3H^2}, \quad \Omega_k = -\frac{k}{a^2 H^2}. \quad (8.3)$$

Then

$$\Omega_\Lambda + \Omega_m + \Omega_k = 1. \quad (8.4)$$

In particular, if the universe is flat, $k = 0$, and therefore

$$\Omega_\Lambda + \Omega_m = 1. \quad (8.5)$$

That is the cleaned mathematical statement of what Susskind wants to say: Ω is the fraction of the expansion equation carried by the various forms of energy. Matter here includes dark matter. If the matter and vacuum terms add to more than one, then $k > 0$ and space is positively curved. If they add to less than one, then $k < 0$ and space is negatively curved. If they add to one, space is flat.

The lecture then turns from notation to evidence, and the order matters. First one measures ordinary matter and dark matter. The latter is not seen directly, but inferred gravitationally from things like stellar orbits in galaxies. Next one measures the Hubble expansion itself. Next one measures the curvature geometrically, by large-scale triangulation. Finally one notices that the universe is not merely expanding, but accelerating. In the lecture's language, the direct evidence for vacuum energy is not H by itself, but the fact that H is tending toward a constant, so that

$$a(t) \propto e^{Ht}. \quad (8.6)$$

If $\dot{a}/a$ becomes nearly constant, then one is on the exponential branch.

This is the first important beat of the chapter. Flatness is not introduced as a prejudice. It is presented as an empirical fact, checked several ways. Matter plus dark matter, the Hubble expansion, and the geometrical measurement of curvature all point in the same direction: however the universe got that way, it is very nearly flat, to within a few percent.

8.2 Why Flatness Is Not Enough: The Need for Structure

Once the evidence is on the table, Susskind asks the natural next question: why is the universe so flat? His answer is inflation. To say that the universe is flat, in the lecture's deliberately physical language, is to say that it is very big, very smooth, and very homogeneous on scales of billions, really tens of billions, of light years. Inflation is the conjectured early period of exponential expansion that stretches the universe into just such a state.

But now one can overdo a good thing. A perfectly smooth universe stays perfectly smooth. Gravity does not manufacture lumpiness out of exact symmetry. If the universe were exactly homogeneous, there would be no galaxies, no stars, no planets, and no observers to ask what Ω means.

Tracing the present universe backward, the lecture places the primordial inhomogeneities at the level of roughly one part in 10^5 by the time of transparency:

$$\frac{\delta\rho}{\rho} \sim 10^{-5}. \quad (8.7)$$

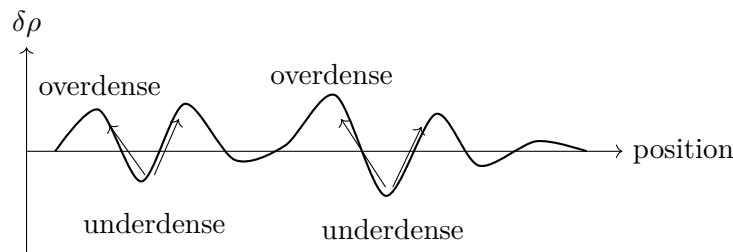


Figure 8.1: A transcript-backed schematic of the lecture's structure-formation story. Gravity deepens the overdense regions and empties out the underdense regions, so the contrast grows.

The universe was therefore extraordinarily smooth, but not perfectly so. There was a slight wiggleness in the density on many scales.

What happens next is the gravitational instability story. Small overdensities attract matter from nearby regions. Those nearby regions then become more underdense, which makes the original overdensities stand out even more strongly. Self-gravity amplifies the contrast rather than erasing it.

This is where the lecture draws a sharp contrast with ordinary matter in a room. If we made a small density perturbation in the air around us, pressure and thermalization would erase it. Air returns to equilibrium. Gravity does the opposite. Under self-gravity a little overdensity tends to grow into a bigger one.

The lecture also reminds us that what we directly observe on the sky is not the mass density itself but temperature anisotropy in the cosmic microwave background. Some of that temperature pattern comes from density fluctuations, some from velocity fluctuations, and the mapping is not trivial. Overdense regions tend to look slightly colder because photons lose more energy climbing out of deeper gravitational wells, though Doppler shifts from the motion of the gas are superposed on top of that. The numerical analysis is complicated. The robust conclusion is not: by the time the universe became transparent, there were already fluctuations of order 10^{-5} .

8.3 Pressure, Matter Domination, and the Onset of Collapse

At this point a student asks exactly the right question: if gravity amplifies overdensities, why did it not do so much earlier? Susskind's answer is immediate: pressure. The lecture returns to the equation-of-state relation

$$p = w\rho, \tag{8.8}$$

and asks us to remember the values

$$w_{\text{rad}} = \frac{1}{3}, \quad w_m = 0, \quad w_{\text{vac}} = -1. \tag{8.9}$$

The qualitative meaning is what matters. Radiation has positive pressure, and positive pressure homogenizes. It pushes disturbances out and tends to smooth the medium. That is why ordinary gases do not spontaneously collapse into lumps. Matter, by contrast, is effectively pressureless in the cosmological approximation. If one imagined a medium with $p = 0$ and no gravity, any inhomogeneity would simply sit there, frozen in. But once gravity is present, the absence of pressure support means there is nothing to prevent an overdensity from pulling matter inward.

So there is a competition:

$$\text{gravity} \quad \text{versus} \quad \text{pressure.} \quad (8.10)$$

During the radiation-dominated era the pressure was so large that it prevented gravity from doing its work. Later, when radiation became dynamically unimportant, gravity won and the lumps began to contract.

Dark matter enters at precisely this stage. The lecture is careful about the order. First the dark matter fluctuations begin to make large halos. The pressure is small, so those fluctuations can grow. Ordinary baryonic matter then falls into the dark halos, but baryons do require true dissipative processes if they are to settle toward the centers and make luminous galaxies. Dark matter does not dissipate in that way, so it remains as the extended halo.

There is also a timing subtlety that the lecture wants us to keep in mind. The matter-radiation crossover is not exactly the same event as transparency or recombination. Dark matter began to contract a little earlier, perhaps by a factor of ten in time, before the universe became transparent. That is why we do not directly see the beginning of the contraction in the microwave background. We see it only after the fact.

And here the lecture leaves us with a useful promise. In ordinary mechanics one expects dissipation to matter whenever energy is to be degraded. Susskind says there is also a cosmological kind of dissipation, one tied to expansion itself, and that we will need it later. This is not yet the inflaton story, but it is already the right intuition.

8.3.1 Question & Answer

Why didn't gravity make galaxies earlier? Because earlier on the universe was radiation dominated, and radiation carries positive pressure. With $p = \rho/3$, the radiation bath resists gravitational clumping and tends to homogenize the medium. Only after matter became more important than radiation did pressure cease to block collapse.

Dark matter could begin this process first. Its pressure was effectively negligible, so its small fluctuations grew into halos before ordinary matter could settle into stars and galaxies. The baryons followed later, and because they can dissipate, they collected nearer the centers of those halos.

8.4 The Inflaton and Its Potential

Only after the flatness problem and the structure problem are both alive does the lecture introduce the mechanism of inflation itself. The basic object is a scalar field ϕ . Its energy density has three pieces:

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + V(\phi). \quad (8.11)$$

The first term is due to time variation, the second to spatial variation, and the third is the potential-energy density.

Susskind motivates this by comparison with more familiar fields. An electromagnetic field has energy from its time dependence and from its spatial gradients. A scalar field has those same two kinds of energy, but in addition it may have a genuine potential $V(\phi)$. That is not exotic. The Higgs field is the standard example of a scalar field with a nontrivial potential. What is unusual here is the shape of the potential needed for inflation.

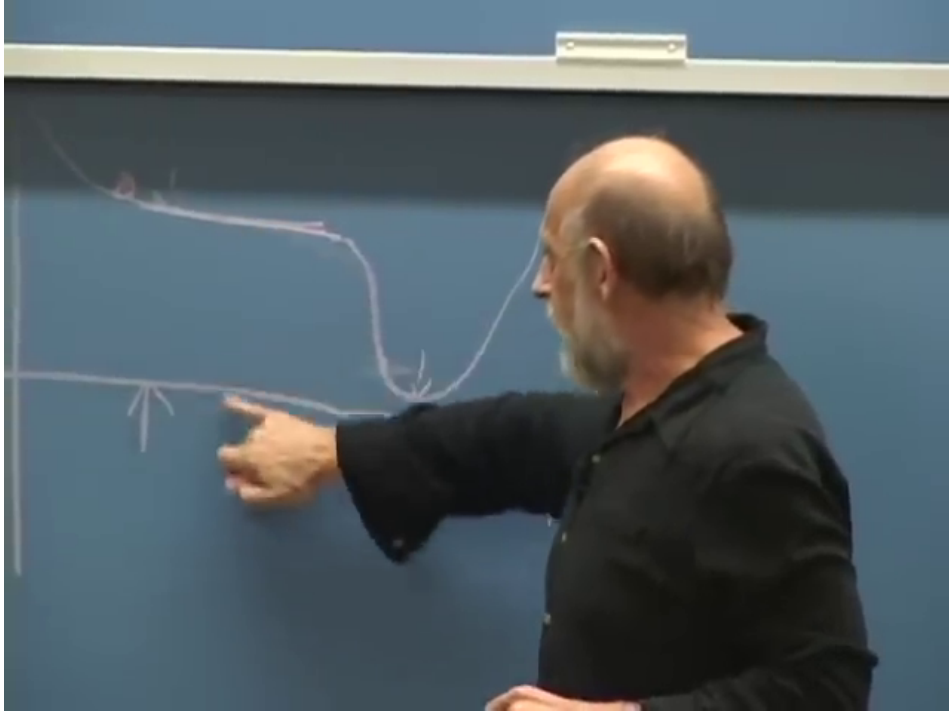


Figure 8.2: Inflationary potential with plateau and minimum. The lecture uses this blackboard sketch to emphasize the broad plateau, the steep drop, and the low residual minimum.

At the level of the lecture, the shape is more important than the microscopic origin. The field is supposed to sit at some value of ϕ on a broad, almost flat plateau. This is not a place in space; it is a value of the field. If the field initially varied a bit from place to place, inflation would stretch those wrinkles to such enormous scales that our observable patch would lie deep inside one nearly uniform region. That is why, for the later derivation, we are entitled to treat the field as approximately homogeneous.

The board sketch is only qualitative, but the qualitative structure is clear enough to preserve. The potential is high and nearly flat over some range, slopes gently downward, then falls sharply, reaches a low minimum, and rises again on the other side.

The lecture repeatedly insists that this is a cartoon, not a derived potential. It is contrived in the modest sense that some parameters have to be tuned. But it captures the dynamics we want. The field begins high on the plateau, rolls slowly, then goes over the edge. The minimum is not exactly zero. In Planck units the present vacuum energy is tiny,

$$V_{\text{today}} \sim 10^{-123}, \quad (8.12)$$

so the minimum must sit at a very small positive value. The plateau during inflation was much higher than this, though still much smaller than the Planck scale.

Now the key physical point: as long as the plateau is flat enough and the field is moving slowly enough, the first two terms in ρ_ϕ are negligible compared with $V(\phi)$. The field therefore behaves as a vacuum-energy source. The lecture writes this only schematically on the board, so we restore the standard normalization in print:

$$H^2 \simeq \frac{8\pi G}{3} V(\phi), \quad a(t) \propto e^{Ht}, \quad (8.13)$$

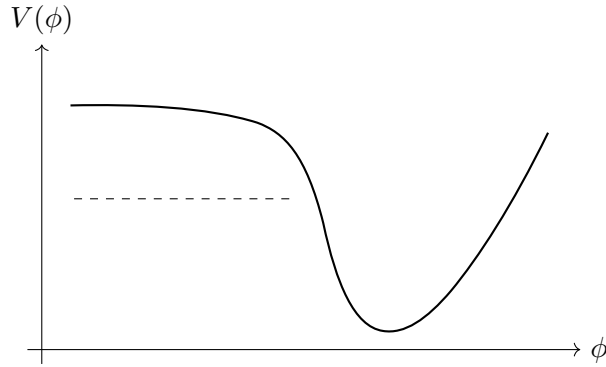


Figure 8.3: A qualitative redraw of the potential. The axis labels are note-writer reconstructions from the transcript; the robust content is the plateau, the cliff, the low minimum, and the rising branch.

with the understanding that the lecture itself only insists on the proportionality

$$H \propto \sqrt{V(\phi)}. \quad (8.14)$$

If $V(\phi)$ were perfectly flat and ϕ just sat still, this would be exactly vacuum-energy expansion.

The lecture's picture is then deliberately stark. All of cosmology, in this cartoon, is the story of a field rolling from high up on a potential toward a low minimum. Almost everything we can observe happened after the field went over the cliff. Going over the edge is where the large vacuum-like energy of the plateau was converted into more familiar energy: radiation, ordinary matter, dark matter, and a very small leftover vacuum energy.

8.5 Slow Roll, Reheating, and Hubble Friction

At this point the lecture raises the next obstacle. If the field falls off the edge, why should it not simply slosh forever? Why should it not overshoot, swing back, and perhaps even climb again? In ordinary mechanics one would ask where the dissipation came from. Susskind's answer is that cosmology provides its own dissipation. It is not genuine microscopic friction, but it behaves mathematically like friction.

To see it, we simplify exactly as the lecture does. We neglect the spatial gradients of the field, because inflation has stretched them out. We then treat $\phi(t)$ as an effective homogeneous degree of freedom. The top line of the board is cropped, so the Lagrangian must be regularized cautiously. The cleaned homogeneous form is

$$L_{\text{eff}} = a^3(t) \left[\frac{1}{2} \dot{\phi}^2 - V(\phi) \right]. \quad (8.15)$$

The factor a^3 is essential. The field terms are energy densities, and to get the effective Lagrangian for the homogeneous mode we multiply by the physical volume.

Now we work through the derivation in the same spirit as the lecture. First,

$$\frac{\partial L_{\text{eff}}}{\partial \dot{\phi}} = a^3 \dot{\phi}. \quad (8.16)$$

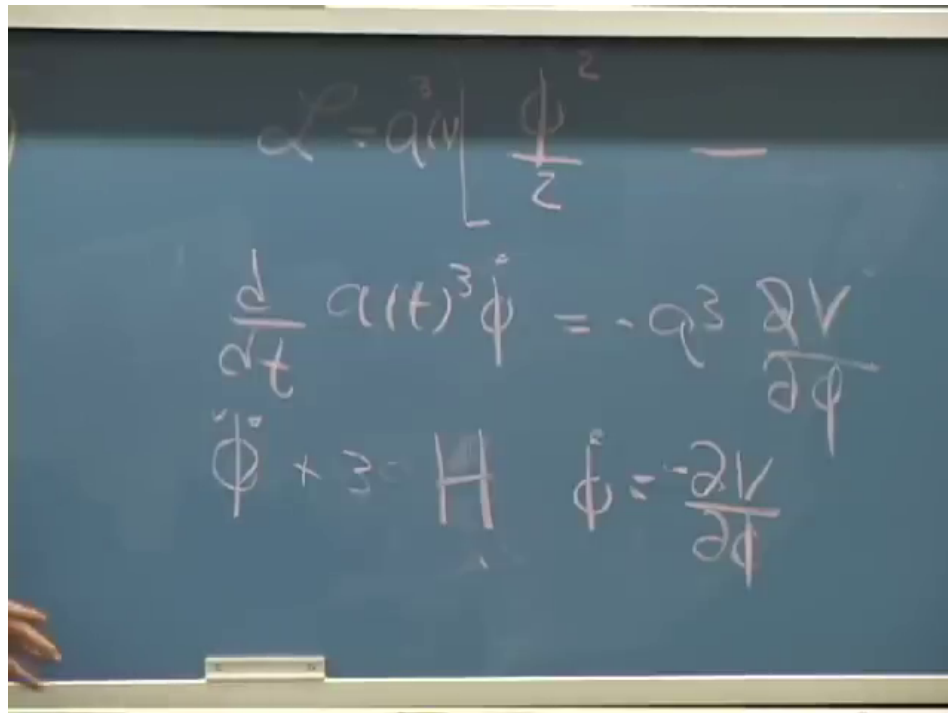


Figure 8.4: Scalar-field equation with Hubble friction. The board shows the Euler–Lagrange form and the final damped equation; the top Lagrangian line is only partial and is regularized in the text.

The Euler–Lagrange equation then gives

$$\frac{d}{dt} (a^3 \dot{\phi}) = \frac{\partial L_{\text{eff}}}{\partial \phi} = -a^3 \frac{\partial V}{\partial \phi}. \quad (8.17)$$

This is one of the secure board equations. Expanding the time derivative,

$$\frac{d}{dt} (a^3 \dot{\phi}) = a^3 \ddot{\phi} + 3a^2 \dot{a} \dot{\phi}. \quad (8.18)$$

Dividing by a^3 and writing $\dot{a}/a = H$, we obtain

$$\ddot{\phi} + 3H \dot{\phi} = -\frac{\partial V}{\partial \phi}. \quad (8.19)$$

The lecture explicitly corrects the sign at the board: the force is minus the derivative of the potential. That corrected sign is the one shown here.

This is the promised cosmological dissipation. The term $3H\dot{\phi}$ is what Susskind calls Hubble friction. It is not true friction. It came from differentiating the factor $a^3(t)$. But dynamically it behaves like the viscous drag on an object moving through a fluid. The lecture’s analogy is a rock falling through water. There is a force term, an acceleration term, and a term linear in the velocity that rapidly brings the motion to terminal speed.

That is the slow-roll idea in the language actually used in the lecture. If the slope is shallow, the terminal velocity is small. If the terminal velocity is small, the field moves only slowly across the plateau. During that long slow slide the potential energy remains almost constant, so the expansion remains almost exponential.

The field does eventually reach the cliff and then the minimum. Once it is near the minimum it oscillates, and those oscillations are just the quanta of the inflaton field. If those quanta are unstable, they decay into photons, electrons, positrons, and other ordinary particles. That is the lecture's reheating picture. The vacuum-like energy on the plateau is converted into the hot matter and radiation content of the post-inflationary universe.

8.5.1 Question & Answer

Why doesn't the inflaton oscillate forever? Because the expansion of the universe itself damps the motion. The term $3H\dot{\phi}$ continually degrades the coherent motion of the field, in much the same way that viscosity degrades the motion of a particle in a fluid. Then, once the field is oscillating near the minimum, those oscillations are the inflaton quanta, and if the quanta are unstable they decay into more familiar forms of matter and radiation. The sloshing is not protected; it is redshifted and then converted.

8.6 Quantum Fluctuations, Horizon Exit, and Scale Invariance

Now the lecture turns the knife once more. If inflation expands the universe by many orders of magnitude, should it not iron out every last irregularity? On the face of it, inflation seems almost too good. It explains flatness and homogeneity, but threatens to erase the very seeds that galaxies need. Susskind's answer is that the surviving irregularities are not leftover classical wrinkles. They are generated anew by quantum mechanics.

The lecture first asks us to forget expansion for a moment. Take a field that has already been ironed nearly uniform. Why should it still fluctuate? Because of the uncertainty principle. One cannot lock both ϕ and its velocity into exact values. The ground state of the field is therefore not dead still. It is agitated by zero-point motion, just as the ground state of a harmonic oscillator carries zero-point energy.

Why do we not ordinarily notice that agitation? Because most of the fluctuations are short wavelength, and short wavelengths oscillate rapidly. We average over them. In ordinary circumstances the field fluctuates too fast for those fluctuations to show up macroscopically.

Cosmology changes the story because the universe expands. We use a comoving coordinate x , which labels positions but is not itself a physical distance. The physical separation between two fixed comoving positions is

$$\lambda_{\text{phys}} = a(t) \Delta x. \quad (8.20)$$

During inflation the scale factor grows nearly exponentially, while the physical Hubble scale is approximately fixed:

$$d_H = \frac{c}{H} \simeq H^{-1} \quad (c = 1). \quad (8.21)$$

Therefore the comoving horizon shrinks:

$$d_{H,\text{coord}} = \frac{1}{aH}. \quad (8.22)$$

The lecture explains the freezing with a rope analogy. A wave oscillates because each bit of rope is pulled by neighboring bits. But if the wavelength becomes as large as the horizon, the relevant

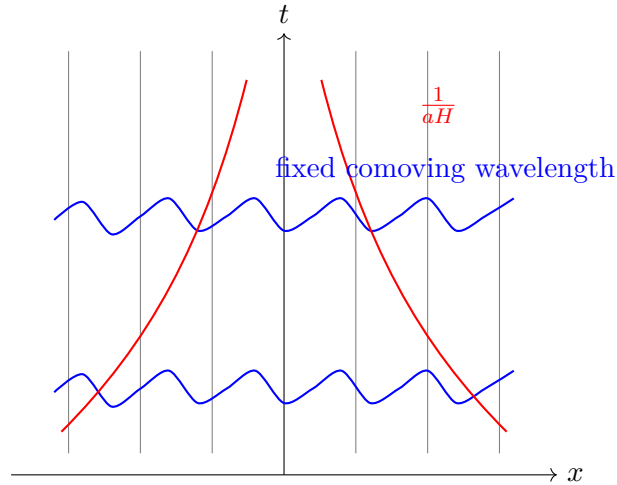


Figure 8.5: A comoving spacetime sketch for mode freezing. The mode keeps roughly the same coordinate wavelength, while the comoving horizon shrinks. Once the mode lies outside the horizon, it freezes.

neighbors lie beyond causal contact. They cannot communicate their pull. The restoring force disappears. Mathematically the result is that once a mode becomes longer than the horizon, it ceases to oscillate:

$$\lambda_{\text{phys}} \gtrsim H^{-1} \quad \implies \quad \text{freeze-out.} \quad (8.23)$$

That happens mode by mode. Longer wavelengths freeze first, shorter wavelengths later, but in an exponentially expanding universe every mode is eventually stretched beyond the horizon. The fluctuations do not vanish. They freeze into classical-looking variations of the field.

This is also where the lecture insists on something subtle and important. It does not matter how long inflation lasts. One might think that enough inflation would wash out everything. But as time goes on, new vacuum fluctuations are continuously generated on short scales, then stretched, then frozen. So the old fluctuations are stretched away, but new ones keep coming in. At a fixed point in space the field undergoes a random walk as one frozen mode after another is added with random phase. At different points in space the random walk is different. That is what builds a statistical pattern of large-scale variation.

The lecture's phrase for the resulting distribution is a scale-invariant spectrum. The exact statement is only approximate, and the lecture does not derive a closed formula for it. The physical claim is that the same freezing process repeats on successive logarithmic scales, so the amplitude comes out roughly the same from one wavelength band to the next.

There is one more step. During inflation there is a temporary, small horizon associated with the vacuum energy on the plateau. After the field falls off the cliff, the horizon grows again. Modes that had been stretched outside this temporary horizon eventually come back in. In the lecture's language they go out of the horizon and later re-enter the horizon. By that time they have been stretched to macroscopic scales, so when they start to slosh again they do so as huge, slowly varying structures.

8.6.1 Question & Answer

If inflation smooths everything out, why aren't the galaxy seeds erased? Because inflation smooths away preexisting classical irregularities, but it does not eliminate quantum zero-point fluctuations. Expansion continually stretches those fluctuations beyond the horizon, where they freeze into real field variations. Inflation therefore removes the old roughness and generates a new one. The galaxy seeds are not leftover dirt from before inflation; they are frozen quantum fluctuations produced during inflation itself.

8.7 What Inflation Predicts, What It Hides, and Why Fine-Tuning Returns

Once those frozen variations exist on the inflaton field, the lecture asks what happens when the field finally reaches the cliff. Here the fluctuations are not magnified in spatial size; inflation has already done that. What is magnified now is their amplitude as density contrasts.

Susskind's way of seeing it is to imagine nearby regions with slightly different values of ϕ . Regions a little farther to the right are a little farther downhill and therefore reach the cliff first. Suppose, for simplicity, that when a region falls off the cliff its vacuum energy is converted into radiation. That radiation dilutes with expansion:

$$\rho_r \propto a^{-4}. \quad (8.24)$$

A neighboring region still on the plateau remains vacuum dominated, and vacuum energy does not dilute. Therefore the earlier-falling region has already redshifted part of its energy away by the time the later-falling region finally makes the transition.

This is the origin of the final density contrast. A slightly later region emerges denser, simply because it waited longer before converting its energy into a form that dilutes. In a schematic note-writer reconstruction of the spoken argument,

$$\delta t \sim \frac{\delta \phi}{\dot{\phi}}, \quad \frac{\delta \rho}{\rho} \propto H \delta t \sim H \frac{\delta \phi}{\dot{\phi}}. \quad (8.25)$$

The lecture does not present this as a finished formal formula, so it should be read only as a compact summary of the logic. Its content is exactly what Susskind emphasizes verbally: the flatter the potential, the smaller the terminal velocity $\dot{\phi}$; the smaller the terminal velocity, the larger the final density variations. Likewise, the higher the plateau, the larger the fluctuations tend to be.

The width of the plateau plays a different role. It mostly controls how long inflation lasts, that is, how many e-foldings occur:

$$N = \ln \frac{a_f}{a_i} = \int_{t_i}^{t_f} H dt. \quad (8.26)$$

The height affects the size of H ; the width and slope determine how long the field stays on the plateau. So the lecture naturally separates two questions: how big are the fluctuations, and how many e-foldings took place?

If the height of the plateau is large enough, another observable can appear: gravitational waves produced during the inflationary era. Their presence would tell us about the overall height of the potential. Their absence already tells us that the scale cannot be too high. The lecture gives only

ballpark numbers here and says so explicitly. In Planck units the interesting window is of order

$$10^{-16} \lesssim V \lesssim 10^{-14}, \quad (8.27)$$

up to order-of-magnitude uncertainty. Above that range, one would already have seen an excessive gravitational-wave contribution in the microwave background. Below it, there may be no realistic chance of seeing inflationary tensor modes at all.

Susskind then pauses over the units. The explanation is the standard natural-units one. The Compton wavelength of a particle of mass m is

$$\lambda_C = \frac{\hbar}{mc}. \quad (8.28)$$

If we set $c = \hbar = 1$, then length has units of inverse mass. Energy density is energy per volume, so it has units

$$[\rho] = (\text{mass})^4. \quad (8.29)$$

That is the sense in which 10^{-14} , 10^{-16} , or 10^{-123} are being quoted.

At this stage the lecture takes a broader view of what inflation has accomplished. Its success is real. It correlates fluctuations on very different wavelength scales using essentially one underlying picture. But exactly because it stretches away earlier irregularities and replaces them with quantum fluctuations generated later, it also makes the pre-inflationary past difficult to recover. The success of inflation is that it washes out what came before. The problem is that the very same washing-out leaves us with little direct access to what came before.

That is why Susskind becomes cautious about the beginning. To say that the universe “started” high on the plateau is not to claim that this is the absolute beginning of everything. It only means that this is as far back as the inflationary description can be trusted in the cartoon. If inflation lasted too long, earlier information is erased. If we were extremely lucky and the number of e-foldings were only barely sufficient, perhaps some trace of an earlier stage might survive. But in general inflation tends to wipe out the past.

The lecture then adds a late geometric caveat. We have been using

$$a(t) \propto e^{Ht} \quad (8.30)$$

throughout, and that is correct for the inflationary branch at late times. But exact de Sitter space is more properly described, in a global slicing, by

$$a(t) \propto \cosh(Ht). \quad (8.31)$$

That solution contracts to a minimum size and then re-expands. Whether the contracting branch should be taken literally in cosmology is left as a question mark in the lecture. The point is not to settle it here, only to avoid mistaking the late-time exponential form for the whole exact story.

Finally the lecture returns to fine-tuning and anthropic reasoning. To first approximation the inflationary model has three parameters: the height V_0 of the plateau, its slope, and its width $\Delta\phi$. The slope is especially important because it controls the terminal velocity and therefore both the fluctuation amplitude and the duration of slow roll. The lecture remarks that many physicists are uneasy if $\Delta\phi$ becomes larger than the Planck mass, and more strongly that the energy density should not exceed order one in Planck units. Given such rough bounds, the raw fraction of parameter space

that yields enough inflation can be small. In the lecture's illustrative estimate, getting roughly 60 to 62 e-foldings is a few-percent tuning, on the order of 2%.

But then the lecture changes the question. There are two fine-tuning puzzles on the table. One is this mild tuning of the inflationary plateau. The other is the present cosmological constant, of order 10^{-123} . The proposed unifying explanation is anthropic: the universe had to allow structure to form.

For a positive cosmological constant, the Newtonian picture is that of a universal repulsion proportional to distance. If the present vacuum energy were only a couple of orders of magnitude larger than observed, it would already prevent the primordial 10^{-5} fluctuations from condensing into galaxies. That is the lecture's retelling of Weinberg's argument: the cosmological constant need not be exactly zero; it only has to be small enough that structure can form.

Negative curvature leads to the same kind of bound. In the Newtonian language reviewed earlier in the course, $k < 0$ corresponds to recession faster than escape velocity. Too much negative curvature means too much outward thrust, and that also prevents galaxies from forming. Inflation dilutes curvature, so one can ask how many e-foldings are required before structure formation becomes possible. The lecture quotes

$$N_{\min} \approx 59.5, \quad N \gtrsim 62 \quad (8.32)$$

as the relevant benchmark. Fewer than about 59.5 e-foldings would leave too much curvature for galaxies to condense; our existence therefore tells us that the inflationary history lay beyond that threshold.

This changes the meaning of probability. If one asks for the raw volume of parameter space that gives sufficient inflation, the answer may be only a few percent. But if one conditions on being in a universe where galaxies exist and observers can ask the question, then the forbidden part of parameter space is discarded at the outset, and the conditional probability can become large. The lecture does not present that as the end of the matter. It presents it, very deliberately, as the place where the great puzzles, the great confusions, and the great debates now live.

8.8 Summary

The lecture begins with a student's question about Ω , because inflation is not introduced in a vacuum; it is introduced as the proposed explanation of an observed fact, namely that the universe is almost flat. Flatness then forces the question of structure, because a universe made too smooth could never make galaxies. That leads to the pressure-versus-gravity story, the special role of dark matter, and finally the inflaton: a scalar field on a shallow potential whose vacuum-like energy drives exponential expansion. The field rolls slowly because expansion itself produces Hubble friction. Inflation does not erase all seeds of structure, because quantum fluctuations are continually created, stretched beyond the horizon, and frozen in. Those frozen fluctuations later re-enter and become the density variations from which galaxies grow. The same framework is impressively successful and deeply frustrating: it explains a great deal by washing out initial conditions, and therefore makes the earlier history correspondingly hard to recover.